The Geography of Protests: Spatial Fragmentation, Exposure, and Participation in the George Floyd Movement

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Abstract

Whether protest movements win votes for their cause or cost them votes is unsettled. This paper studies the question by treating protests as spatial events. I ask whether residents of neighborhoods near protest sites, who are exposed to a protest, react differently when they vote from those of neighborhoods where participants come from. Combining 7,920 geocoded George Floyd protest events with mobile phone location data, I measure protest participation from each block group. I then exploit differences in exposure and participation within the same zip code to estimate the effect of each on the change in vote shares between 2016 and 2020. Neighborhoods near protest sites saw increases in Republican vote share, a backlash that decays with distance and is larger near violent protests. Neighborhoods whose residents attended protests saw increased Democratic vote share. The effects coexist: the backlash is large but concentrated in a few neighborhoods, the mobilization small but spread across many neighborhoods. Participation moved about 137,000 votes toward the Democrats, while exposure near protest sites cost them about 87,000; the net is not distinguishable from zero. On these estimates, violence cost the movement about half of its potential local gains. The protests' direct local effects are unlikely to have decided the 2020 election.

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1 Introduction

Protests are a common way in which people express political views in democratic societies. In the United States, the Crowd Counting Consortium recorded about 72,000 protest events between January 2017 and the end of 2020. The protests that followed the murder of George Floyd in May 2020 were the largest of those four years: according to the New York Times, between 15 and 26 million people took part in the first weeks of the protests, which could make them the largest movement in the history of the United States (Buchanan et al., 2020). Recent research in the social sciences has asked whether protests change public opinion and voting, and has found that they sometimes do (Madestam et al., 2013; Wasow, 2020; Gethin and Pons, 2026). What is less clear is the direction: the same movement can win votes for the cause it advances and cost it votes, a trade-off that the evidence so far does not settle.

I study this trade-off by treating protests as spatial events. Protests take place at physical locations, and the people who attend travel there from where they live. A protest therefore reaches the residents of two kinds of neighborhoods: those where it takes place, whether the residents want it or not, and those where its participants come from. Those two groups of residents need not react in the same way, so that a protest may win votes in one kind of neighborhood and lose them in the other. Exposure to a protest nearby might generate a backlash against the movement, or sympathy for its cause. Participation in a protest might generate mobilization for the movement, or wear down the appetite for protesting. Because studies that compare counties or congressional districts see only the sum of the two reactions, this paper asks how protests affect voting where they take place and where their participants live.

The George Floyd protest movement of 2020 offers a good setting to answer this question for four reasons. First, the wave was large and spread out. Between 27 May and 2 November 2020, 7,920 protests took place at 4,815 venues in 368 metropolitan areas, and more than half of the neighborhoods where they took place had seen no protest in the three preceding years. Second, the 2020 presidential election came five months after the protests began, so that a change in opinion could be expressed at the polls. Third, the movement was nationally salient. It was in the news for weeks, and its cause was the same in every city. Fourth, the cause quickly took a partisan side, so that a reaction to the protests could be traced to a vote for one party or the other.[1]

I assemble a novel dataset at the level of the census block group, the smallest unit for which the Census publishes demographic estimates. The dataset combines three types of information: protest venues, protest participation, and voting in presidential elections. The venues are the locations of the George Floyd protests recorded by the Crowd Counting Consortium, which I geocode to the street address from the media reports behind each entry. Participation is measured from the real-time location pings of 12.9 million mobile devices in 2020. I assign the home of each device

[1] In June 2020, 67 percent of American adults supported the Black Lives Matter movement, 91 percent of Democrats and 40 percent of Republicans; by September, support among Republicans had fallen to 19 percent while support among Democrats was little changed, at 88 percent (Pew Research Center, 2020a,b).

to the block group where it spent most of its nights, and define a resident as attending a protest when their device is observed at a venue while a protest is held there. To my knowledge, this is the first paper to use these data to measure protest participation in a months-long movement at the national scale. The voting data are the block-group returns of the presidential elections of 2012, 2016, 2020 and 2024, allocated from precinct returns in proportion to population.

The George Floyd protests did not take place in a few central locations, but simultaneously at multiple venues within the same city.[2] As a result, many neighborhoods were close to a protest, and the people who attended came from all over the city. I exploit this variation in exposure and participation to study the effects of protests on voting in the neighborhoods that received a protest and in those that sent participants to them. For exposure, I compare block groups that had a protest nearby with block groups of the same zip code that did not. For participation, I compare block groups that sent residents to a protest with block groups of the same zip code that sent none. In both cases, the outcome is the change in a block group's vote between the 2016 and 2020 presidential elections. Comparing block groups within the same zip code holds fixed the effect of the protests through the media and national politics, which is common to all block groups of a zip code, and isolates the direct effect of protests, through exposure or participation.

Exposure generated backlash against the movement. Block groups with at least one protest within one kilometer shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code without one, holding fixed whether a block group had protests nearby before the George Floyd wave. Near violent protests—those with reported arrests, injuries or property damage—the point estimates are larger and the shift is detectable out to three kilometers rather than two.

Participation generated mobilization for the movement. Block groups where at least one resident attended a protest shifted 0.07 percentage points more toward the Democratic candidate than block groups of the same zip code that sent none, holding fixed how many of their residents regularly visited the protest venues and how mobile their residents were before the George Floyd protest wave. The shift comes from participation in nonviolent protests and from residents who would not have been at the venue without the protest. Heavier attendance at violent protests, and attendance by residents who regularly visited the venue and would have been there anyway, for work or other reasons, moved the vote away from the Democratic candidate instead.

Neither exposure nor participation is random: protests do not take place in random locations, and residents choose whether to attend. My design does not require that they be randomly allocated. The identifying assumption is that the vote in the block groups near a George Floyd protest, or in those that sent residents to protests, would have moved like the rest of their zip code had the protests not taken place, given their demographic characteristics before the wave. Two tests support this assumption. First, a placebo exercise separately estimates the effect of exposure and

[2]The New York metropolitan area alone had 291 venues, and in New York, Los Angeles, San Francisco and Chicago alike most venues hosted a single event (Figure 1).

of participation on the change in vote between the 2012 and 2016 presidential elections: neither coefficient is distinguishable from zero. Second, a difference-in-differences regression follows the same block groups across the four presidential elections from 2012 to 2024, comparing those within 2.5 kilometers of a protest with the rest of their zip code: the Republican vote share of the two groups did not move apart between 2012 and 2016, and diverged in 2020 and 2024. These tests support the design, but they cannot rule out that some other event taking place between 2016 and 2020 and unrelated to the protests moved the vote of the block groups near a venue, or of those that sent residents to protests, and not the vote of the rest of their zip code. Such an event would have to be specific to those block groups, absent before 2020 and present again in 2024. Because this shock would have to be so narrowly confined in space and time, I find it implausible that one exists.

When the backlash and mobilization effects are estimated together in one regression, they coexist and pull in opposite directions. However, they differ in reach. Per neighborhood, the backlash effect is larger, but it is confined to the relatively small number of neighborhoods located within a few kilometers of a protest, while the mobilization effect is smaller but spread across many neighborhoods, because most of them sent people to the George Floyd protests. The magnitude of the two effects can be compared with each other, and with other shocks to voting studied in the political economy literature, through their persuasion rates, which I calculate following DellaVigna and Gentzkow (2010). Because this rate accounts for how many residents each treatment reached, on this scale the mobilization effect is the larger of the two: its rate is at the lower end of the range estimated for media interventions, while the rate of the backlash effect is on the order of a candidate's whole television advertising campaign and below the arrival of Fox News on local cable (Spenkuch and Toniatti, 2018; DellaVigna and Kaplan, 2007).

At the county level, Gethin and Pons (2026) find that the George Floyd protests increased support for the Democratic Party. Their design compares counties with more and less protest activity, so it measures the effect of the protests on a county as a whole. At that level, the backlash in the neighborhoods near a protest and the mobilization in those that sent participants to it are added together, and only their sum can be seen. My results are complementary to theirs: by comparing neighborhoods within the same zip code, I separate the two effects and show that the sum hides a backlash concentrated near the protest venues. At the same time, the effect of the protests on the vote through the media and national politics, which is common to all neighborhoods of a zip code, is outside my design.

Participation moved about 137,000 votes toward the Democrats and exposure within three kilometers of a protest site about 87,000 toward the Republicans. Exposure beyond three kilometers adds an imprecisely estimated 54,000 votes toward the Democrats, so the net of the three—about 104,000 votes toward the Democrats—is not distinguishable from zero. The direct local effects of the George Floyd protests were too small and spread out to decide the 2020 election. Of the six states where the 2020 presidential election was closest, Georgia is the only one where the estimated

shift is distinguishable from zero and comes close to the final margin, but even there falls short of it.

The George Floyd protest movement differed from recent protest movements in the United States in its high level of violence and its geographic fragmentation. These two facts motivate two counterfactual exercises.[3] First, I simulate what the effects of protest exposure would have been at different levels of violence. Had no protest turned violent, the net shift of the two effects would have been about twice as large as the one observed. On these estimates, violence cost the movement about half of its potential local gains. However, I estimate that one protest in four would have needed to turn violent for the backlash effect to outweigh the gains of the movement, a level of violence much higher than observed. Second, moving every protest to a venue that had hosted protests before the George Floyd wave would not have detectably changed the aggregate effect of the protests on the 2020 vote. On these estimates, it was the violence of the protests, and not where they took place, that cost the movement votes.

Finally, North Carolina's individual voter files show the channels behind the backlash effect. Near violent protests, the voter roll became less Democratic because new registrants were less Democratic than the roll they joined, while the turnout of those already registered did not detectably change; near nonviolent protests, turnout fell, most among occasional voters. The mobilization effect is estimated too imprecisely in the state to distinguish an effect of the national size from zero.

Related literature. This paper contributes to four streams of literature in political economy. First, the paper builds on work on the effects of protests on public opinion, voting and policy (Madestam et al., 2013; Wasow, 2020; Enos et al., 2019; Hungerman and Moorthy, 2023; Caprettini et al., 2026; Gethin and Pons, 2026; Klein Teeselink and Melios, 2025; Reny and Newman, 2021; Mazumder, 2018; González and Larreboure, 2021; Leon-Ablan and John, 2024; Wallace et al., 2014).[4] Most of this work operates at coarse geographic levels, like counties or congressional districts, where the effects of protests through mass media, party politics and direct exposure are bundled together.[5] I contribute to this literature by separating the direct effect of protests on voting in the neighborhoods where protests take place and in those where their participants come from.

Caprettini et al. (2026) also treat protests as spatial events and find that Nazi marches in 1932 Hamburg persuaded voters in some neighborhoods along their route while repelling them in others. My paper departs from theirs in three ways. First, I study protests whose organization was largely decentralized, while they focus on marches organized by a party as campaign propaganda. Second, thanks to my granular data on phone location I can measure participation at the block group level

[3] Both exercises rest on two strong assumptions: participation is held at its observed level in every scenario, because the attendance response to a protest's violence is not modeled, and the geography of the protests is unchanged, with violent protest days relabeled as nonviolent at random until the violent share reaches its target.

[4] On the George Floyd wave, Gethin and Pons (2026) find that, of fourteen waves of American protests since 2017, it is the one that moved votes at the county level, and Klein Teeselink and Melios (2025) that the counties with more protests voted more Democratic in 2020.

[5] An exception is Enos et al. (2019), who find that support for liberal policy rose in the precincts near the 1992 Los Angeles riot in the ballot measures that followed.

and estimate its effect on voting, while they study only exposure along the routes of the marches. Third, my analysis covers hundreds of metropolitan areas in a contemporary setting, while they study one city. To my knowledge, my paper is the first to separate the effects of exposure and participation on voting at the neighborhood level, across thousands of protests and hundreds of metropolitan areas.

Second, the paper speaks to the literature on protest participation and collective action (surveyed by Cantoni et al., 2024). This work has studied the decision to protest as a strategic choice (Cantoni et al., 2019; Chenoweth et al., 2022), the role of social media, mobile phones and social networks in coordinating protests (Enikolopov et al., 2020; Manacorda and Tesei, 2020; González, 2020), and the persistence of engagement after a first protest (Bursztyn et al., 2021). The closest to mine in data is Sonin et al. (2023), who use device traces to profile who traveled to the Capitol on 6 January 2021. In these studies participation is either the outcome or an individual treatment. Instead, I treat participation as a neighborhood-level treatment and estimate its effect on the vote where participants live.

Third, the paper relates to the literature, surveyed by DellaVigna and Gentzkow (2010), studying how media (DellaVigna and Kaplan, 2007; Enikolopov et al., 2011; Adena et al., 2015; Martin and Yurukoglu, 2017; Spenkuch and Toniatti, 2018) and in-person contact (Gerber and Green, 2000; Pons, 2018; Kalla and Broockman, 2018) move votes. I contribute to this stream of work by adding protests as a further local, in-person shock to the vote.

Fourth, this paper adds to the evidence that where people live shapes how they vote (Cantoni and Pons, 2022; Autor et al., 2020) and to the literature that has studied the urban economics of unrest (DiPasquale and Glaeser, 1998; Collins and Margo, 2007; Brooks et al., 2026).

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 documents where the George Floyd protests took place and how fragmented they were within cities. Section 4 estimates the effect of exposure to a protest nearby on the vote, and Section 5 the effect of participation. Section 6 puts the two effects together, compares their size and estimates their net effect on the 2020 presidential election. Section 7 uses North Carolina's voter files to ask through which channel each shift arrived. Section 8 concludes.

2 Data

I put together a dataset at the level of the census block group, the smallest Census unit with published demographic estimates, which links three sources: the locations and dates of the George Floyd protests, geocoded from the record of the Crowd Counting Consortium; the participation of each block group's residents in those protests, measured from the real-time location data of mobile devices; and the returns of the presidential elections from 2012 to 2024, reported by precinct and allocated to block groups. A fourth source, North Carolina's voter files, records who was registered

and who voted, separately from the vote itself; Section 7 uses it to ask through which channel the vote effects arrive. This section describes each source and what it measures. Appendix A describes how each source was compiled, from the selection of the protest events to the construction of the North Carolina measures, and Appendix B validates the participation measure.

2.1 Protest events and venues

The protest events come from the Crowd Counting Consortium (Chenoweth et al., 2025), a project of the Harvard Kennedy School and the University of Connecticut that has recorded political crowds in the United States since January 2017. The Consortium finds events mainly in the press. Research assistants read the reports and code each event, and the Consortium publishes only the records it can confirm. One record is one event. It carries the date, the city and the state, the location within the city, when the reports gave one, the type of action, the organizers, a summary of the participants' claims, the crowd size, and whether arrests, injuries, property damage or the police's use of chemical agents were reported.

I use the Consortium's compiled file for 2017 to 2020, about 72,000 events. The George Floyd wave includes every event dated between 27 May and 2 November 2020 whose claims concern Black Lives Matter, racial justice or police violence; counter-protests are excluded. The Consortium's own coordinates locate only the city or town, so I geocode each event's location. I keep the events whose coordinate is precise to a street address, a named place or a street, and drop those located only to a town or a district. The selection, reported step by step in Appendix A.1, leaves 7,920 events at 4,815 venues. The same procedure gives the 36,974 protests from January 2017 to 26 May 2020, which the estimates use to control for the places that had hosted protests before.

The wave's events fall in 4,025 block groups of 368 metropolitan areas.

An event is violent when the Consortium's record reports any arrest, injury to participants or to police, property damage, or the use of tear gas or other chemical agents by the police. That is 823 of the 7,920 events, 10.4 percent. The classification, which is built by the source, is the one used wherever the paper splits protests by violence.

2.2 The device panel and the participation measure

Participation is measured from the real-time location pings of mobile devices in the United States, from 31 December 2019 to 31 December 2020. Each device is homed to the block group where it spent most of its nights in September and October 2020, after the wave's peak and before the election. The panel holds 12.9 million homed devices in 173,544 block groups of the 368 metropolitan areas. Table A1 reports the panel, the attendance it detects and the voting sample. The panel's coverage per resident is nearly flat across income, race, education and renter share and falls with density, and the block groups it covers match the resident population on seven of eleven characteristics (Figures A1 and A2).

I define a resident of a block group as attending a protest when a device homed there is observed within 200 m of a venue, between noon and midnight local time, on a day that venue hosted a George Floyd protest. Presence requires at least two pings inside that radius spanning at least ten minutes, so that a device passing by does not count; pings with a reported location accuracy worse than 100 m are dropped. Summed over the wave, the panel records 385,891 such attendances at 200 m, from 108,679 block groups. The participation measure of a block group has two parts: whether any of its residents attended, the extensive margin of participation, and the number of attendances as a share of the devices homed in the block group, the intensive margin. Residents who were regulars at those venues before the protests, present there on ordinary days for work or other reasons, are counted as attending. Their ordinary presence at the same venues in the three weeks before the first event enters every regression as a control, so that block groups are compared at the same level of ordinary presence. Section 5 writes the specification out.

The measure has two limits. It is a relative index of participation across block groups and not a count of participants (Figure A4). It can be built only for protests whose date and venue are known. Appendix B shows what the device data can and cannot see (Figure A3) and what the measure tracks (Figure A5).

2.3 Election returns and block-group characteristics

The electoral outcomes are the returns of the presidential elections of 2012, 2016, 2020 and 2024 at the block-group level. For 2016, 2020 and 2024 they are precinct returns that I allocate to block groups in proportion to the population of each piece of the intersection of a precinct with the block groups. Appendix A.3 describes the method, its coverage and how it compares with two recent papers that applied a similar procedure. The 2016 and 2020 returns are the precinct files of the Voting and Election Science Team (Voting and Election Science Team, 2018, 2020). The 2024 returns are the precinct files of the New York Times (The New York Times, 2025), completed with the Redistricting Data Hub files for four states. The 2012 returns are the presidential results of the panel of Longuet-Marx (2026), which reports them at the block-group level. They cover 42 states. Table A2 reports the coverage by year and source.

The outcome throughout the paper is the change between two elections in a party's share of all votes cast for president in the block group, in percentage points, so that the estimates describe how a block group's vote moved between two elections and not whether it leans toward one party. The denominator is all votes cast, not the two-party vote. The exposure sections use the change in the Republican share and the participation sections the change in the Democratic share.

The block-group characteristics come from the 2014–2018 American Community Survey five-year estimates: the shares of residents who are white, Black, Hispanic and Asian, median age, median household income, the unemployment rate, the share of residents aged 25 and over with an associate degree or higher, and the renter share. Population and density are the 2010 census counts. To these I add the distance to the central business district. They enter the regressions as controls and are

the dimensions along which I examine whether the effects vary. Every regression also carries the share of residents with a bachelor's degree or higher, on its own and interacted with the four racial shares.

The participation estimates of Section 5 are fitted on the block groups of the 368 metropolitan areas with vote returns in 2012, 2016 and 2020 and a device panel: 135,691 block groups in 333 metropolitan areas, of which 85,399 sent at least one resident to a protest. The sample is urban and not representative of the national electorate, but it is a large part of it: the ballots cast in the sample in 2020 are 68.5 percent of all ballots cast nationally that year. The exposure estimates of Section 4 do not need the device panel and are fitted on a different set of block groups; each section states its own sample.

2.4 The North Carolina voter file

North Carolina's State Board of Elections publishes the state's voter registration file and its voter history file, and Section 7 uses them to ask through which channel the vote effects arrive. I use the registration file as it stood on the day of each general election from 2012 to 2024. Each record is one registered voter, with a permanent identifier that follows the person across snapshots, the registration status, the party of registration, the residential address, and the person's age, sex, race and ethnicity. The history file lists, for each registered voter, the elections in which a ballot was cast. It begins with the general election of 2016. Neither file records how anyone voted.

A person is on the roll at a snapshot when registered with an active or inactive status.[6] Every residential address is placed in its block group with the Census Bureau's geocoder.

From the two files I build three measures at the block-group level: turnout, the composition of the roll, and the movements that change that composition. Turnout is measured over a window between two elections, for the people who were on the roll at both of them: at each of the two elections, it is the share of those same people who cast a ballot, so that the two rates are computed on one fixed set of people and their difference is not moved by who joined or left the roll in between. The composition of the roll is the share of registered voters in each of four classes, Democratic, Republican, unaffiliated, and a residual of the other parties, at each election. The movements are the five ways in which that composition changes between two elections, counted person by person: people newly register, leave the roll, move in from elsewhere in the state, move out, or change the party they are registered with; the five add up exactly to the change in a party's share. Appendix A.4 defines the movements in full and the sorting of voters by their record before the protests.

None of the measures above is a vote share. The vote shares of Section 7 come from the election returns of Section 2.3; the voter files say whether a change in them came through who voted, who was on the roll, or who changed party.

[6] The state marks a registrant inactive after a period without voting or an unanswered address confirmation, so inactive status is partly defined by not voting. The notes of Section 7 carry that caveat where turnout is measured.

The North Carolina analysis uses the North Carolina block groups of the national samples: 3,655 in the exposure specification and 3,649 in the participation specification. The individual-level estimates use the 3,606,754 people registered at both the 2016 and the 2020 snapshots and living in one of those block groups in 2016. It is neither the national sample nor a random sample of the state, and block groups outside it are unobserved rather than untreated. Section 7 reports the estimates, and Appendix G the specifications and the exhibits in full.

3 Protest Venues and Spatial Fragmentation

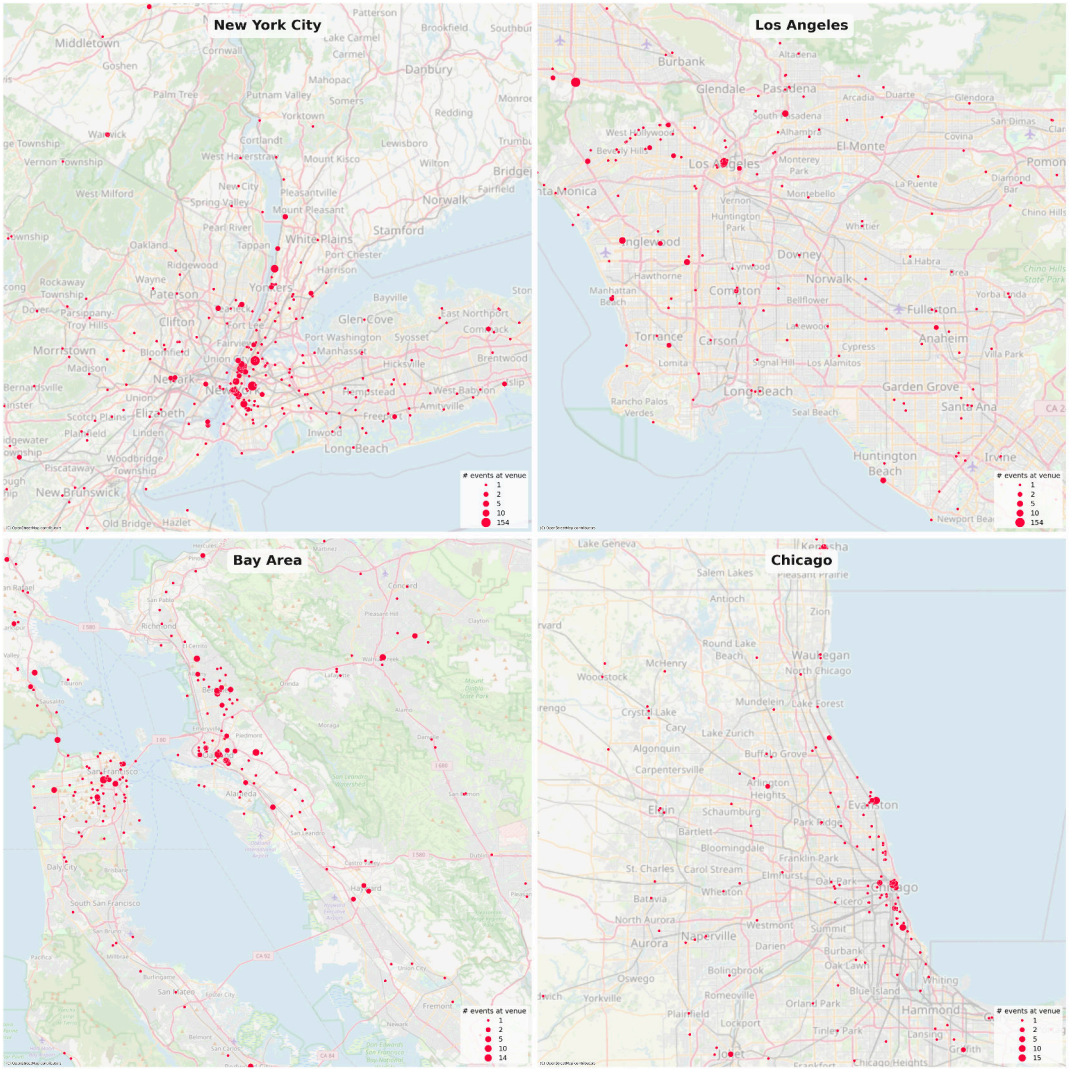
The George Floyd protests did not gather in a few central squares. Figure 1 maps the venues of four cities over the wave, and each venue is sized by the number of events it hosted. In each city a few large markers stand out and many small ones are scattered through the towns and neighborhoods around them; most venues hosted a single event. Reported attendance was concentrated in the same way: in each of the four cities a single venue accounts for between a tenth and just over a third of the total crowd reported, while the smaller protests were spread across more than a hundred venues (Figure A6).

Table 1 places the wave beside the protests of the three preceding years and beside the other protests of the same months. Its 7,920 events took place in 4,025 block groups, 1.97 events per block group against 2.63 for the protests of January 2017 to May 2020. The wave was spatially fragmented: its events were spread over more places than the protests before it, and half of its block groups, 51.4 percent, did not host a protest in the past or in the same months. The crowds were not larger than those of earlier protests: the mean reported crowd is 464 people against 831. What set the wave apart is violence. The Crowd Counting Consortium codes 10.4 percent of its events as violent, against 2.2 percent of the earlier protests and 2.7 percent of the other protests of the same months, roughly four times; the share of events with arrests is more than three times larger, and the share with property damage larger still.

The block groups that hosted the wave also differ from those that hosted earlier protests. Against both comparison groups, wave events took place in block groups that are denser, have a higher Black population share, lower median income and a lower share of residents with an associate degree or higher, and a slightly higher Democratic vote share in 2016 (Table A3).

Two features of the wave matter for what follows. First, the venues were numerous and scattered across space: not only within a city, but also within a zip code, some block groups had a protest close by and others did not. Second, one event in ten turned violent, so that among the block groups with a protest nearby, some were near a violent one and most were not. The first feature gives the variation in exposure that Section 4 uses; the second gives the split between violent and nonviolent exposure that runs through the paper.

Figure 1: Protest venues by number of events in four cities



Notes: Each panel maps the protest venues of one metropolitan area around its central city. A marker is one venue, sized by the logarithm of the number of George Floyd protest events recorded there. The record holds 291 venues and 660 events in the New York metropolitan area, 178 and 400 in the Los Angeles one, 241 and 377 in the San Francisco Bay Area and 171 and 228 in the Chicago one; each panel shows the city and its surroundings, so some venues of the metropolitan area lie outside the panel. The busiest venue of the Bay Area, with 20 events, is one of those: the largest size in that panel's legend is 14 events.

4 Protest Exposure

This section asks what a protest nearby did to the vote of the block groups around it. Exposure to protests could in principle move the vote in either direction. Votes moving toward the Republican candidate would suggest a backlash against the movement, while votes moving toward the Democratic candidate would suggest growing sympathy with the movement. Section 4.1 states the comparison and the specification; Section 4.2 reports the estimate for exposure to any protest and Section 4.3 for exposure to a violent one; Section 4.4 presents a second design, which follows the

Table 1: The George Floyd wave against earlier and contemporaneous protests

	George Floyd (5/27–11/2/20)	Pre-period (1/2017–5/2020)	Non-George Floyd (5/27–11/2/20)
Panel A: Counts			
N events	7,920	36,974	4,606
N unique block groups	4,025	14,058	2,692
Events per block group	1.97	2.63	1.71
Panel B: Event characteristics (event-weighted means)			
Mean crowd size	464	831	328
% violent	10.4	2.2	2.7
% with arrests	7.0	2.0	1.7
% with property damage	4.4	0.2	0.6

Panel C: Also hosted (% of the column's block groups)	All George Floyd block groups	With a violent George Floyd event
A pre-period protest (1/2017–5/2020)	44.8	59.5
A violent pre-period protest	5.4	15.7
A non-George Floyd protest, same months	18.6	34.8
A violent non-George Floyd protest	1.2	4.6
A violent George Floyd event	13.5	—

% of George Floyd block groups that did not host a protest in the past or in the same months: 51.4%

Notes: The table compares the George Floyd wave with protests in the three preceding years and with other protests in the same months. In Panels A and B, column 1 is the George Floyd wave: the Black Lives Matter events; column 2 is every protest event from January 2017 to May 2020; column 3 is every other protest event of the same months. Events are the Crowd Counting Consortium records geocoded to a venue. Panel A counts events and distinct block groups and reports their ratio. Panel B reports event-weighted means: the media-reported crowd size; the share of events coded violent in the Crowd Counting Consortium record; the share with arrests; the share with property damage. The violence rate of the wave is the share of its 7,920 events, each event counted once, coded violent. Panel C has columns of its own: two populations of the wave's block groups, all of them and those that hosted a violent George Floyd event. Its rows are the other protests those block groups may also have hosted, the last of them a violent event of the wave itself, and a cell is the percentage of that column's block groups that hosted at least one. That last row is not reported in the second column, which is the population defined by having one. The panel's last line is the percentage of the wave's block groups that did not host a protest in the past or in the same months, that is, in neither of the other two groups. The characteristics of the venues' block groups are in Table A3.

same block groups across four elections and tests formally for a pre-trend; Section 4.5 asks whether the effect differs across kinds of block group; Section 4.6 follows the exposed block groups to 2024; and Section 4.7 collects the robustness checks.

4.1 Specification

The question is whether block groups near a protest venue changed their vote differently, between the 2016 and the 2020 presidential elections, from block groups of the same zip code that were farther away. Distance is measured in ten rings of one kilometer around the block group's population-weighted centroid, and a block group is exposed in a ring when at least one George Floyd protest had its venue there (Butts, 2023). A block group with protests in several rings is counted in each of them, so the coefficient on a ring compares the block groups with a protest in that ring, whatever

else they have around them, with the block groups without one.[7] The average block group of the sample had 16.5 protests within 10 km of its centroid, of which 1.71 were violent.

The estimating equation is

$$\Delta Y_i = \sum_{r=1}^{10} \beta_r D_{ir} + \sum_{r=1}^{10} \gamma_r P_{ir} + X_i'\delta + \alpha_{z(i)} + \varepsilon_i, \tag{1}$$

where ΔY_i is the change in the Republican share of all votes cast in block group i between the 2016 and 2020 presidential elections, in percentage points; D_{ir} equals one when at least one George Floyd protest of 27 May to 2 November 2020 had its venue in ring r around the centroid of block group i, ring r spanning the distances from $r-1$ to r kilometers; P_{ir} is the same indicator for the protests of January 2017 to May 2020, so that a block group near a place that already hosted protests is compared with block groups near such places; X_i are the demographic controls, measured before the wave: population, density, racial composition, median age, median household income, the share with an associate degree or higher, the unemployment rate, the renter share, the distance to the central business district, and the share with a bachelor's degree or higher, on its own and interacted with racial composition; and $\alpha_{z(i)}$ is a fixed effect for the zip code of block group i, so that every comparison is between block groups of the same zip code. The regression is weighted by votes cast in 2020, and standard errors are clustered by metropolitan area. The coefficients of interest are the β_r: each is the shift in the Republican share associated with a protest in that ring, relative to block groups of the same zip code with no protest in it, holding the other rings fixed. The same equation with the 2012–2016 change as the outcome is the pre-period placebo test: if the block groups near a venue were already moving toward the Republicans before the wave, the β_r of that regression would show it.

The sample is the block groups of the metropolitan areas with at least one protest venue that have vote returns in 2012, 2016 and 2020 and every control, resulting in 136,644 block groups in 344 metropolitan areas and 11,457 zip codes. The design does not require exposed and unexposed block groups to look alike, only that their votes would have moved alike within the zip code, given the controls. Table A4 nevertheless compares, within zip codes, the block groups near a violent protest with those near only nonviolent ones on fifteen characteristics: the two groups do not differ in their 2016 vote shares or in their 2012–2016 trend. Whether anything the table does not measure was already moving the vote is what the pre-period placebo test asks.

4.2 Exposure and the vote

Figure 2 plots the ten ring coefficients for exposure to any protest, with the pre-period placebo on the same axes; Table A5 reports the coefficients of the first five rings. Block groups with a

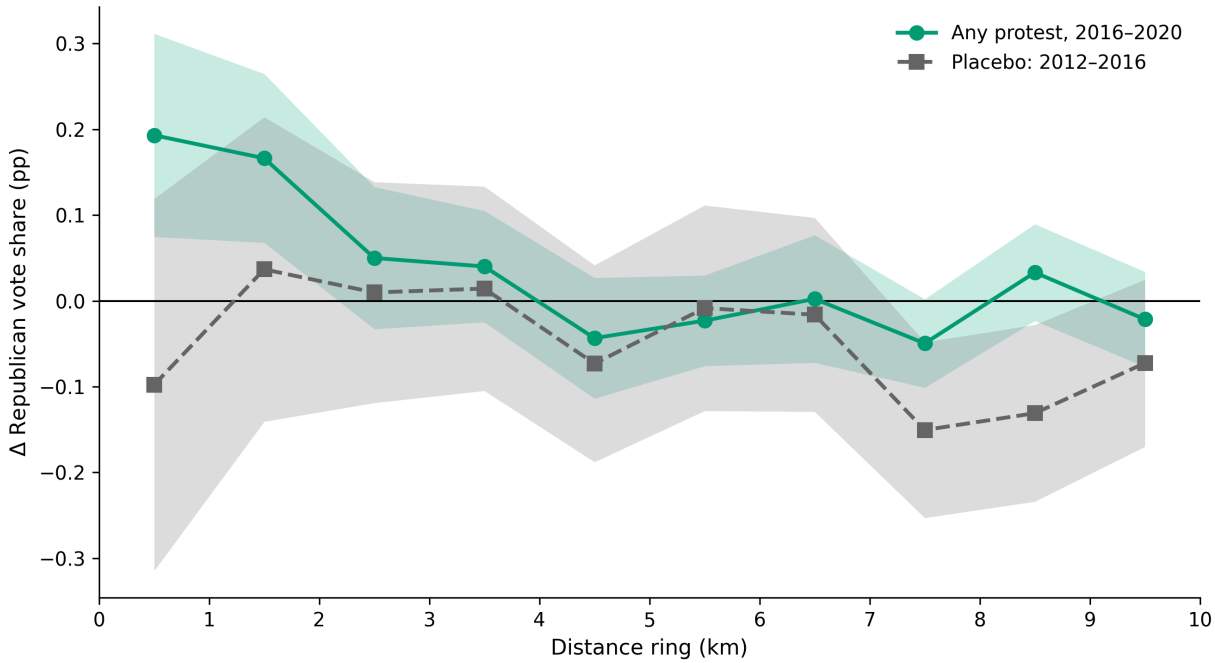
[7] Under the nearest-protest definition most block groups sit in the first three rings and the outer rings are thin, because a block group with a protest nearby is never counted in an outer ring even when it also has protests there; the definition used here keeps every protest within 10 km.

protest within one kilometer shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code with no protest in that ring, and block groups with a protest between one and two kilometers away shifted 0.17 points more; both are significant at the 1 percent level, and the five inner rings jointly reject zero ($p < 0.01$). Standard errors that allow the errors of block groups up to 50 km apart to be correlated (Conley, 1999), in place of clustering by metropolitan area, leave every coefficient of the first five rings at the same significance level, for any protest and for violent protests (the bracketed rows of Tables A5 and A6). Beyond two kilometers the coefficients are small and their intervals cover zero. The shift is therefore detectable out to about two kilometers at this sample size.

There are two concerns about reading this shift as the effect of the George Floyd protests. First, the block groups near a venue may not be affected by these protests per se, but may be the places where protests of any kind always take place, and such places may move for reasons of their own. The specification meets this concern with the ten indicators for a protest of January 2017 to May 2020 in each ring, so that a block group near a venue of the wave is compared with block groups near places that had hosted protests before. Splitting those earlier protests by their political orientation, left-wing, right-wing or neither, and entering the three sets of indicators in every combination leaves the coefficient of every ring inside the 95 percent confidence interval of the estimate above, for any protest and for violent protests (Table A7); the 0–1 km coefficient for any protest falls to between 0.15 and 0.16 whenever the indicators for the protests coded neither enter, against 0.19. Controlling for the retail share of jobs in the block group also leaves the estimates inside that confidence interval. Second, the block groups near a venue may have been moving toward the Republicans already, independently of the protests. Two tests answer this concern. The pre-period placebo of Figure 2 re-estimates the same equation on the 2012–2016 change and finds no coefficient distinguishable from zero at any ring within 7 km. The difference-in-differences design of Section 4.4 follows the same block groups across the four elections from 2012 to 2024 and finds that exposed and unexposed block groups of a zip code did not move apart between 2012 and 2016. Section 4.5 adds a check on the dimension along which selection into this wave would likely show, the Black population share, and detects no gradient.

Table A8 shows the 0–1 km estimate as each class of control enters: it falls to about two-fifths of its size when the demographic controls enter and moves little after that, including when the block group's own 2012–2016 change is added as a control. The rise in the Republican share is a fall in Democratic votes rather than a rise in Republican ones: in the same table the Democratic share falls in the 0–1 km ring and the logarithm of Democratic votes falls, while the logarithm of Republican votes moves little; turnout, measured against the voting-age population, falls as well, though that outcome is the least reliable of the five. Whether the missing Democratic votes are abstentions, a change in who lives in the block group, or voters switching is not identified here; Section 7 asks that question with the North Carolina voter file.

Figure 2: Exposure to any protest and the change in the Republican vote share, by distance ring



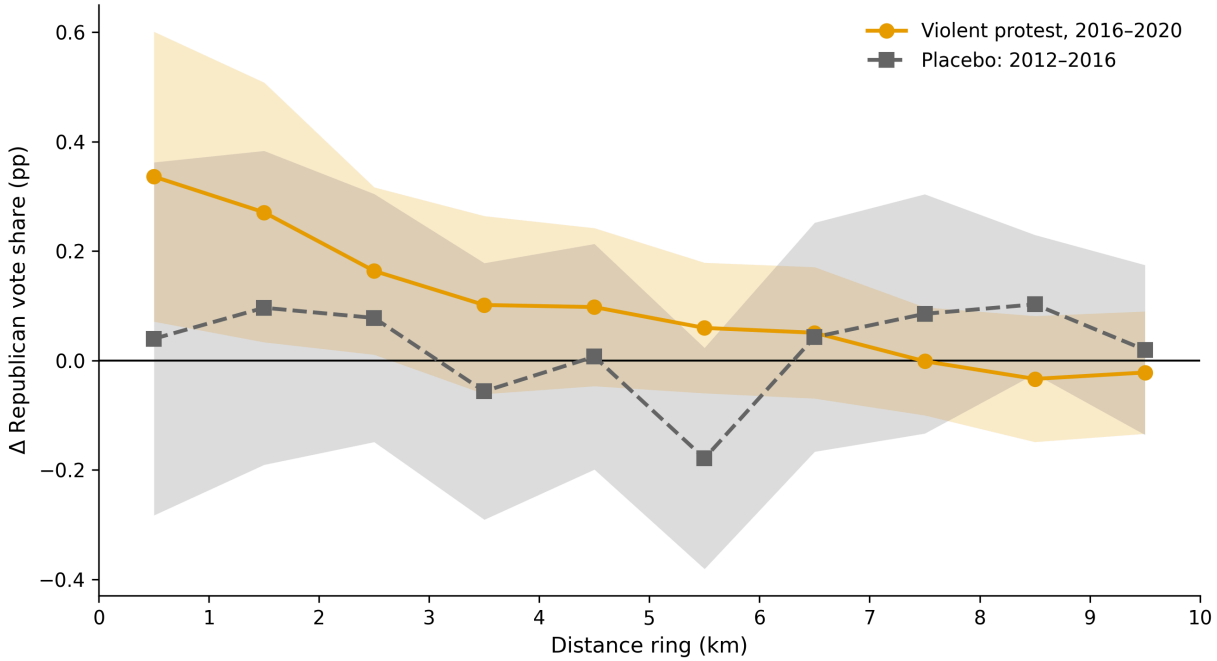
Notes: The figure plots, for each of ten one-kilometer distance rings around a block group, the estimated shift in the Republican vote share associated with at least one protest in that ring. Circles are the coefficients for the 2016–2020 change; squares are the same coefficients with the 2012–2016 change as the outcome, the placebo; shaded bands are 95% confidence intervals; markers sit at the midpoint of each ring. The regression is of the change in the Republican share of all votes cast between the 2016 and 2020 presidential elections, in percentage points, on ten indicators, entered jointly, for at least one George Floyd protest with its venue in each ring, with ten indicators for a protest of January 2017 to May 2020 in each ring, the demographic controls, and zip code fixed effects, so that every comparison is between block groups in the same zip code; the regression is weighted by votes cast in 2020. The demographic controls are population, density, racial composition, median age, median income, the share with an associate degree or higher, unemployment, renter share, distance to the central business district, and the share with a bachelor's degree or higher, on its own and interacted with racial composition. The sample is the block groups of metropolitan areas with a protest venue that have vote returns in 2012, 2016 and 2020; $N = 136{,}644$. Standard errors are clustered by metropolitan area; 344 clusters. The coefficients and standard errors for the first five rings are in Table A5.

4.3 Exposure by protest violence

Figure 3 repeats the estimation with exposure to a violent protest as the treatment; Table A6 reports the first five rings. The point estimates are larger at every inner ring: 0.34 percentage points within one kilometer, 0.27 between one and two kilometers and 0.16 between two and three, each significant at the 5 percent level, so that the shift is detectable out to three kilometers rather than two. The placebo is not distinguishable from zero at any ring.[8] Table A9 shows the violent 0–1 km estimate as the controls enter: without fixed effects it is not distinguishable from zero, and once the demographic controls enter it is significant and of similar size in every column that follows; the logarithm of Democratic votes is negative throughout. The logarithm of Republican votes, in

[8]The violent and the all-protest profiles are not statistically distinguishable: the violent estimates are larger but less precise, since 2,494 block groups have a violent protest within one kilometer against 13,609 with any protest, and the difference between violent and all-protest exposure is not statistically established.

Figure 3: Exposure to a violent protest and the change in the Republican vote share, by distance ring



Notes: The figure repeats Figure 2 with exposure to a violent protest as the treatment: circles are the coefficients on ten one-kilometer ring indicators, each equal to one when at least one violent George Floyd protest had its venue in the ring, for the 2016–2020 change in the Republican vote share, squares the same coefficients for the 2012–2016 change (the placebo), shaded bands 95% confidence intervals. The outcome, the demographic controls, the zip code fixed effects and the weights are those of Figure 2, and the earlier-protest ring indicators are matched to the family: they count violent earlier protests. The sample is that of Figure 2, $N = 136{,}644$. Standard errors are clustered by metropolitan area; 344 clusters. The point estimates are larger than those for any protest at every inner ring, but the two are not statistically distinguishable: the difference between violent and all-protest exposure is not statistically established. The first five rings are in Table A6.

the same table, is not distinguishable from zero in any column.

The backlash is present for all protests, the nonviolent ones included, and its point estimates are larger for the violent ones. Figure A7 estimates the violent and the nonviolent profiles side by side: nonviolent exposure shifts the vote toward the Republicans in the two inner rings, and the violent estimates are larger at every inner ring.

4.4 Difference-in-differences approach

The ring design compares the change in the vote between two elections across block groups of the same zip code. A second design follows the same block groups across four presidential elections, 2012, 2016, 2020 and 2024, and asks whether the block groups within 2.5 kilometers of a protest venue moved differently, in each election relative to 2016, from all other block groups of the same zip code (Linden and Rockoff, 2008). Its purpose is the pre-period: where the ring design's placebo re-estimates the same equation on the 2012–2016 change, this design puts the pre-period inside one regression, so that a pre-trend would appear as a 2012 coefficient different from zero. It is estimated

separately for exposure to a violent protest and for exposure to a nonviolent protest.

The estimating equation is

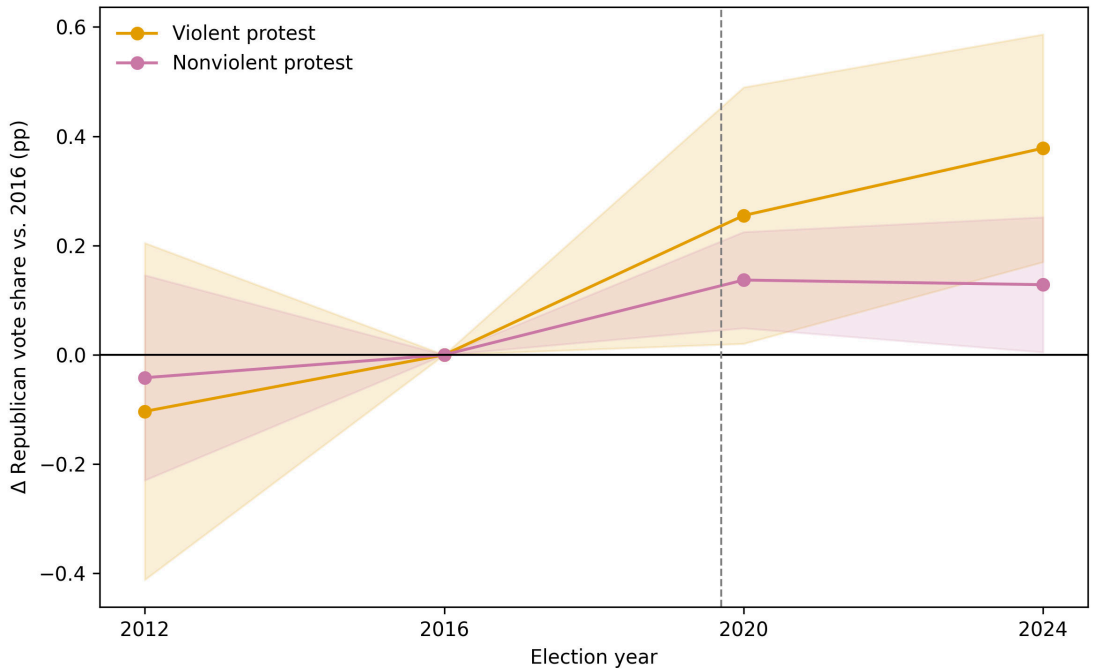
$$Y_{it} = \sum_{t \neq 2016} \beta_t \left(E_i \times \mathbf{1}[\text{year} = t] \right) + \sum_{t \neq 2016} \gamma_t' \left(C_i \times \mathbf{1}[\text{year} = t] \right) + \alpha_i + \delta_{z(i),t} + \varepsilon_{it}, \tag{2}$$

where Y_{it} is the Republican share of all votes cast in block group i in election year t, one of 2012, 2016, 2020 and 2024, in percentage points; E_i equals one when at least one George Floyd protest had its venue within 2.5 km of the block group's population-weighted centroid; $\mathbf{1}[\text{year} = t]$ is an indicator for election year t; C_i is the same set of controls as in equation (1), standardized within the panel; α_i is a block group fixed effect, so that each block group is compared with itself over time; and $\delta_{z(i),t}$ is a zip code by election year fixed effect, so that the comparison is within zip code and election. The coefficients of interest are the β_t, the difference between exposed and unexposed block groups in year t relative to the same difference in 2016; β_{2012} is the pre-trend test. The panel is balanced, 135,572 block groups with a Republican share in all four elections; the regression is weighted by votes cast in 2020 and standard errors are clustered by metropolitan area.

This specification differs from the ring design in three ways. It works on the level of the vote share in a panel rather than on its change; it defines exposure by one cutoff, 2.5 km, rather than by ten rings; and its controls are standardized within the panel and interacted with the year, so that its coefficients are not comparable in size with the ring estimates. The assumption it rests on is also different. The ring design asks that the votes of exposed and unexposed block groups of a zip code would have moved alike between 2016 and 2020 had there been no protest; the panel design absorbs everything fixed about a block group and asks only that the two groups would have moved alike across elections; the 2012 coefficient checks that on the elections before the wave.

Figure 4 plots the coefficients and Table A10 reports them, with two further columns in which the comparison group is restricted to the block groups whose nearest protest lies between 2.5 and 10 km. The 2012 coefficient is not distinguishable from zero for either family: exposed and unexposed block groups did not move apart between 2012 and 2016. The 2020 and 2024 coefficients are positive and significant at the 5 percent level for both families, about 0.26 in 2020 and 0.38 in 2024 for violent exposure, 0.14 and 0.13 for nonviolent exposure. The violent 2024 point estimate is larger than its 2020 one. Figure A8 moves the 2.5 km cutoff from 0.5 to 5 km: the 2020 estimate is fairly stable across cutoffs and the 2012 placebo is not distinguishable from zero at any of them for either family. A sensitivity analysis of the kind proposed by Rambachan and Roth (2023) bounds each violation of parallel trends after 2016 by a multiple of the difference between exposed and unexposed block groups estimated for 2012, the one election before the wave and so a single, imprecisely estimated benchmark; the smallest multiple at which the confidence set includes zero is 0.25 for the 2020 coefficient and 0.50 for the 2024 one under violent exposure, and 0.60 and 0.10 under nonviolent

Figure 4: Event study of the Republican vote share around exposure within 2.5 km, 2012–2024



Notes: The figure is an event study of the Republican vote share in block groups within 2.5 km of a protest against all other block groups, across four presidential elections. A marker is the coefficient for an election year relative to 2016, which is normalized to zero; bands are 95% confidence intervals; the dashed line marks the George Floyd protests. Circles trace exposure to a violent protest, the other series exposure to a nonviolent protest; the two are separate regressions. The regression is of the Republican share of all votes cast in the block group in each of the four elections, in percentage points, on the exposure indicator interacted with an indicator for each election year other than 2016 and on the controls interacted with the same indicators, with block group fixed effects and zip code by election year fixed effects, so that exposed and unexposed block groups are compared in the same zip code and election. Exposure equals one when at least one George Floyd protest of the stated family (violent, or nonviolent) had its venue within 2.5 km of the block group. The controls are the demographic controls of Figure 2, log population, log density and distance to the central business district, and the ten earlier-protest ring indicators of that figure, matched to the family. The panel is balanced: block groups with a Republican share in all four elections, 135,572 block groups and 542,288 block group by year observations. The regression is weighted by votes cast in 2020; standard errors are clustered by metropolitan area; 339 clusters. The 2012 coefficient is the pre-period test. Controls here are standardized within the panel, so these interaction terms are not comparable in magnitude with the cross-sectional tables. The plotted coefficients are columns (1) and (3) of Table A10.

exposure.

4.5 Exposure across block-group characteristics

Does the backlash differ by the kind of block group? The question is asked for two reasons: whether the shift is a general response to a protest nearby or is carried by a few kinds of block group, and whether the block groups near a venue were selected along race, the one dimension this movement was about. For each of five characteristics, block groups are split into four quartiles of the characteristic measured relative to the other block groups of their zip code, so that the bottom quartile is the block groups in the bottom quarter of their zip code. The ring indicators of the specification are interacted with the quartiles. Figures 5 and A9 plot the 0–1 km coefficient by quartile, with its

placebo, for any protest and for violent protests. Whether the four coefficients differ is decided by a joint test of the quartile interactions, not by the individual intervals.

Income is the one characteristic that carries a pattern. For violent protests the backlash is largest in the block groups in the richest quarter of their zip code, the second quarter sits below the first, so the pattern is not monotone, and the joint test rejects ($p < 0.01$) while its placebo does not (Figure A9); for any protest the four estimates are alike and the joint test does not reject. Race is the dimension along which selection into exposure would likely show, given the nature of this protest movement. The wave's venues sat in block groups with a higher Black population share than the venues of earlier protests (Table A3), and if the block groups near a venue were already on a different electoral path for reasons tied to this movement, the estimate would differ across the Black share more than across any other characteristic. Figure 5 splits block groups by their Black population share relative to their zip code: the backlash is positive in every quartile under both treatments, the placebo is not distinguishable from zero in any of them, the point estimates do not rise with the Black share, and the joint tests do not reject ($p = 0.60$ for any protest, 0.62 for violent protests). The figure is uninformative about a gradient: it neither establishes one nor rules one out. Along the one dimension where this wave would be expected to produce one, none is detected. Education, density and age carry no detectable pattern either (Figure A9): the point estimates are positive in every quartile and no joint test rejects. The results reassure on both counts: the backlash is not the response of one kind of block group, its point estimate being positive in every quartile of every characteristic, and the selection check on race detects no gradient.

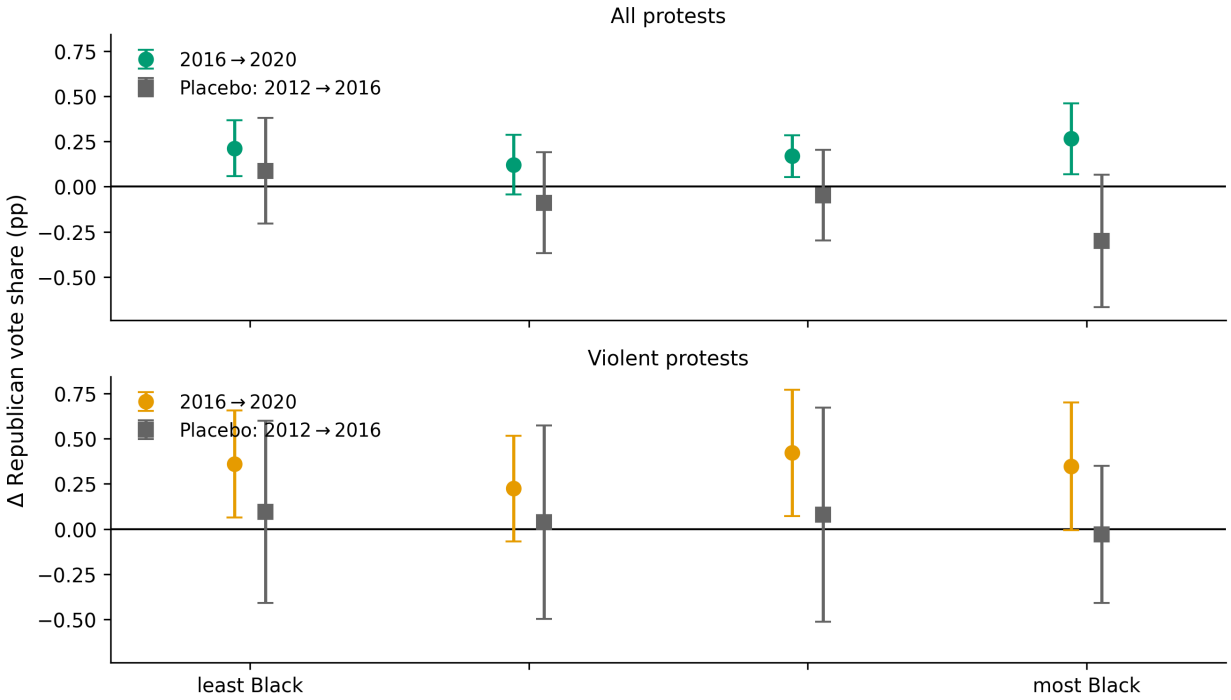
4.6 Persistence to 2024

Does the backlash persist? The ring design answers the question when its outcome window is moved to 2024. Table A11 re-estimates the ring profiles with the 2020–2024 change and the cumulative 2016–2024 change as outcomes, on the 135,572 block groups with a 2024 return. For violent exposure the cumulative point estimates at the inner rings are larger than the 2016–2020 ones, and the joint test over the five inner rings rejects where the 2016–2020 one does not; the 2020–2024 coefficients on their own are positive at the four innermost rings and not distinguishable from zero. For nonviolent exposure the cumulative coefficients at the inner rings are of the order of the 2016–2020 ones, the joint test over the five inner rings rejects in both windows, and the 2020–2024 coefficients sit near zero. The backlash was still present in 2024. The difference-in-differences design reads the same way: in Figure 4 the 2024 coefficient is positive for both families, and for violent exposure it is larger than the 2020 one. For violent exposure the point estimates in both designs are consistent with an effect that grew rather than merely persisted.

4.7 Robustness

Three exercises ask whether the ring estimates depend on choices made in building them; each is reported in full in Appendix D.7. First, the venues are relocated at random (Table A12): random

Figure 5: Exposure by quartile of the Black population share



Notes: Block groups are split into four quartiles of their Black population share, measured relative to the other block groups of the same zip code (the share demeaned within zip code), so that the leftmost group is block groups in the bottom quarter of their zip code and the rightmost those in the top quarter; the quartiles hold equal numbers of block groups. Each circle is the 0–1 km coefficient for that quartile, from one regression in which the ten ring indicators of Table A5 (top panel, any protest) or Table A6 (bottom panel, violent protest) are interacted with the four quartile indicators, with the quartile indicators, the previous-protest and demographic controls of Figure 2, zip code fixed effects and weights by votes cast in 2020; the outcome is the change in the Republican share of all votes cast between the 2016 and 2020 presidential elections, in percentage points. Squares are the same coefficients when the 2012–2016 change is the outcome (the placebo). Whiskers are 95% confidence intervals. $N = 136{,}644$ in every fit; standard errors are clustered by metropolitan area; 344 clusters. Whether the four quartile coefficients differ is decided by a joint test of the quartile interactions: the joint test does not reject for either treatment ($p = 0.5954$ for any protest, 0.6208 for violent protests), so the panel is uninformative about a gradient.

placement does not replicate the headline, and the observed decay across rings is steeper than the decay random geometry itself produces.[9] Second, the rings are redrawn so that every ring covers the same area as the innermost one (Table A13): the 0–1 km estimates stay positive and significant for violent and for nonviolent exposure; the equal-area carving is a robustness column beside the one-kilometer rings, not a replacement, and neither carving is a truer estimate. Third, each of the 42 states and each of the 30 largest metropolitan areas is dropped in turn (Figure A10): none of the 72 drops moves the violent 0–1 km estimate outside the full-sample confidence interval.

[9]Permutation p-values are the yardstick because the conventional t-test is not one here: under random placement the null profile is not centered at zero, so that test reports significance at 0–1 km on a large share of draws for any protest and on a smaller share for violent protests (Borusyak and Hull, 2023).

5 Protest Participation

This section asks what sending residents to a protest did to the vote of the block groups they came from. Participation could move the vote in either direction: the vote moving toward the Democratic candidate would suggest a mobilization of the neighborhoods whose residents took part, while votes moving to the Republican candidate would suggest that taking part wore down the appetite for protesting in those same neighborhoods. Section 5.1 states the comparison and the specification; Section 5.2 reports the estimate for participation at any protest, states the identification concern and reads the checks that answer it; Section 5.3 splits participation by whether the protest attended turned violent and by whether the attender was a regular at the venue; Section 5.4 asks whether the estimate differs across kinds of block group; Section 5.5 follows the same block groups to 2024, and Section 5.6 collects the robustness checks.

5.1 Specification

The question is whether block groups whose residents went to a George Floyd protest changed their vote differently, between the 2016 and the 2020 presidential elections, from block groups of the same zip code whose residents did not go, or went in smaller numbers. A resident participates when a device homed in the block group is detected at a protest venue during a protest. Participation enters in two parts: whether any resident of the block group attended, and how many did, measured as attendances per device homed in the block group, in logs. The first part compares the block groups that sent anyone with those that sent nobody; the second compares, among the block groups that sent anyone, those that sent more with those that sent fewer. Residents who were regulars at those venues, present there on ordinary days as well, are counted as attending, and their ordinary presence at the same venues in the three weeks before the first event enters as a control, so that two block groups are compared at the same level of ordinary presence.

The estimating equation is

$$\Delta Y_i = \beta_D \, D_i + \beta_L \, (L_i - c) \, D_i + A_i'\theta + X_i'\delta + M_i'\lambda + S_i'\kappa + \alpha_{z(i)} + \varepsilon_i, \tag{3}$$

where ΔY_i is the change in the Democratic share of all votes cast in block group i between the 2016 and 2020 presidential elections, in percentage points; D_i equals one when at least one resident of block group i attended a George Floyd protest, the extensive margin of protest participation; $L_i = \ln(a_i/n_i)$ is the intensive margin, the log of the participation rate, where a_i counts the attendances of block group i over the wave and n_i is the number of panel devices homed in the block group; c is the mean of L_i among the block groups that sent anyone, so that β_D is read at the typical participation rate of a sending block group rather than at a rate of one. A_i controls for the residents' ordinary presence at the same venues, measured between 4 and 26 May 2020 on days those venues hosted nothing and entered in the same two-part form as the treatment; X_i are the demographic controls of Section 4.1, measured before the wave: population, density, racial composition, median

age, median household income, the share with an associate degree or higher, the unemployment rate, the renter share, the distance to the central business district, and the share with a bachelor's degree or higher, on its own and interacted with racial composition; M_i are four measures of how far residents' devices ranged on an ordinary day before the pandemic; S_i are indicators for deciles of the number of panel devices homed in the block group; and $\alpha_{z(i)}$ is a fixed effect for the zip code of block group i, so that every comparison is between block groups of the same zip code. The regression is weighted by votes cast in 2020, and standard errors are clustered by metropolitan area. The coefficients of interest are β_D, the shift in the Democratic share associated with sending anyone, relative to block groups of the same zip code that sent nobody, and β_L, the further shift associated with sending more among the block groups that sent anyone. The same equation with the 2012–2016 change as the outcome is the pre-period placebo test: if the block groups that sent residents to the wave were already moving toward the Democrats before it, β_D in that regression would show it.

The sample is the block groups of the metropolitan areas with at least one protest venue that have vote returns in 2012, 2016 and 2020, every control and a device panel, resulting in 135,691 block groups in 333 metropolitan areas and 13,603 zip codes, of which 85,399 sent at least one resident. Among them participation is thin: the median sending block group recorded 2 attendances over the wave. The design does not require sending and non-sending block groups to look alike, only that their votes would have moved alike within the zip code had nobody gone, given the controls.

5.2 Participation and the vote

Figure 6 plots the estimate, with two placebos on the same axes; Table A14 reports the coefficients. Block groups whose residents turned up at protest venues during protest hours, holding fixed how many of them were regulars at those venues, shifted 0.07 percentage points more Democratic in presidential vote share between 2016 and 2020 than otherwise-similar block groups in the same zip code that sent none or fewer; sending more added no detectable shift.

Participation in a protest is not random: residents choose to go, and the block groups that sent them differ from those that did not. The threat to the identifying assumption of Section 5.1 is that the block groups that sent residents were already moving toward the Democrats, and would have voted more Democratic in 2020 without the protests. Four checks answer it. The first is the pre-period placebo, column (2) of Table A14: the same specification with the 2012–2016 change as the outcome. The block groups that sent residents to the wave's protests were not detectably moving toward the Democrats before it; the interval of that coefficient covers zero. The second test, column (3) of Table A14, asks whether it is the kind of block group that sends people to protests, rather than participation in this wave, that carries the shift: the treatment is replaced by attendance, measured in the same way, at the protests of 2019 and of 1 January to 26 May 2020 held at venues that later hosted a George Floyd protest. That attendance predicts no detectable shift in the 2020 vote, and its coefficient differs from the column (1) estimate: the shift is specific

to participation in the George Floyd protests and not to protest-going as such.[10] Split by the political orientation of the earlier protests (Table A15), the extensive margin of earlier attendance is not distinguishable from zero at the 5 percent level for the protests coded left-wing or for those coded right-wing; the intensive margin is negative and distinguishable from zero for both, so on that margin the split placebo is not clean at the 5 percent level. Adding earlier attendance as a control, at left-wing protests, at right-wing ones, at both, or at all earlier protests, leaves both margins inside the 95 percent confidence interval of the estimate in every combination (Table A16); attendance at the earlier protests coded neither left- nor right-wing cannot be separated in the attendance data. Those columns are estimated on the 132,482 block groups where earlier attendance is observed, and on the extensive margin at least half of each column's move from the estimate is that change of sample and not the control. Controlling for the retail share of jobs in the block group also leaves both margins inside that confidence interval. The third is the controls. Table A18 shows the estimate as each class of control enters. Comparing block groups within zip codes removes most of the raw association, and the extensive margin then falls to about two-fifths of its within-zip-code size as the demographic, racial-composition, device-panel and ordinary-presence controls enter; the pre-period placebo is far from clean in the first two columns, where the 2012–2016 change is itself strongly related to participation, and clean once the demographic controls enter, which argues against reading the first two columns as estimates and for the full specification, whose controls do identification work. The intensive margin is not distinguishable from zero in any column and its sign changes across them. The fourth is race, the dimension along which a trajectory tied to this wave would likely show: Section 5.4 splits the estimate by the Black population share and finds it positive in every quarter, with no detectable gradient along that dimension.

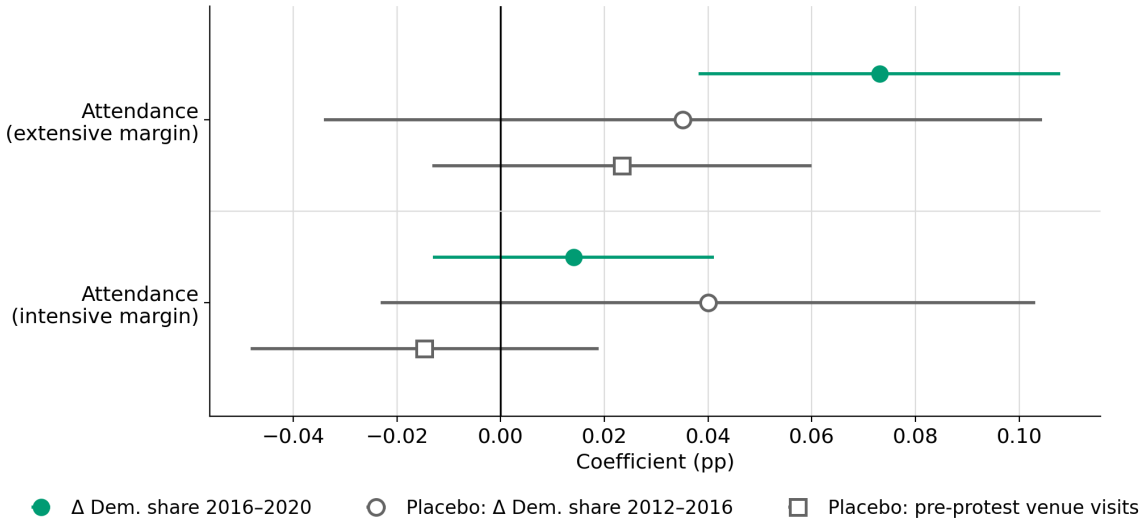
Table A19 estimates the specification on four other outcomes. The Republican share moves in mirror image, and the two shares are not exact mirrors only because the denominator is all votes cast; turnout, Democratic votes and Republican votes per voting-age person show no detectable change. What this means is that the shift is visible in the two parties' shares of the vote and not in the number of votes either party received per voting-age person, which the data measure too imprecisely to sign. Whether the shift came from more Democratic votes, from fewer Republican ones, or from a change in who lives and votes in the block group is therefore not identified.

5.3 Participation by protest violence and by habitual attendance

Figure 7 splits participation in two ways; Table A17 reports the coefficients. Each split estimates its four terms, two extensive margins and two intensive margins, in one regression, so that each half is estimated holding the other fixed. The first splits it by whether the protest attended turned violent. Attendance at violent protests is the smaller part: 40,723 attendances from 24,333 block groups, against 261,701 from 79,966 at nonviolent protests, a ratio of 0.16 in attendances. Block

[10] The Wald test rejects the equality of the two ($p = 0.03$). The Wald row of the table also tests the pre-period placebo against the column (1) estimate rather than against zero and does not reject ($p = 0.33$), so the pre-period comparison alone does not separate the two.

Figure 6: Participation and the change in the Democratic vote share



Notes: Each marker is an estimated coefficient and each whisker a 95 percent confidence interval, in percentage points of the Democratic share of all votes cast for president. The upper group is the extensive margin, whether the block group sent anyone to a protest, and the lower group the intensive margin, how much it sent. Filled circles are the estimate; open circles are the same specification on the 2012–2016 change; open squares replace the treatment with attendance at earlier protests at the same venues. The regression is of the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections on the two terms, with the venue-traffic controls, the demographic controls, the neighborhood mobility controls, indicators for deciles of the block group's panel devices, and zip code fixed effects, so that every comparison is between block groups in the same zip code; the regression is weighted by votes cast in 2020. Regulars at protest venues are counted, and the venue-traffic controls hold fixed their presence at the same venues in the three weeks before the first event. The intensive margin is the log of attendances per panel device, zero where nobody attended. The demographic controls are listed in the note to Figure 2; the mobility controls are four measures of how far residents' devices ranged on an ordinary day before the pandemic. The sample is the block groups of metropolitan areas with a protest venue that have vote returns in 2012, 2016 and 2020 and a device panel; $N = 135{,}691$ for the estimate and the first placebo, 85,399 of which sent at least one resident, and 132,482 for the second placebo. Standard errors are clustered by metropolitan area; 333 clusters. The coefficients, standard errors and the tests of each placebo against the estimate are in Table A14.

groups that sent anyone to a nonviolent protest shifted 0.07 percentage points more Democratic. For violent protests the estimate is −0.02 percentage points and is not distinguishable from zero. The two differ: Wald p = 0.02. Among block groups that sent people, sending more is associated with no further shift for nonviolent protests (0.01, not significant) and with a negative shift for violent ones (−0.09). These also differ: Wald $p < 0.01$.[11] This result suggests that mobilization takes place only in the block groups that sent residents to nonviolent protests, and is not present when the protests turned violent. Moreover, the more residents a block group sent to a violent protest, the further its vote moved away from the Democratic candidate. I read this as a backlash entering through the back door of participation when protests turn violent: not only the residents of the block groups close to a violent protest reacted against it; so did the residents of the block groups that sent people to it. The mobilization at nonviolent protests appears in the extensive margin,

[11] The extensive-margin difference reaches the 5 percent level at 100, 200 and 400 m but not at 50 m; the intensive-margin difference reaches it at 200 and 400 m only; and at 50 m the nonviolent intensive margin inverts, to −0.17.

whether anyone went; the reaction to violent protests appears in the intensive margin, how many went: the shift is larger the more residents of the block group participated. This result may be suggestive of peer effects in the origin neighborhood, where a negative reaction to a violent protest starts to take place only if more than one resident happens to participate. I read the split under the hypothesis that most residents at a protest that turned violent had not chosen a violent protest, so that attending one does not mark a block group as different in advance. Reverse causality is not excluded, but it is harder to square with these estimates: block groups already moving toward the Democrats would be expected to move further that way the more residents they sent, and at violent protests the intensive margin has the opposite sign.

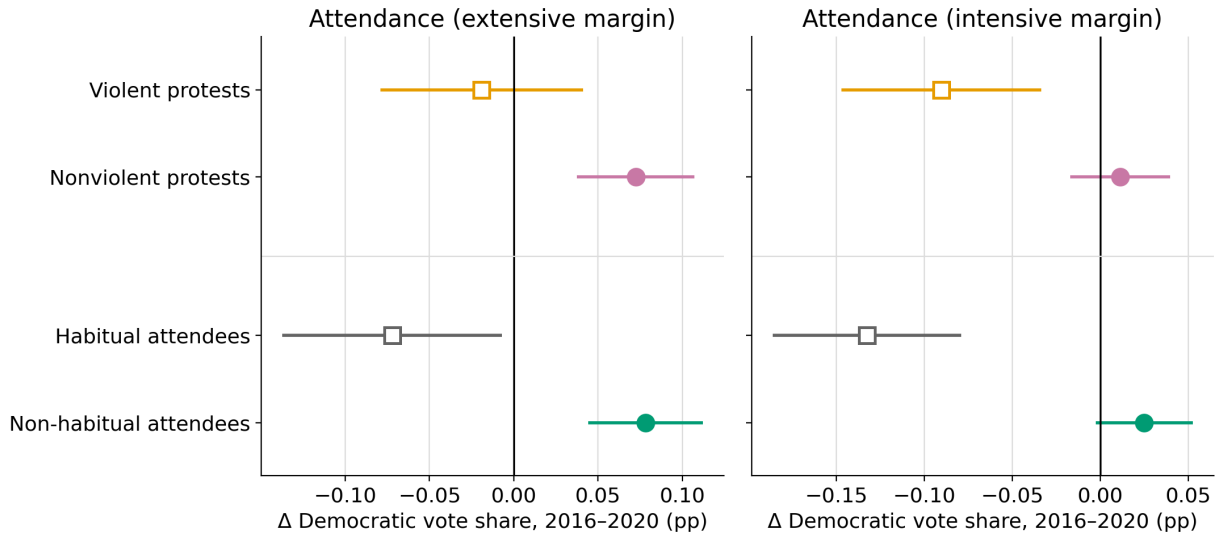
The second splits attendance by whether the attender was a regular at that venue: a device seen there, in the same hours, on the ordinary days of 2020: on at least five such days, or on more than a tenth of all the days it was seen anywhere. I treat regulars as incidental participation: residents who would have been at the venue regardless of the protest, because they work there or go there often for other reasons; the other attenders are residents who went there for the protest. Regulars are about half of the attendance the measure counts, so this is a split between two large groups and not between a group and a remainder. The shift comes from the attenders who were not regulars: their extensive margin is 0.08 and significant at the 1 percent level, while the regulars' extensive margin is -0.07 and their intensive margin -0.13, both distinguishable from zero with that sign, and the two halves differ on both terms (Wald $p < 0.01$). This is a fact about who attended, not about which protest. So the mobilization takes place only in the block groups that sent residents who would not have been at the venue otherwise. The block groups whose residents were there anyway, the incidental participation, reacted negatively to being there and moved away from the Democratic candidate. I interpret this again as a backlash entering through participation: Section 4 studied the residents of the block groups around a venue, while this is a backlash among people who live elsewhere and go to the venue regularly for work or other reasons.

5.4 Participation across block-group characteristics

Does the shift differ by the kind of block group? For each of five characteristics, block groups are sorted into four quartiles of the characteristic, ranked across the whole estimation sample rather than within the zip code as in Section 4.5, and the extensive margin of equation (3) is interacted with the quartile; Figures 8 and A11 plot the four quartile extensive-margin estimates with their placebos. Whether the four differ is decided by a joint test of the quartile interactions, not by the individual intervals.

Density is the one characteristic on which the joint test rejects (Figure A11, $p < 0.01$): the least dense quarter carries the largest estimate and is the only quarter whose interval clearly excludes zero, the two middle quarters sit near zero, and the densest quarter's estimate is larger than theirs with an interval that reaches zero, so the four are not monotone in either direction. A density pattern is not a test of any mechanism this paper asserts: the treatment is attendance, and whatever contact

Figure 7: Participation split by protest violence and by whether the attender was a regular



Notes: Each marker is an estimated coefficient and each whisker a 95 percent confidence interval, in percentage points of the Democratic share of all votes cast for president; the left panel is the extensive margin and the right panel the intensive margin. The upper two rows come from one regression that splits attendance by whether the protest attended turned violent; the lower two rows come from a second regression that splits it by whether the attender was already a regular at that venue, the habitual attenders. Each half enters as its own extensive margin and its own intensive margin, so a block group whose residents attended both kinds of venue-day carries both pairs of terms; the halves are not mutually exclusive categories. The outcome, the controls, the zip code fixed effects and the weights are those of Figure 6. $N = 135{,}689$ for the violence split and 135,691 for the habitual split. Standard errors are clustered by metropolitan area; 333 clusters. The treatment is split and the venue-traffic controls are not. The coefficients, standard errors and the Wald tests of each split are in Table A17.

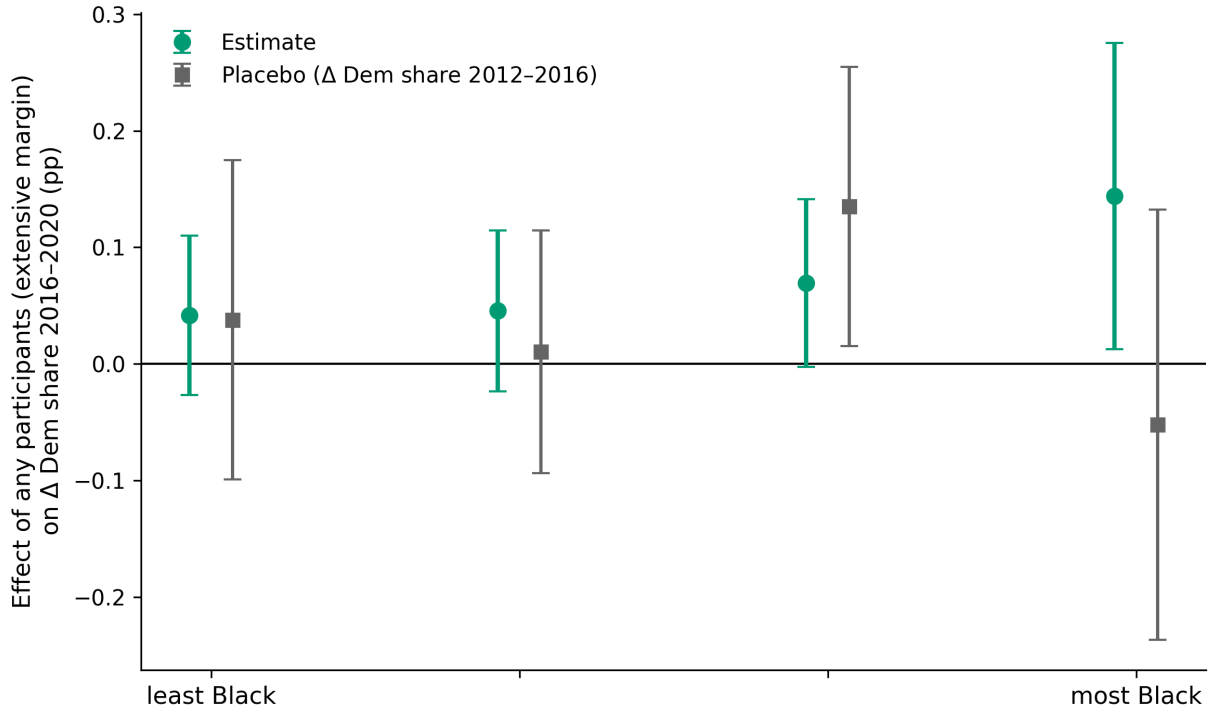
took place did so at the protest and not in the attender's neighborhood.

Race is again the dimension along which selection into participation would likely show, given the nature of this protest movement. The check is run to meet the identification concern of Section 5.2: if the estimate were only the trend of the kinds of block group that sent residents, the shift should differ across the Black population share more than across any other characteristic, and it should vanish where a Democratic trend would have to be absent.[12] Figure 8 splits block groups by their Black population share. The estimate is positive in every quarter; the most-Black quarter carries the largest point estimate, 0.14, and is the only quarter whose interval excludes zero, so the shift does not vanish where a Democratic trend would have to be absent. The joint test of the quartile interactions does not reject. The figure is uninformative about a gradient: it neither establishes one nor rules one out. Along the one dimension where a trajectory tied to this wave would be expected to produce one, none is detected.[13] The result reassures on the concern the check was run for: the shift does not vanish where a Democratic trend would be weakest, and no gradient is detected along

[12] In a regression of the outcome on the controls alone (Table A20), the Black population share carries a negative partial slope on the 2016–2020 change in the Democratic share within zip codes, at the mean level of education, so a shift carried by a Democratic trend would be weakest in the most-Black block groups.

[13] $\chi^2(3) = 1.36$, $p = 0.71$. The placebo is not clean in the third quarter, where its interval excludes zero, and that quarter's estimate is read with this in mind; the placebo interactions jointly do not reject ($p = 0.20$).

Figure 8: Participation by quartile of the Black population share



Notes: Block groups are sorted into four quartiles of the Black population share and the specification of Table A14 is estimated with the extensive margin interacted with the quartile. Circles are the quartile extensive-margin estimates; squares are the same estimates when the 2012–2016 change is the outcome instead, the placebo; whiskers are 95 percent confidence intervals; the axis is in percentage points of the Democratic share of all votes cast. Quartiles are of the raw Black population share from the American Community Survey, ranked across the whole estimation sample and cut into four equal quarters, so the leftmost group is the least-Black quarter of the sample's block groups and not the least-Black quarter of each zip code. The outcome, the two treatment terms, the controls, the zip code fixed effects and the weights are those of Table A14. $N = 135{,}691$; standard errors are clustered by metropolitan area, 333 clusters. Only a joint test of the quartile interactions can decide whether the effect varies with the Black population share, and it does not reject, $\chi^2(3) = 1.36$, $p = 0.714$, so the panel is uninformative about a gradient: it neither establishes one nor rules one out.

the dimension where selection into this wave would show.

Income, education and age carry no detectable pattern (Figure A11): the point estimates fall from the poorest to the richest quarter and from the least to the most educated quarter without a reversal, the youngest quarter carries the largest estimate, and no joint test rejects at the 5 percent level ($p = 0.057$, 0.051 and 0.066), so each panel is uninformative about a gradient rather than evidence against one; no placebo joint test rejects. The income and education patterns have the same shape and the two characteristics are correlated, so they are not two independent findings.

5.5 Persistence to 2024

Table A21 reports equation (3) estimated with three outcome windows, the 2016–2020 change, the 2020–2024 change and the cumulative 2016–2024 change, on the 134,570 block groups that have a vote return in 2024 as well, in 328 metropolitan areas. The participation effect did not subside

between 2020 and 2024. The 2020–2024 change is positive and significant (0.05), the Republican-share mirror moves with it (−0.05), and the cumulative 2016–2024 shift is 0.12 against 0.07 over 2016–2020, which suggests the effect may have grown rather than merely persisted.[14] Sending more added no detectable shift in any of the three windows.

5.6 Robustness

Five exercises ask whether the estimate depends on choices made in building the participation measure and the specification; each is reported in full in Appendix E.6. First, across all 512 combinations of nine controls for ordinary presence at the venues, the extensive margin is positive and significant at the 1 percent level (Table A22). Second, the detection radius is moved from 50 to 400 m (Table A23). The intensive margin's sign is not radius-robust: it is negative and significant at 50 m (−0.16) and at 100 m (−0.06), where tight-radius exposure misclassification is strongest, and not distinguishable from zero at 200 m (0.01) or 400 m (0.02). The extensive margin is positive and significant at all four radii. Third, with terms for the neighboring block groups, the block group's own extensive margin stays positive and significant in every column (Table A24). Fourth, under five denominators of the participation rate the extensive margin is positive and significant; the intensive margin is positive under all five and significant under four, and is a null only under the denominator the paper uses (Table A25). Fifth, no single state or large metropolitan area moves the extensive margin outside its full-sample confidence interval (Figure A12). An instrument for the extensive margin, the log distance to the nearest protest venue, was also tried; its two-stage estimate is too imprecise to be the estimate of record and is not reported.

6 Aggregate Net Effect

Sections 4 and 5 estimated two treatments separately, exposure to a protest nearby and participation in one, but both apply to the same units. A block group can be near a venue and home to attenders at once, so what it received is a bundle of two treatments pulling in opposite directions at the same time. This section puts the two together in four steps. Section 6.1 places them in a single joint regression, to test whether they coexist on the same block groups at the same time. Section 6.2 compares them in magnitude, as average effects on the block groups each treatment reached, and expresses them as persuasion rates beside the rates published for media and campaign interventions. Section 6.3 turns the coefficients into votes with a back-of-the-envelope calculation and asks whether the direct local effects of the protests could have decided the 2020 election, and Section 6.4 asks what the wave's violence cost. Appendix F holds the joint regressions, the decomposition tables and the counterfactual exercises in full.

[14] The increase is treated as suggestive rather than established, because 200 m is the only one of four detection radii at which it reaches the 5 percent level. Cumulatively from 2016 to 2024 the effect is positive and significant at all four radii and for both parties, so persistence is the radius-robust finding; the increase is not.

6.1 The two treatments in one regression

The joint regression is the union of the two separate specifications, equations (1) and (3): the participation terms of the second and the exposure terms of the first enter together, with the controls of both, in the same notation:

$$\begin{aligned}\Delta Y_i = \beta_D\, D_i + \beta_L\,(L_i - c)\, D_i + \sum_{r=1}^{10} \beta_r\, D_{ir} + \sum_{r=1}^{10} \gamma_r\, P_{ir} \\ + A_i'\theta + X_i'\delta + M_i'\lambda + S_i'\kappa + \alpha_{z(i)} + \varepsilon_i,\end{aligned} \tag{4}$$

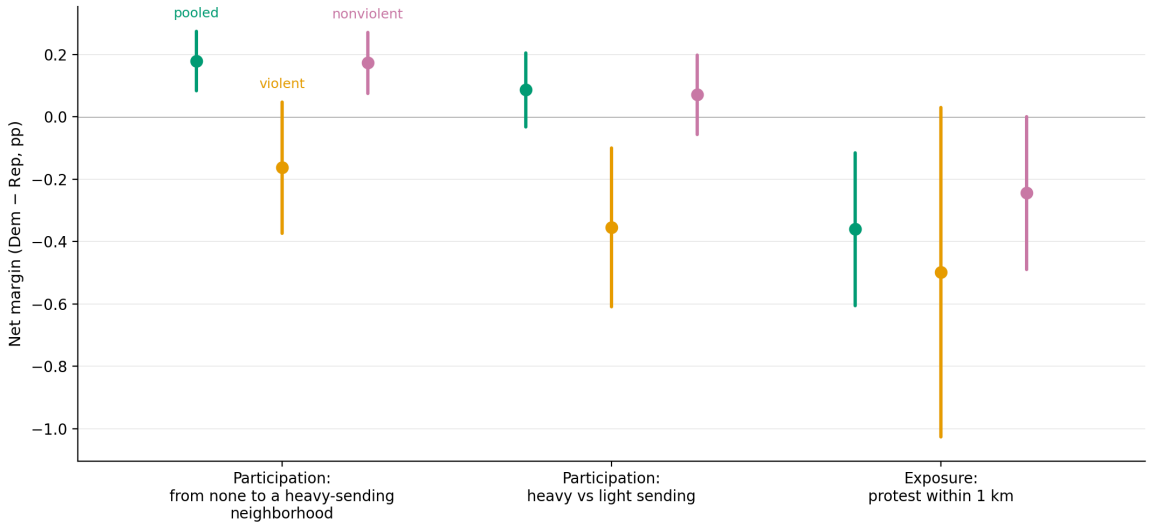
where ΔY_i is now the change in the net margin of block group i, the change in the Democratic share of all votes cast minus the change in the Republican share between the 2016 and 2020 elections, in percentage points; D_i and $(L_i - c)\, D_i$ are the extensive and intensive margins of protest participation defined in Section 5.1, with c the mean of L_i among the sending block groups of this sample; D_{ir} are the ten exposure rings of Section 4.1 and P_{ir} the same rings for the protests of January 2017 to May 2020; A_i, X_i, M_i and S_i are the controls of the two sections, and $\alpha_{z(i)}$ the zip code fixed effects. The regression is weighted by votes cast in 2020, clustered by metropolitan area, and estimated on the 133,406 block groups on which both treatments are observed together, in 333 metropolitan areas.

Figure 9 draws three quantities from it on one scale, for all protests and split by protest violence: sending residents at a high rate against sending nobody, heavy against light sending among the block groups that sent anyone, and a protest venue within one kilometer. The two treatments coexist and pull in opposite directions. Holding exposure fixed, the extensive margin of participation shifts the net margin 0.15 percentage points toward the Democrats, significant at the 1 percent level; holding participation fixed, a protest venue within one kilometer shifts it 0.36 points toward the Republicans, also at the 1 percent level; heavier sending among the block groups that sent anyone adds no detectable shift (Table A26). The exposure profile survives: the shift toward the Republicans is detectable in the first two rings, which jointly reject zero, and from the third ring outward no ring is distinguishable from zero at the 5 percent level (Tables A26 and A27). Split by protest violence (Table A27), the readings of Sections 4.3 and 5.3 hold in one regression: the participation shift sits at nonviolent protests, and heavier participation at violent protests moves the margin toward the Republicans.

6.2 The size of the two effects

A coefficient is the shift for one block group that received the treatment, and the two treatments reached very different numbers of block groups. To put them on one footing, each is turned into an average effect over the block groups it reached: the shift the regression attributes to a treated block group, averaged over all the block groups with that treatment. Both averages come from the joint regression split by protest violence (Table A27). For exposure, the average over the block groups with a violent protest within one kilometer is 0.62 percentage points on the Republican

Figure 9: Participation, exposure and the change in the net margin, by protest violence



Notes: Each marker is an estimated shift in the net margin, the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections minus the change in the Republican share, in percentage points; a marker above zero is a shift toward the Democrats. Whiskers are 95 percent confidence intervals. The three groups along the horizontal axis are three quantities, not three samples: sending at the 90th percentile of the participation rate, taken over all block groups and the same for every series, against sending nobody; the 90th against the 10th percentile of the rate among the block groups that sent anyone; and at least one protest venue within one kilometer of the block group. Within each group the three series are all protests, violent protests and nonviolent protests. The regression is of the net margin on the participation extensive and intensive margins and on ten indicators for a protest venue in each one-kilometer ring, with the controls below and zip code fixed effects, weighted by votes cast in 2020; the all-protests series comes from the pooled fit of Table A26, where the equation is written out, and the violent and nonviolent series from the split fit of Table A27. The controls are those of Figure 2 and Table A14 together: the earlier-protest ring indicators, which in the split fit are split by protest violence as the exposure rings are, the demographic controls, the venue-traffic controls, the neighborhood mobility controls and the indicators for deciles of the block group's panel devices. The sample is the 133,406 block groups on which both joint fits are estimated. Each whisker is built as in Table A31, on 333 metropolitan-area clusters. On exposure the two ring families are not statistically distinguishable, over all ten rings jointly ($F(10, 332) = 1.41$, $p = 0.174$) or at the one ring drawn ($p = 0.399$), so the gap between the violent and nonviolent markers of the third group is a difference in point estimates.

share; for participation, the average over the block groups that sent anyone to a protest is 0.09 on the Democratic share (Table A28). The backlash effect is the larger and the less precisely estimated; the mobilization effect is smaller and precise. The difference in precision follows the number of block groups each treatment reached: block groups with a violent protest within one kilometer are few, while block groups that sent a resident are most of the sample (Sections 4.3 and 5.1).

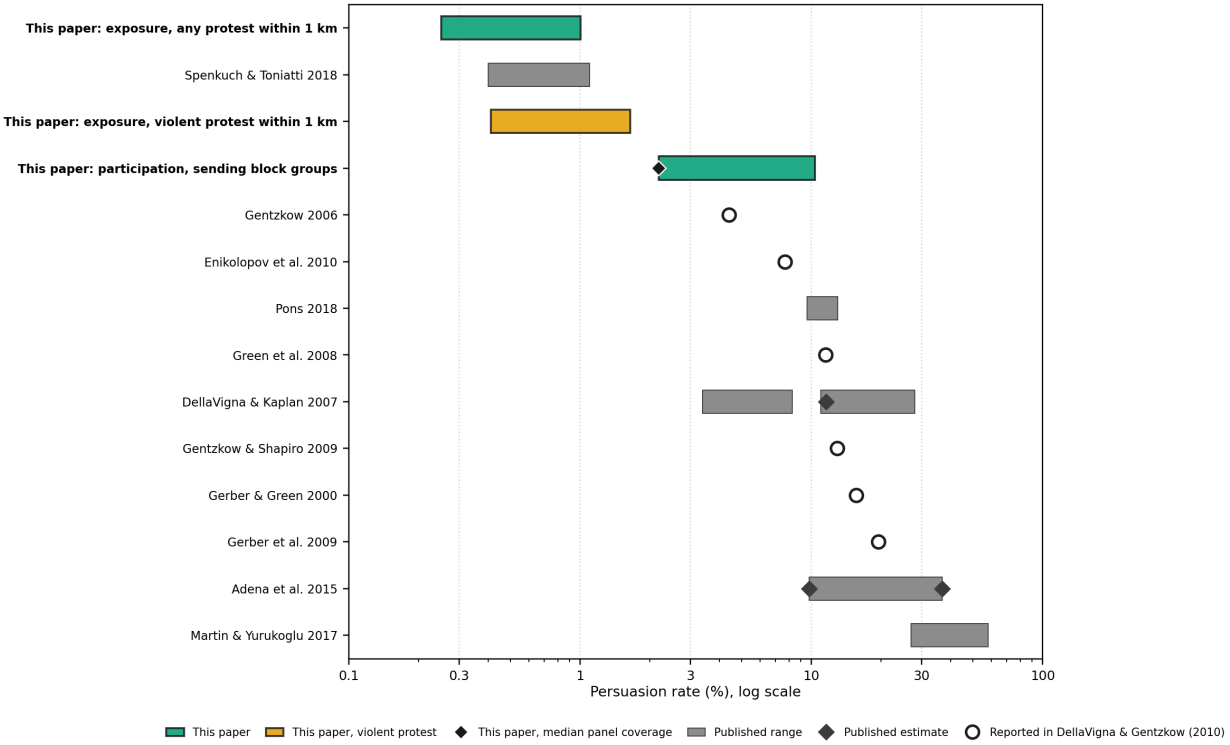
To give the two effects a scale, each is expressed as a persuasion rate, the measure used to compare media and campaign interventions that reach different shares of their audience (DellaVigna and Gentzkow, 2010): among the people a treatment reached who were not already voting the way it pushes, the share whose vote it changed. The rate divides the estimated shift in the vote share by the reach of the treatment and by the share of voters left to persuade. The shifts are the estimates of Sections 4 and 5: the 0–1 km coefficients for exposure and the extensive margin for participation.

For exposure the reach is not observed, since no data record who among the residents within one kilometer saw a protest, and the rate is reported under three assumptions: that every resident within one kilometer was reached, which gives the lower bound on the rate, that one half were, and that one quarter were. For participation the reach is measured, as attendances per voting-age resident of the sending block groups, divided by the share of a protest's crowd that the device panel detects. The persuasion rate of exposure to any protest is 0.25 percent at the lower bound, 0.50 percent if one half of the residents were reached and 1.00 percent if one quarter were; for exposure to a violent protest the three rates are 0.41, 0.82 and 1.65 percent. The persuasion rate of participation is 2.19 percent at the panel's median coverage of a crowd, and 10.39 percent at the 75th percentile of that coverage. Figure 10 sets these rates beside published ones. The rates of exposure are of the order of the rate of a candidate's whole television advertising campaign (Spenkuch and Toniatti, 2018) and below every other published rate in the figure, the arrival of Fox News on local cable (DellaVigna and Kaplan, 2007) and door-to-door canvassing (Pons, 2018) among them; the rate of participation is above the rates of exposure and at the lower end of the range of the media and canvassing rates. The order of the two is the reverse of their order as average effects, because the two treatments differ in reach: a protest within one kilometer is assumed to reach between a quarter and all of a block group's residents, while a sending block group's attendances are between 1.5 and 7.2 for every hundred of its voting-age residents. The comparison is one of orders of magnitude and not of precise positions: each published rate rests on its own paper's definition of who was reached, and the paper's own rates move with the reach assumed.

6.3 The net effect in votes

What do the two add up to in votes? A back-of-the-envelope calculation applies the coefficients of the joint regression to each treated block group of the sample and sums the implied vote shifts. So it speaks about the electorate in the sample, 133,406 block groups in metropolitan areas, and not about the national electorate (Figure 11; Table A31 is the full decomposition, with every interval). Participation moved about 137,000 votes toward the Democrats, a contribution estimated precisely, with an interval clear of zero. Exposure within three kilometers of a venue moved about 87,000 votes toward the Republicans, also with an interval that excludes zero; beyond three kilometers the exposure contribution is not distinguishable from zero, and neither is the exposure contribution as a whole. The net of the two is 104,428 votes toward the Democrats, one tenth of a percentage point of the net margin over the sample's electorate, and it is not distinguishable from zero. These are the votes moved by direct exposure to protests and by participation in them, in the metropolitan areas of the United States, between block groups of the same zip code; whatever protests did to the vote through the media, an effect recent work has looked for at coarser levels (Gethin and Pons, 2026), is common to a zip code and is netted out by its fixed effect, and rural areas are outside the sample. The net is imprecise because the specification carries the exposure rings out to ten kilometers, and Section 4 showed the coefficients beyond three kilometers imprecisely estimated, so the votes those rings imply are imprecise too, while the contribution within three kilometers is

Figure 10: The paper's persuasion rates against published persuasion rates

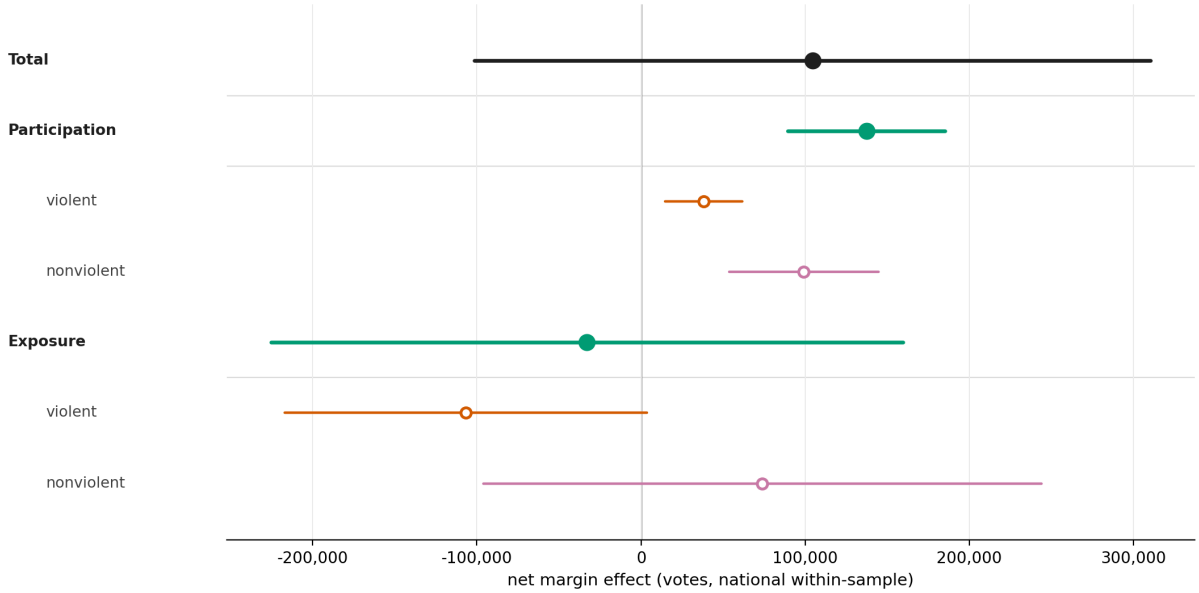


Notes: The figure sets the persuasion rates implied by this paper's estimates beside persuasion rates published for media and campaign interventions. The horizontal axis is a persuasion rate in percent, on a logarithmic scale: among the people a treatment reached who were not already voting the way it pushes, the share whose vote it changed (DellaVigna and Gentzkow, 2010). The rate is $f = 100 \times (y_T - y_C)/[(e_T - e_C)(1 - y_0)]$, where $y_T - y_C$ is the effect on the vote share, $e_T - e_C$ the reach, the share of the treated group that the treatment reached net of the comparison group's, and y_0 the share already voting that way. The three colored boxes are this paper's. For exposure, $y_T - y_C$ is the 0–1 km coefficient of Table A5 (any protest) or of Table A6 (violent protest), and y_0 the mean 2016 Republican share of the block groups with such a protest within one kilometer. Who among their residents saw the protest is not observed, so the reach is assumed, and the box runs from a reach of one, every resident within one kilometer reached, which gives the lower bound on the rate, through one half, to a reach of one quarter. For participation, $y_T - y_C$ is the extensive margin of Table A14, y_0 the mean 2016 Democratic share of the block groups that sent anyone, and the reach is measured: attendances per voting-age resident of the sending block groups, divided by the share of a protest's reported crowd that the device panel detects. The box runs from the median of that share over the 4,211 events with a reported crowd size, 0.04, to its 75th percentile, 0.19, and the black diamond marks the median. Gray boxes are the ranges that five papers publish, each across its own audience definitions, elections or samples: Spenkuch and Toniatti (2018), for a candidate's whole advertising campaign; DellaVigna and Kaplan (2007), with one box for the audience measured by recall and one for the audience measured by viewing diaries; Pons (2018), for the two rounds of the election; Adena et al. (2015), for 1930 and 1933; and Martin and Yurukoglu (2017), for three elections, from a structural model. Gray diamonds are single published rates. Open circles are rates listed in Table 1 of DellaVigna and Gentzkow (2010), drawn under that table's labels; the diamond on the row of DellaVigna and Kaplan (2007) is that table's rate for the paper. Each published rate uses its own paper's definition of reach. The boxes are ranges over assumptions or over published estimates, not confidence intervals.

precise on its own.[15] Over 2016 to 2024 rather than 2016 to 2020 (Table A32), the participation contribution is larger, 188,008 votes, and the exposure contribution within three kilometers is larger

[15]With the exposure rings drawn two and a half kilometers wide instead of one (Table A30), the participation contribution barely moves and stays precise, while the exposure contribution changes sign and straddles zero, so the aggregate exposure contribution is not robust to how the rings are drawn.

Figure 11: The aggregate net effect in votes, by treatment and protest violence



Notes: Each marker is a number of votes and each whisker a 95 percent confidence interval on that number. The quantity is the net margin: votes shifted toward the Democrats minus votes shifted toward the Republicans, so a marker to the right of zero is a net shift toward the Democrats. The aggregate is built by applying the estimated coefficients to the treated block groups of the estimation sample and summing the implied vote shift over them; it is a statement about the electorate in the sample, 133,406 block groups in metropolitan areas, and not about the national electorate. The rows are the total and the two treatments, participation and exposure, each split by protest violence. The coefficients are those of the joint regressions of Tables A26 and A27, of the change in the net margin between the 2016 and 2020 elections, in percentage points, on the participation extensive and intensive margins and on ten one-kilometer exposure rings, with the controls and zip code fixed effects of Figure 9, weighted by votes cast in 2020. Each interval is built as in Table A31, clustered by metropolitan area, 333 clusters; the whiskers are that interval in votes, the percentage-point endpoints multiplied by the summed 2020 vote weight. The total is not distinguishable from zero whatever its point estimate, and the point total is not evidence that the movement gained votes on net. The difference between the violent and the nonviolent distance profiles is not statistically established on this fit's own test over all ten rings ($F(10, 332) = 1.41$, $p = 0.174$), so the pattern under exposure is a description of point estimates and not a demonstrated difference. Table A31 is the full decomposition behind this figure, with the distance bands.

too, while the exposure contribution as a whole and the total are again not distinguishable from zero.

Could the direct local effects of the protests have decided the presidential election by themselves? Unlikely, on these estimates. Table A33 applies the same coefficients to the treated block groups of six states where the election was closest and sets the estimated net shift against each state's final margin. The national coefficients of record are applied to each state's treated block groups, so the per-state split shows where the treated block groups are and how violent that state's protests were. In every state the participation contribution is distinguishable from zero and the exposure contribution is not. The shifts are small against the margins and spread across the states: only in Georgia is the net shift on the scale of the final margin, 5,753 votes against 12,670, and Georgia's is the one shift distinguishable from zero at the 5 percent level; in the other five states the shift is well below the margin, and the three closest states together shifted 7,681 votes against a combined

margin of 43,884.

6.4 The cost of the wave's violence

A striking feature of the George Floyd movement against earlier protest movements was its violence: 10.4 percent of its events turned violent, against 2.2 percent of American protests in the three preceding years, and Section 4 showed the shift toward the Republicans larger near violent protests. The last exercise asks what that violence cost the movement in votes. It re-runs the aggregate of Section 6.3 under alternative shares of violent protests, to answer three questions: what the net shift would have been had no protest turned violent, what it would have been with only as many violent protests as in earlier movements, and at what share of violent protests the backlash would have outweighed the mobilization gains. The exercise rests on two strong structural assumptions. First, participation is held at its observed level in every scenario, since the attendance response to a protest's character is not observed and is not modeled. Second, the geography of protests is unchanged, violent protest days being relabeled as nonviolent at random until the violent share reaches a target and the distance-ring exposures recomputed under the new labels. On these estimates the violence is associated with a loss of Democratic votes. With no violence at all the net shift would have been 0.20 percentage points against the 0.10 observed, about twice, so on these estimates violence is associated with the loss of about half the net shift the wave would otherwise have produced in the neighborhoods it reached; at the rate of earlier movements most of that gain would already have been there (Table A34).[16] Figure A13 traces the same exercise over every violent share: the point path crosses zero, the share at which the backlash outweighs the mobilization gains, at 25.1 percent, one protest in four, far above the 10.4 percent that turned violent.[17]

A last counterfactual varies where the protests happened rather than what they were (Table A35): every protest is moved to a venue that had hosted protests before the wave, with attendance either unchanged or scaled to the venues that remain, and the total changes by 0.01 and -0.02 percentage points, both with intervals that cover zero. The two counterfactuals together say that on these estimates it is the violence of the protests, and not the geography of their venues as such, that would have changed what the movement got from the vote: moving every protest to a venue that had hosted protests before leaves the net shift where it is, while removing the violence would have about doubled it.

7 Mechanisms: Evidence from North Carolina

Sections 4 and 5 found two shifts in the vote between the 2016 and 2020 presidential elections, on the national sample: toward the Republican candidate in block groups near a protest venue, more

[16] That difference is 0.10 percentage points, with a 95 percent confidence interval of -0.002 to 0.207 and $p = 0.054$; the interval includes zero, so the data do not distinguish the two scenarios.

[17] The interval on that crossing is wider than the range of shares the figure draws, so the crossing is a landmark on the point path and not an estimated threshold.

so near a violent one; and toward the Democratic candidate in block groups that sent residents to a protest. This section uses the North Carolina voter files of Section 2.4 and Appendix A.4 to ask through which channel each shift arrived. North Carolina publishes its registration file as it stood on the day of each past election, together with the history of who voted in each. Therefore turnout, the composition of the roll and the movements that change it can be followed person by person on the block groups where the vote is measured.[18] A vote share alone cannot separate these channels: the same change in a party's share arrives whether the people already registered voted at different rates than before, whether the roll changed because people newly registered, were removed or moved, or whether people on the roll changed party, and Sections 4.2 and 4.3 could say only that the votes missing near a protest were Democratic ones.[19]

The first question is whether the national results appear in this state at all. Table A36 re-runs the exposure specification of Section 4.1 and the participation specification of Section 5.1 on urban North Carolina, the exposure rings split by protest violence, beside their national estimates. The shift toward the Republicans near a violent protest reproduces: within one kilometer the Republican share rose 1.11 percentage points more than in block groups of the same zip code without a violent protest in that ring, against 0.34 nationally.[20] The participation shift has no detectable counterpart: the extensive-margin estimate is not distinguishable from zero where the national one is.

Every channel is estimated in the joint specification of Section 6, with both treatments, the same controls and zip code fixed effects, the exposure rings split by protest violence; Table A37 estimates each channel again under each arm's own specification, and every estimate distinguishable from zero keeps its sign and rough size there. At the level of the person, the history file sorts registrants by their record before the protests: habitual voters, who voted in both the 2016 general election and the 2018 midterm; occasional voters, who voted in one of the two; and registrants who voted in neither.

The turnout channel is the one the voter file measures most directly: whether the people already registered in a block group voted. Figure A14 plots the change from 2016 to 2020 in the turnout of the people on the roll at both elections, by distance to a violent and to a nonviolent protest. Near nonviolent protests turnout fell, within one kilometer by 1.54 percentage points; near violent protests the estimate within one kilometer is -0.15 percentage points with an interval that covers zero. The fall near nonviolent protests is present for registered Democrats, Republicans and unaffiliated voters alike, each with an interval below zero (the right panel of the figure; Appendix G.2). Which registrants stopped voting? Figure A15 sorts the same people by their voting history and, within

[18] To my knowledge, no other state's published file combines these: many states do not register voters by party at all, several restrict the file to their own residents, parties or candidates, and almost none keeps the file as it stood at past elections.

[19] Few of the state's counties hold a sample block group, so the asymptotic standard error is not trusted on its own: every star in this section and in Appendix G comes from a wild cluster bootstrap on the county clusters (Cameron et al., 2008), and the tables print the asymptotic standard error beside each coefficient.

[20] Table A36 prints the coefficients, with the joint specification of Section 6 as a third block; its national cells are the paper's own estimates. The North Carolina point estimate is about three times the national one, on a much wider interval; Appendix G.1 reads that comparison.

the block group, compares each group's change in turnout with that of the habitual voters. Within one kilometer of a nonviolent protest the turnout of occasional voters fell 4.38 percentage points further than that of habitual voters, and that of registrants who had not voted 1.50 further; near a violent protest the gaps are 3.83 and 1.64, and the four joint tests over the ten rings all reject.

The registration channel is who is on the roll and with which party. Near nonviolent protests the Democratic share of registered voters did not detectably change (Table A37); near violent protests it fell, by 1.14 percentage points within one kilometer, at the 10 percent level on the bootstrap. Figure A16 splits that change into the five movements, each estimated as its own outcome on the same specification so that the five sum to the total: new registration accounts for the whole of it, 1.17 of the 1.14 points, and is the only movement distinguishable from zero, so the decline is a matter of who newly joined the roll and not of anyone already on it leaving or changing party. The share fell because the newcomers, though more Democratic than the new registrants of the sample as a whole, are less Democratic than the roll they joined.

The worry this raises is that the block groups near the venues of 2020 were already changing hands before the protests. Table A38 tests it directly, taking the same five movements over the earlier window from the 2012 to the 2016 election and regressing each on the 2020 treatment.[21] Within one kilometer, the ring that carries the registration result, no movement loads on the 2020 treatment and neither does the total, so the check is clean at the ring that matters; at two to four kilometers it is not, and the estimates at those distances are not read as effects of the 2020 protests (Appendix G.4).[22]

Read together, the North Carolina voter files paint the following picture on the channels that moved alongside the vote shifts of Sections 4 and 5. Near violent protests, the roll became less Democratic through new registrants while the turnout of the people already registered did not detectably change. Near nonviolent protests turnout fell, most among occasional voters and registrants who had not voted before, while the composition of the roll did not detectably change. The participation shift has no detectable counterpart in the state's vote and none in its roll, the attendance terms of every channel being not distinguishable from zero at the 5 percent level, so on that arm the state reports a null and nothing more.

8 Conclusion

This paper studies the direct local effects of the George Floyd protests of 2020 on the presidential election five months later. The movement was large and exceptionally violent, its cause quickly took a partisan connotation, and its protests were spread within cities: within a zip code some block

[21] The voter history file begins with the 2016 election (Appendix A.4), so the turnout channel has no earlier window on which the same check could be run.

[22] One fact is a limit on the clean ring: every one of the 50 block groups with a violent protest within one kilometer is in the upper half of the sample's distribution of residential turnover over 2012 to 2016, and none is in the lower half, so at this distance proximity to a violent protest cannot be separated from being a high-turnover neighborhood.

groups had a protest close by and others did not.

This paper assembles a novel dataset at the level of the census block group. It combines the venues of the George Floyd protests, geocoded from the media reports behind the Crowd Counting Consortium's record, protest participation measured from the real-time location pings of 12.9 million mobile devices, and the block-group returns of the presidential elections of 2012 to 2024. It then uses the spatial fragmentation of the protests as a source of variation. For exposure, it compares block groups that had a protest nearby with block groups of the same zip code that did not; for participation, block groups that sent residents to a protest with block groups of the same zip code that sent none. In both cases the outcome is the change in the block group's vote between the 2016 and 2020 elections. The comparison within the zip code holds fixed what reached every block group through the media and national politics.

Exposure generated backlash. Block groups with a protest within one kilometer shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code with no protest in that ring, and near violent protests the point estimates are larger. Within 7 km of a future venue, block groups were not already moving toward the Republicans, and a difference-in-differences design over four elections shows no detectable pre-trend. The backlash shows no detectable gradient along the Black population share, the dimension where selection into this wave would likely show, and was still present in 2024.

Participation generated mobilization. Block groups whose residents turned up at protest venues during protest hours shifted 0.07 percentage points more Democratic than otherwise similar block groups of the same zip code that sent none or fewer, and sending more added no detectable shift. Those block groups were not already moving toward the Democrats, and earlier attendance at the same venues predicts no detectable shift. The mobilization sits at nonviolent protests and with attenders who were not regulars there: sending more residents to a violent protest moved the vote further from the Democratic candidate, as did incidental attendance by residents who go to those venues regularly. I read both as a backlash entering through participation. The shift did not subside between 2020 and 2024.

In one joint regression the two treatments coexist and pull in opposite directions. Averaged over the block groups each reached, exposure to a violent protest within one kilometer moved the Republican share by 0.62 percentage points and participation moved the Democratic share by 0.09 percentage points. Expressed as a persuasion rate, the effect of exposure is of the order of that of a candidate's whole television advertising campaign and below the rate estimated for the arrival of Fox News on local cable. In votes, participation moved about 137,000 toward the Democrats and exposure within three kilometers about 87,000 toward the Republicans; the net is not distinguishable from zero, and the direct local effects are unlikely to have decided the election. On these estimates, what cost the movement votes was its violence, not where its protests took place. Without violence the net shift would have been about twice the one observed, while moving every protest to a venue used before the wave leaves it where it is.

North Carolina's voter files show which channels moved alongside the shifts. Near violent protests the roll became less Democratic through new registrants while the turnout of those already registered did not detectably change; near nonviolent protests turnout fell and the roll did not detectably change.

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A Data and sample

This appendix describes how the four sources of Section 2 were compiled: the protest record, the device panel and its home rule, the allocation of precinct returns to block groups, and the North Carolina voter file.

A.1 The protest record

The Crowd Counting Consortium records marches, protests, demonstrations, rallies, strikes, riots and similar actions, and not meetings, panel discussions or town halls; its record begins with the Women's March of January 2017. From April 2017 its main input was a daily crawl of more than 3,400 local and national news outlets, run first by the Count Love project and later by the Consortium itself; social media were read without a systematic search, the public could submit reports, and for large protests held in many places at once the organizers' websites were used. The crowd size as the sources described it is converted into a low and a high count by fixed rules. In 2023 alone, the Consortium recorded more than 29,500 events, with crowds that summed to more than nine million people (Ulfelder, 2024).

The compiled file for 2017 to 2020 holds 72,181 events, of which 21,049 are dated between 27 May and 2 November 2020. I drop the events held online, the events without a location within the city or without a state, the events whose location is ambiguous, and the duplicate records of one event, which leaves 17,312. Five events cannot be geocoded. I keep the events whose coordinate is a street address, the center of a named place or street, or a point interpolated along a street, and drop the 3,170 whose coordinate is only approximate, at the level of a town or a district. Of the rest, 1,611 fall outside every metropolitan area and 4,606 are not related to Black Lives Matter, which leaves the 7,920 events of the wave. They took place at 4,815 venues; 871 venues hosted more than one event, a venue can host more than one event on the same day, and the busiest hosted 154. Of the 823 violent events, arrests are the most common of the five markers, reported at 555 of them.

A.2 The device panel and the home rule

A device's home is the block group where it spent most of its nights in September and October 2020, taken from each night's median location, and a device is homed only if it is observed on at least ten nights in that window with at least half of them in that block group. A home over January and February 2020 is computed as a comparison: 3.6 million of the 12.9 million devices have a home in both windows, and 15.5 percent of those moved block group between them.

A block group typically covers a few city blocks. The Census Bureau targets between 600 and 3,000 residents when it draws a block group, and 92.37 percent of the block groups of the country fall in that range. I use the 2010 census block groups as published in the 2018 TIGER/Line release throughout. Block groups were not redrawn between the 2010 and the 2020 censuses, but outside Puerto Rico 73 block groups of the 2010 release and 72 of the 2018 release carry an identifier the

other release does not, from renumbering in four states and from two county renames in South Dakota and Alaska; the 2012 returns, which arrive on the 2010 identifiers, are matched across that seam by spatial overlay (Appendix A.3), and the other sources use the 2018 identifiers directly.

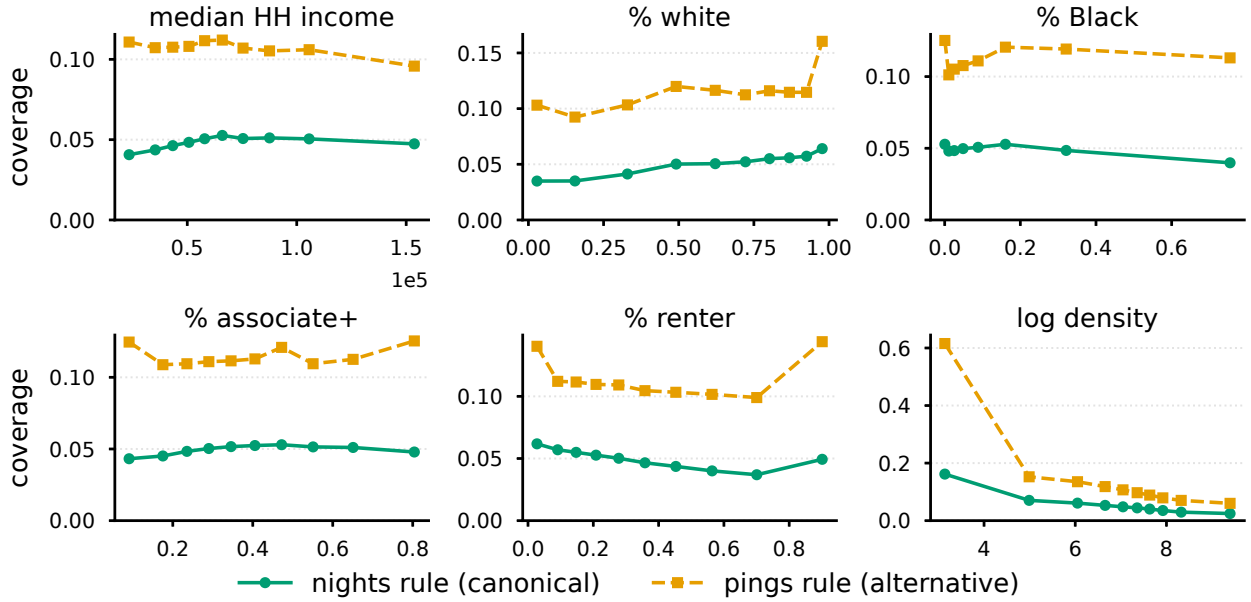
The ordinary presence that enters every participation regression as a control is measured at the same venues between 4 and 26 May 2020, the three weeks before the first event, on days those venues hosted nothing. The four measures of how far residents' devices ranged on an ordinary day, the mobility controls, are taken over January and February 2020, before the pandemic. Every aggregate left the secure server under the disclosure rules stated in the note to Table A1: counts are built by block group and day, a day-level count of one to four is withheld, and a block group with fewer than ten homed devices is withheld. Table A1 reports the panel, the attendance it detects and the voting sample; Figures A1 and A2 compare the panel's coverage and composition with the resident population.

Table A1: The device panel, detected attendance, the voting sample and the disclosure rules

	Value
Homed devices in the panel	12,867,138
Block groups in the panel	173,544
Metropolitan areas covered	368
Attendances, George Floyd window, by venue detection radius	
50 m	50,185
100 m	137,882
200 m (primary)	385,891
400 m	1,031,983
Block groups with any detected attendance (200 m)	108,679
Voting sample (participation estimation sample)	
Block groups in the sample	135,691
... with any detected attendance (200 m)	85,399
2020 ballots cast in the sample	108,552,069
... as a share of all 2020 ballots cast nationally	68.5%
Disclosure regime (applied before export)	
Block group × day cells built	64,896,796
Day-level small-count suppression	counts of 1–4 withheld (minimum five devices)
Thin-panel rule	block groups with < 10 homed devices withheld
Block group × day rows withheld as thin	1,231,946
Participation events, all resolutions	587,009
... share sitting in withheld day cells	85.0%

Notes: The table describes the device panel the participation measure is built from, the attendance it detects, the voting sample the participation estimates use, and the disclosure rules applied before any aggregate left the secure server. A metropolitan area is a Census core-based statistical area; the panel covers the 368 that contain at least one protest venue. A device's home is the block group where it spent most of its nights in September and October 2020 (Appendix A.2). An attendance resident-day is one homed device observed within the stated distance of a protest venue on that venue's event date, during the midday window; the counts are summed over the George Floyd wave. The block groups with any detected attendance are those with at least one attendance resident-day at 200 m, counted over the whole panel in the attendance block and over the voting sample in the voting block. The voting-sample rows describe the participation estimation sample, the block groups on which the participation estimates of Section 5 are fitted. The disclosure rows describe the rules applied inside the secure server: counts are built at the block group by day level; a day-level count of one to four is withheld; a block group with fewer than ten homed devices is withheld; a participation event is one device-day of detected attendance at any radius. The participation measure is built from totals over the whole wave, not from the day-level cells. The composition of the homed panel against the Census is in Figure A2.

Figure A1: Device coverage per resident by block-group characteristic



Notes: Each panel is a binned scatter of device coverage against one block-group characteristic. Coverage is the number of homed devices divided by the block group's population; each marker is the mean coverage among the block groups in one bin of the characteristic on the horizontal axis, with bins of equal count. The solid series homes each device to the block group where it spent most of its nights (the nights rule of Appendix A.2, used throughout the paper); the dashed series homes it to the block group holding most of its location pings (the pings rule, shown as an alternative). Median household income is in dollars; % white and % Black are population shares; % associate+ is the share of residents aged 25 and over with an associate degree or higher; % renter is the share of renter-occupied housing units; log density is the natural logarithm of residents per square kilometer. The vertical scale of the density panel is wider than that of the other five. The sample is every block group in the 368 metropolitan areas with a protest venue.

Figure A2: Composition of the homed-device panel against the Census benchmark

		nights rule (canonical)		*pings rule (alternative)*	
Control	**Census**	**Mean**	**Gap/SD**	**Mean**	**Gap/SD**
% White	0.585	0.651	0.21	0.622	**0.12**
% Black	0.130	0.117	-0.06	0.131	**0.00**
% Hispanic	0.191	0.152	-0.17	0.163	**-0.12**
% Asian	0.060	0.049	-0.11	0.051	**-0.09**
Median age	39.1	39.3	0.02	39.2	**0.01**
Median HH income ($)	71,933	73,742	0.05	71,156	**-0.02**
Unemployment	0.061	0.056	-0.07	0.059	**-0.03**
% associate or higher	0.409	0.417	0.04	0.407	**-0.01**
% renter	0.364	0.316	-0.18	0.345	**-0.07**
log population	7.425	7.475	0.07	7.422	**-0.00**
log density	6.822	6.325	-0.28	6.383	**-0.25**
Devices placed: nights 12,890,502 · pings 27,458,445. Block groups covered: 177,205 / 177,468 of 177,543.					

Notes: The figure compares, row by row, the composition of the block groups where devices are homed with the composition of the resident population. Each row is one block-group characteristic. The Census column is the population-weighted mean of the characteristic across block groups; the Mean columns are the device-weighted means, in which each block group counts in proportion to the devices homed there, under the two home rules; Gap/SD is the device-weighted mean minus the population-weighted mean, divided by the standard deviation of the characteristic across block groups. The nights and pings rules are those of Figure A1. Shares are fractions of the block group's population (% White, % Black, % Hispanic, % Asian), of residents aged 25 and over (% associate or higher), of the labor force (unemployment) or of housing units (% renter); median household income is in dollars; log population and log density are natural logarithms of residents and of residents per square kilometer. The sample is every block group in the 368 metropolitan areas with a protest venue; the bottom line reports the devices placed and the block groups covered under each rule, counted at the homing step, before the thin-panel rule of Table A1 withholds the block groups with fewer than ten homed devices. A gap within 0.15 standard deviations is read as a match.

A.3 How precinct returns are allocated to block groups

Precincts are the finest unit at which election returns are published, and their boundaries neither follow Census boundaries nor stay fixed between elections. The paper's unit is the block group, so the returns of 2016, 2020 and 2024 are moved from precincts to block groups by the procedure described here; the 2012 returns arrive at the block-group level from another source, described at the end.

The precinct geometries are the 2016 and 2020 files of the Voting and Election Science Team, 177,194 and 170,200 precincts covering the fifty states and the District of Columbia, and for 2024 the precinct file of the New York Times completed with the Redistricting Data Hub files for Alaska, New Hampshire, Rhode Island and Vermont, 162,766 precincts in 49 states and the District of Columbia; Oklahoma and South Dakota are absent from 2024 because their precinct totals do not reconcile with the certified state totals within two percent. The block-group geometries are the 217,739 block groups of the 2018 TIGER/Line release. Each precinct polygon is intersected with the block-group polygons in an equal-area projection, and every piece of the intersection is given a population: the block group's population in the 2010 census multiplied by the share of the block group's area that the piece covers. A precinct's votes for each candidate are then split across its pieces in proportion to these piece populations, so that each precinct's votes are conserved by construction. The method assumes that population is spread evenly within a block group and that a precinct's vote shares are the same everywhere within the precinct. Where a precinct's overlapping population is zero, its votes are split by area instead, which happens for 67, 71 and 47 precincts in the three years; a piece with no population inside a precinct with population receives no votes; and a block group that overlaps no precinct receives zero votes, which happens for 278, 298 and 3,790 block groups in 2016, 2020 and 2024. Precincts whose polygon does not survive the cleaning of the geometries, 768, 1,106 and 239 in the three years, and precincts that overlap no block group, 4, 7 and 1, are dropped before the allocation, and pieces that fall in a different state from their precinct are removed after it. The votes allocated to block groups are 99.976, 99.973 and 99.992 percent of the votes in the three source files; the shortfall is the dropped precincts and the cross-state pieces, and in almost every state it is a few hundred votes or fewer. The largest shortfalls in proportion are in the District of Columbia, whose whole boundary is shared with two states, and in absolute terms in New York, where most of the gap is precincts that overlap no block group. A block group's presidential total is the sum over all candidates, a party's share is its votes over that total, and a block group with no votes in a year has no share rather than a share of zero.

The 2012 returns are not allocated here. They are the presidential results of the panel of Longuet-Marx (2026), in which the returns of the precincts that reported votes for the House of Representatives are apportioned to the block groups of 2010 through the census blocks inside each precinct, with each block weighted by the share of its area inside the precinct times the block's population. From that file I keep the block groups with at least fifty votes and match them to the 2018 identifiers; all but 51 match by identifier, and those 51, in block groups renumbered between the two releases,

are matched by spatial overlay. The 2012 returns cover the 42 states of Table A2, 174,286 block groups; the eight states and the District of Columbia without them are outside every estimation sample.

Two recent papers put precinct returns on census block groups and describe how. Longuet-Marx (2026) works at the level of the census block, as above, and so requires population to be spread evenly only within a block; this paper works at the level of the block group and assumes it there. Sonin et al. (2023) intersect the 2016 precinct files of the Voting and Election Science Team with the block-group polygons and give each block group the weighted average of each candidate's votes over the precincts that overlap it, with weights equal to the share of each precinct's area inside the block group, without a population term; they check the method on the 2020 returns, allocated the same way, by aggregating the block-group votes back to counties and states, where the vote shares correlate at 0.999 and 0.998 with the official returns. This paper weights by population rather than by area alone, and checks conservation exactly, precinct by precinct and state by state, rather than by correlation. As a check, I redid the 2016 and the 2020 allocations with the population of each piece taken from the 2010 census blocks inside it, the rule of Longuet-Marx (2026): over the block groups of the exposure sample the vote shares of the two allocations correlate at 0.998 in each year, the median block group moves by 0.09 percentage points in 2016 and by 0.08 in 2020, about one block group in five moves by more than 0.5 points and one in eight by more than 1 point; the change in the Republican share between the two elections, the outcome of Section 4, correlates at 0.989 across the two allocations, with a median move of 0.07 points and one block group in seven moving by more than 0.5 points.

Table A2: Block-group vote returns by election year, source and coverage

Year	Source	States covered	Block groups covered	% of universe
2012	(Longuet-Marx, 2026)	42 of 51	174,286	80.0%
2016	Voting and Election Science Team	51 of 51	216,784	99.6%
2020	Voting and Election Science Team	51 of 51	216,788	99.6%
2024	New York Times + Redistricting Data Hub (AK, NH, RI, VT)	49 of 51	213,161	97.9%

Notes: The table reports, for each election year, the source of the returns, the number of states and block groups with returns, and the share of all block groups covered. The 2016, 2020 and 2024 precinct returns are allocated to block groups. The 2016 and 2020 returns are the precinct files of the Voting and Election Science Team. The 2024 returns are the precinct files of the New York Times, completed with the Redistricting Data Hub files for Alaska, New Hampshire, Rhode Island and Vermont. The 2012 returns are the presidential results of the panel of Longuet-Marx (2026), which reports them at the block-group level; they cover only the precincts that returned votes for the House of Representatives. The universe is every block group in the fifty states and the District of Columbia on the 2018 Census boundaries; a block group is covered in a year when that year's returns allocate votes to it. The missing states are given by their postal abbreviations: in 2012 Alaska, Arkansas, the District of Columbia, Hawaii, Louisiana, Nebraska, Nevada, South Dakota and Washington; in 2024 Oklahoma and South Dakota. The paper's estimation samples require returns in 2012, 2016 and 2020, so they exclude the eight states and the District of Columbia without 2012 returns; the 2024 returns enter the persistence estimates.

A.4 The North Carolina voter file

The registration file is read as it stood on 6 November 2012, 8 November 2016, 3 November 2020 and 5 November 2024. Besides the fields named in Section 2.4, each record carries the date of registration and the county. The history file is a rolling window that begins with the general election of November 2016, so the elections of 2012 and 2014 are not in it. Registration and voting are two different things in these files, and the paper keeps them apart. Registration is being on the roll, as a Democrat, a Republican, unaffiliated, or with another party; a registered voter stays on the roll until removed, and can change the party of registration without voting. Voting is casting a ballot in a given election, and the history file records that a ballot was cast and nothing about its content.

Of the registered voters of 2016, the geocoder places 88.6 percent at an exact address match and 7.6 percent at a non-exact match; for 0.9 percent it ties between candidate addresses and for 2.9 percent it finds no match, and these are the registrants dropped. Counts of registered voters by party in a block group are taken over the geocoded individuals, and a party's registration share is its registrants over all registered voters in the block group in that year. Turnout over a window is placed in the block group where the person was registered at its start. The five movements that change a party's registration share between two elections are counted person by person: people newly register, leave the roll, move in from elsewhere in the state, move out, or change the party they are registered with; the five add up exactly to the change in the share, and party changes are counted among the people at the same address at both ends. At the level of the person, the history file sorts voters by their record before the protests: habitual voters, who voted in both the 2016 general election and the 2018 midterm; occasional voters, who voted in one of the two; and registrants who voted in neither.

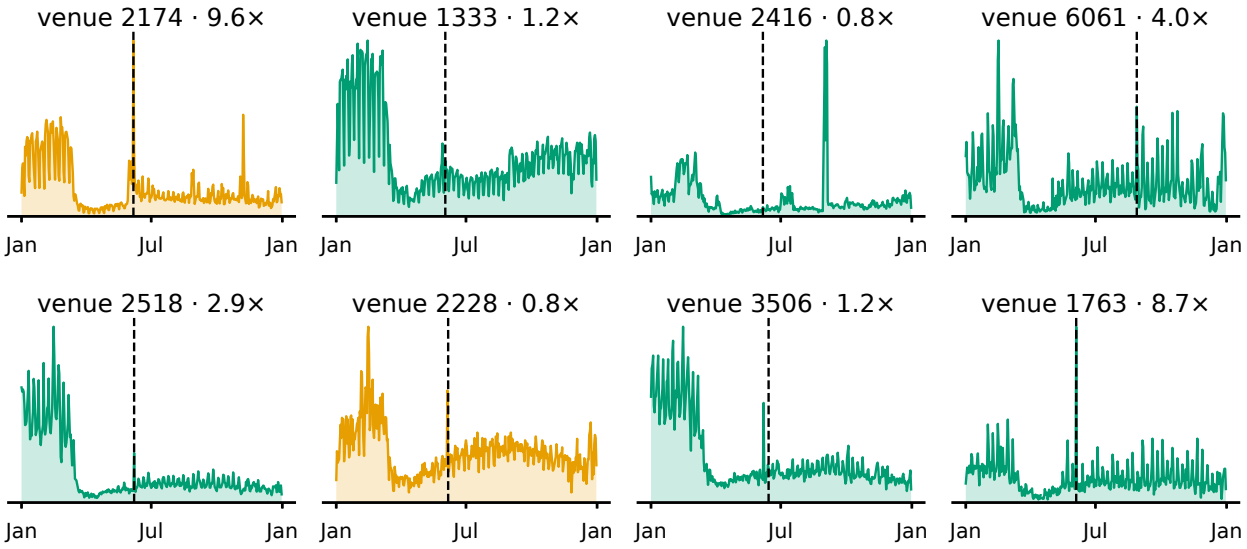
B Validation of the participation measure

This appendix validates the participation measure of Section 2.2. Section B.1 shows what the raw device counts can and cannot see, and Section B.2 what the measure built from them tracks.

B.1 What the device data can and cannot see

Figure A3 shows why the measure is built from known dates and venues rather than from raw venue counts: at most venues the protest is invisible in the raw device counts, buried under ordinary traffic.

Figure A3: Daily device counts at eight protest venues over 2020

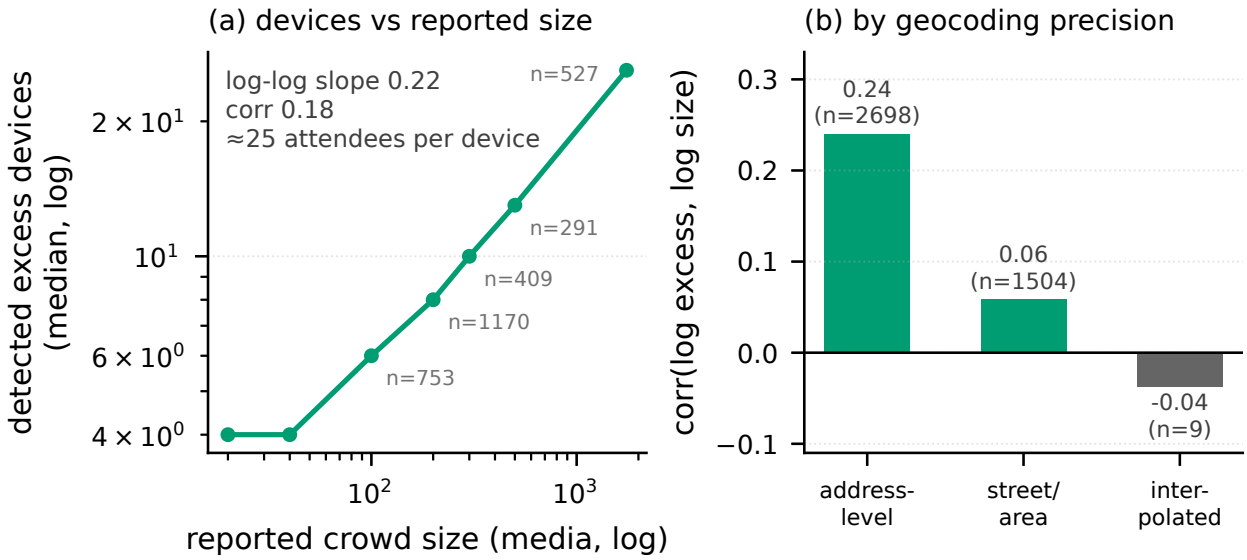


Notes: Each panel is one protest venue. The line is the number of distinct devices observed within 200 m of the venue during the midday window, from noon to midnight local time, on each day of 2020; the dashed vertical line is the venue's event date; orange marks a venue whose event was violent, teal a nonviolent one. The panel title gives the venue's identifier in the event record and the ratio of the event-day count to the venue's median daily count over the year. The eight venues are illustrations, chosen among venues with large media-reported crowds. Daily counts of one to four devices are withheld under the disclosure rule.

B.2 What the measure tracks

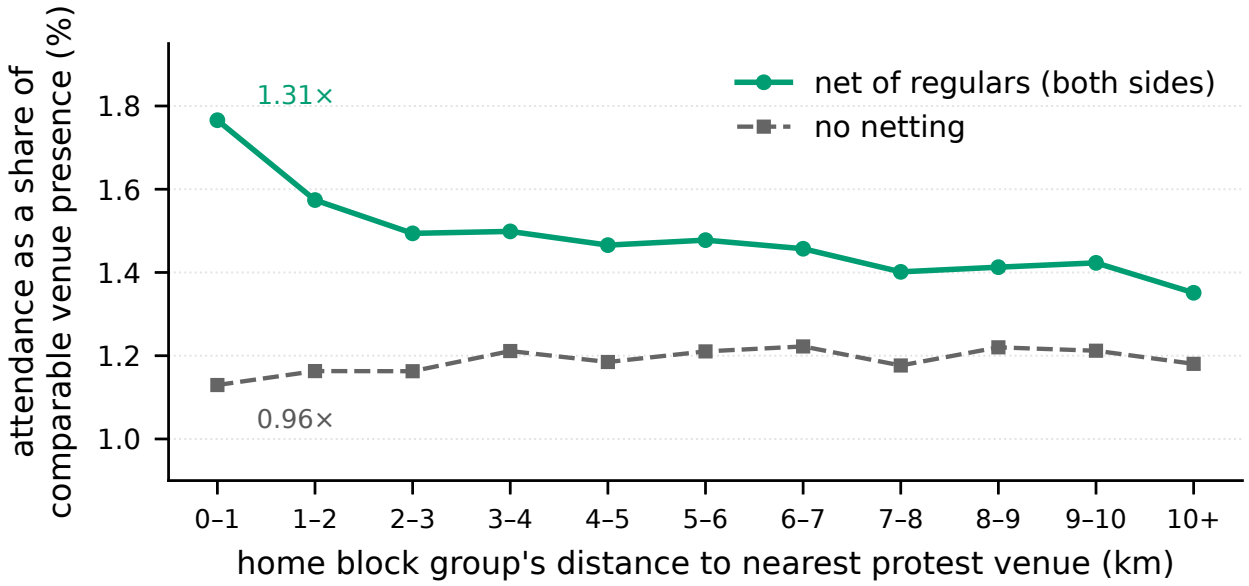
The two figures of this section show what the measure tracks. Detected attendance rises with the crowd size reported in the media, but the panel observes a sample of devices and large crowds spill beyond 200 m, so the measure is a relative index of participation and not a headcount (Figure A4); and net of regulars, attendance is more local than ordinary traffic (Figure A5).

Figure A4: Detected attendance against media-reported crowd size



Notes: The figure compares detected attendance with media-reported crowd size across venues. Detected excess is the number of distinct devices observed within 200 m of the venue during the midday window on the event date, minus the venue's median count on its non-event days. Panel (a): venues are grouped into bins of reported crowd size; each marker is the median detected excess in the bin, with the number of venues in the bin; both axes are logarithmic. The printed slope is from a regression of log excess on log reported size across venues, the printed correlation is between the same two variables, and the attenders-per-device figure is the median ratio of reported size to detected excess. Panel (b) is the correlation between log excess and log reported size within three groups of venues, by the precision of the venue's geocoded coordinate: an address-level coordinate, a street- or area-level coordinate, and an interpolated coordinate; the number of venues is printed under each bar. The sample is every venue with a reported crowd size.

Figure A5: Event-day presence relative to ordinary presence, by distance from home to the nearest venue

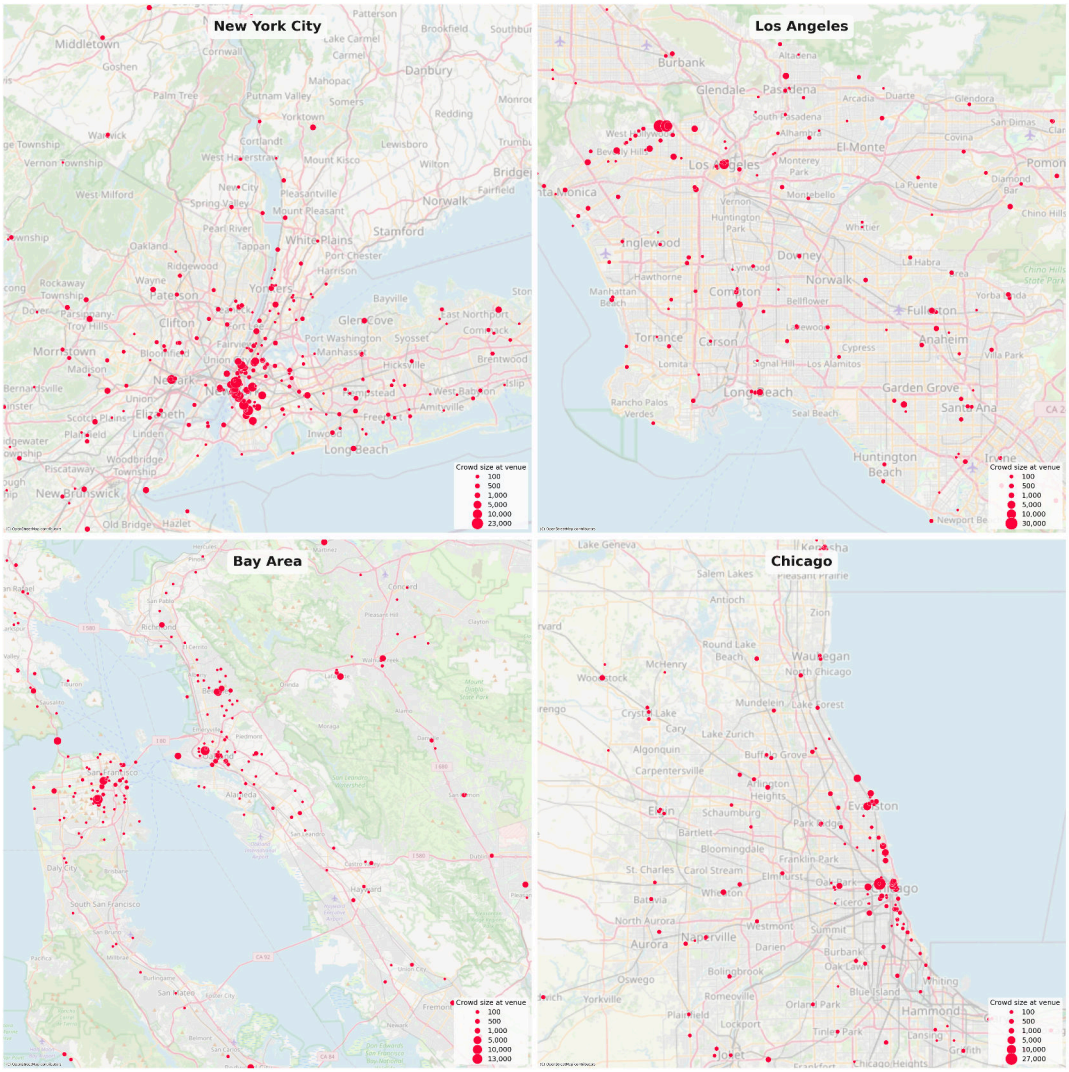


Notes: The figure asks whether residents near a venue attend its protest more than their ordinary presence there would predict. The horizontal axis is the distance from the resident's home block group to the nearest protest venue, in one-kilometer rings, with block groups beyond 10 km pooled. The vertical axis is event-day presence by residents of the ring at venues within 200 m, as a share of the same residents' presence at the same venues on non-event days over the wave, in percent. Presence is counted in device-days, one per device, venue and day. Regulars are devices observed at a venue on at least five non-event days. The solid series removes regulars from both the event-day count and the ordinary-day count; the dashed series applies no netting. The printed ratio is the nearest ring over the farthest ring for each series. The sample is every homed device in the panel, over the wave window of the presence data, 4 May to 16 November 2020.

C Protest venues and spatial fragmentation

This appendix completes Section 3. Figure A6 maps the crowds reported at the venues of the four cities of Figure 1, and Table A3 compares the block groups that hosted the wave's venues with those that hosted earlier and contemporaneous protests.

Figure A6: Protest venues by reported attendance in four cities



Notes: Each panel maps the protest venues of one metropolitan area around its central city, the same four as Figure 1 and in the same order. A marker is one venue; marker size increases with the square root of the total media-reported crowd size at the venue, summed over its events of the George Floyd wave, on the scale in each panel's legend; an event's crowd size is the Crowd Counting Consortium estimate, and where the record lists an event more than once the largest estimate is kept. The venue with the largest total drew 23,200 in New York City, 30,000 in Los Angeles, 13,200 in the Bay Area and 27,024 in Chicago, out of panel totals of 203,463, 119,172, 65,486 and 78,069.

Table A3: Characteristics of the block groups hosting protests, the wave against earlier and contemporaneous protests

	George Floyd (27 May–2 Nov 2020)	Pre-period (1 Jan 2017–26 May 2020)	Other (27 May–2 Nov 2020)
Population	1,480	1,501	1,451
Density (per km^2)	2,846	2,489	2,449
Republican vote share 2016 (%)	26.0	27.0	28.9
Democratic vote share 2016 (%)	67.7	66.4	64.6
Median household income	$65,177	$69,389	$67,038
% Black	15.1	11.5	11.7
% Hispanic	14.1	14.5	13.8
% associate degree+	50.7	53.3	51.4
Median age	38.0	37.4	37.7
Distance to metropolitan-area centroid (km)	28.1	30.0	28.5
Distance to principal city (km)	17.5	16.9	16.5
Inside principal city (%)	90.9	92.3	90.8

Notes: The table reports event-weighted means of the characteristics of the block group containing each protest venue, for the same three groups of events as Table 1: each event contributes its venue's block group once. Population is residents; density is residents per square kilometer; the Republican and Democratic vote shares are shares of all votes cast in the 2016 presidential election, in percent; median household income is in dollars; % Black and % Hispanic are population shares; % associate degree+ is the share of residents aged 25 and over with an associate degree or higher; median age is in years; the two distances are from the block group to the metropolitan area's centroid and to the center of its principal city, in kilometers; inside principal city is the share of events whose venue lies in the metropolitan area's principal city. The sample is the events of Table 1 whose venue falls in a block group with these characteristics available.

D Protest exposure

This appendix reports the exposure estimates of Section 4 in full: how the exposed block groups compare with their neighbors, the ring coefficients, the outcomes and the controls ladder, the second design, the heterogeneity dimensions not shown in the text, persistence, and the robustness exercises.

D.1 Ring construction and balance

Table A4 compares, within zip codes, the block groups near a violent protest with those near only nonviolent ones on fifteen characteristics. The 2,314 block groups that were the only sample block group of their zip code contribute nothing to the estimates and are dropped, and the block groups whose centroid falls outside every tabulation area, 0.4 percent of all block groups, have no zip code and are excluded.

Table A4: Balance between block groups near violent and near only nonviolent protests, within zip codes

	Near-violent mean	Near-nonviolent mean	Diff. (within zip code)
% White	51.4	49.5	0.6
			(0.9)
% Black	17.7	19.2	-1.3
			(0.9)
% Hispanic	18.9	19.5	-0.4
			(0.4)
% Asian	8.2	8.1	1.1***
			(0.3)
Median age	36.1	37.0	-0.3
			(0.3)
Median household income	\$65,953	\$65,774	-\$4,473**
			(\$2,011)
Unemployment rate (%)	7.2	7.4	-0.0
			(0.2)
% associate or higher	50.5	44.6	0.4
			(0.9)
% Renter	67.5	56.7	6.3***
			(1.6)
Population	1,269	1,266	73**
			(32)
Density (per km^2)	10,479	7,049	570*
			(319)
Distance to city center (km)	9.7	16.3	-0.3***
			(0.1)
Δ Republican share 2012–16 (pre-trend, pp)	-3.77	-2.72	0.21
			(0.25)
Republican vote share 2016 (%)	17.6	23.1	0.0
			(0.3)
Democratic vote share 2016 (%)	76.4	71.4	-0.2
			(0.3)
Observations	3,438	14,679	18,117
Joint-balance F (within zip code)			5.53
p-value			0.000

Notes: The sample is the 18,117 block groups with at least one George Floyd protest venue within 1 km, votes cast in 2020, residents, a zip code and a metropolitan area; the 3,438 near-violent block groups have at least one violent protest within 1 km and the 14,679 block groups near only nonviolent protests have none. Columns 1 and 2 are the unweighted group means. Column 3 is the within-zip-code difference: the coefficient from a regression of the characteristic on an indicator for a near-violent block group with zip code fixed effects, so that every comparison is between block groups in the same zip code, weighted by votes cast in 2020. The racial shares are population shares; the unemployment rate is in percent of the labor force; % renter is the share of renter-occupied housing units; distance to city center is the distance from the block group to the central business district, in kilometers; Δ Republican share 2012–16 is the change in the Republican share of all votes cast between the 2012 and 2016 elections, in percentage points; the other characteristics are as defined in the note to Table A3. Each difference uses the block groups with the characteristic and another sample block group in their zip code, fewer than 18,117. Standard errors, in parentheses beneath each difference, are clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The joint-balance F tests all fifteen differences at once, from a within-zip-code regression of the near-violent indicator on the standardized characteristics. A row without a star, such as the three political rows the exposure design rests on, says that the data cannot distinguish the two groups on that characteristic within zip code, not that they are identical.

D.2 The ring coefficients, the lean controls and nonviolent exposure

Tables A5 and A6 print the first five ring coefficients of Figures 2 and 3, with Conley standard errors beside the clustered ones; Table A7 splits the earlier-protest controls by political orientation (Section 4.2), and Figure A7 sets the violent and the nonviolent profiles side by side (Section 4.3).

Table A5: Exposure to any protest, ring coefficients for the first five rings

	(1) 0–1 km	(2) 1–2 km	(3) 2–3 km	(4) 3–4 km	(5) 4–5 km
Any protest (0/1)	0.193***	0.166***	0.050	0.040	-0.043
	(0.060)	(0.050)	(0.042)	(0.033)	(0.036)
	[0.063]	[0.056]	[0.041]	[0.034]	[0.035]
Demographic controls	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644
R^2	0.085	0.085	0.085	0.085	0.085

Notes: Each column reports the coefficient on one distance ring from a single regression with ten one-kilometer ring indicators entered jointly; columns (1) to (5) are the first five rings. The specification and sample are those of Figure 2, which plots all ten rings with the 2012–2016 placebo; Observations is N from the fit. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. Conley standard errors (Conley, 1999) in brackets, which allow the errors of any two block groups whose population-weighted centroids lie within 50 km of each other to be correlated, with equal weight at every distance up to 50 km and none beyond; the stars are from the clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Exposure to a violent protest, ring coefficients for the first five rings

	(1) 0–1 km	(2) 1–2 km	(3) 2–3 km	(4) 3–4 km	(5) 4–5 km
Any violent protest (0/1)	0.336**	0.271**	0.164**	0.102	0.098
	(0.135)	(0.121)	(0.078)	(0.083)	(0.074)
	[0.133]	[0.130]	[0.075]	[0.080]	[0.070]
Demographic controls	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644
R^2	0.084	0.084	0.084	0.084	0.084

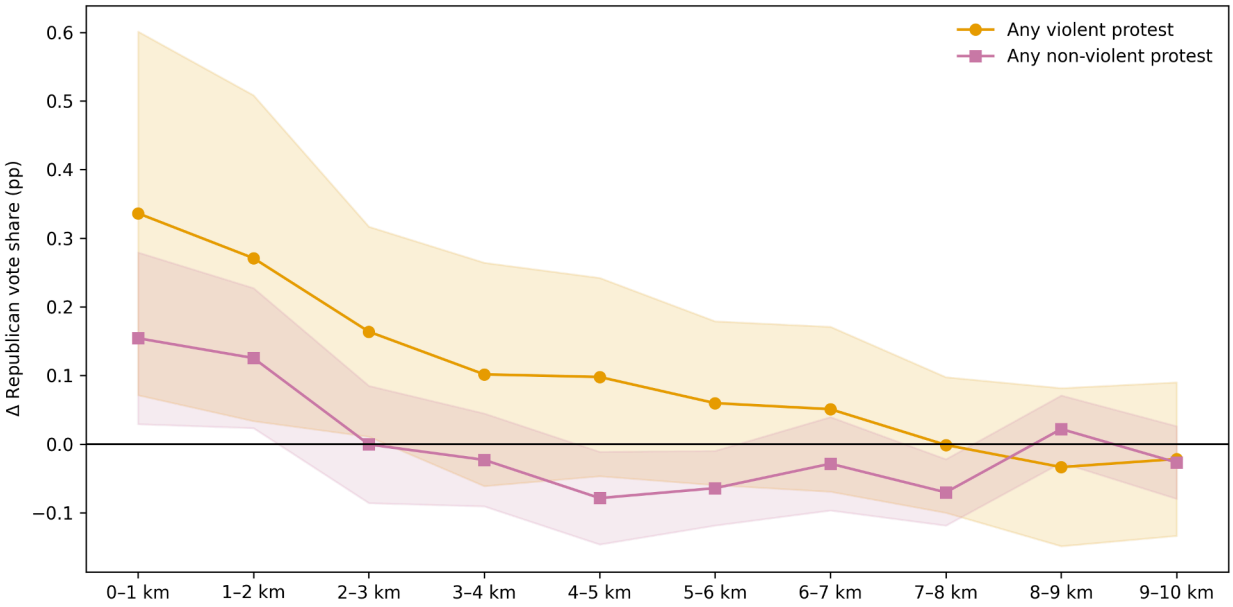
Notes: The ten ring indicators enter one regression jointly and columns (1) to (5) are the first five rings. As in Figure 2, with exposure to a violent protest as the treatment: Any violent protest (0/1) is one when at least one violent George Floyd protest had its venue in the ring. Observations is N from the fit. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. Conley standard errors (Conley, 1999) with the 50 km cutoff of Table A5 in brackets; the stars are from the clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The point estimates are larger than those for any protest in Table A5 at every ring shown, but the two are not statistically distinguishable: the difference between violent and all-protest exposure is not statistically established. Figure 3 plots all ten rings with the 2012–2016 placebo.

Table A7: Exposure to any and to a violent protest, earlier-protest controls split by political orientation

	Dep. variable: Δ Republican share 2016–2020 (pp)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Previous protest controls	Pooled	Left	Right	Neither	L + R	L + N	R + N	L + R + N
Panel A: Any protest (0/1)								
0–1 km	0.193***	0.195***	0.187***	0.154**	0.191***	0.162***	0.156**	0.162***
	(0.060)	(0.060)	(0.065)	(0.061)	(0.062)	(0.059)	(0.063)	(0.061)
1–2 km	0.166***	0.163***	0.140***	0.110**	0.161***	0.133***	0.114**	0.134***
	(0.050)	(0.050)	(0.054)	(0.046)	(0.051)	(0.044)	(0.048)	(0.046)
2–3 km	0.050	0.046	0.016	-0.006	0.045	0.024	-0.000	0.027
	(0.042)	(0.043)	(0.044)	(0.041)	(0.042)	(0.040)	(0.041)	(0.041)
3–4 km	0.040	0.032	0.005	-0.015	0.032	0.013	-0.009	0.015
	(0.033)	(0.034)	(0.034)	(0.031)	(0.034)	(0.032)	(0.032)	(0.032)
4–5 km	-0.043	-0.052	-0.069*	-0.093***	-0.050	-0.072**	-0.085**	-0.068**
	(0.036)	(0.036)	(0.037)	(0.033)	(0.036)	(0.034)	(0.035)	(0.034)
R^2	0.085	0.085	0.085	0.085	0.085	0.085	0.085	0.085
Panel B: Any violent protest (0/1)								
0–1 km	0.336**	0.330**	0.368***	0.355***	0.330**	0.320**	0.353***	0.321**
	(0.135)	(0.135)	(0.136)	(0.136)	(0.136)	(0.137)	(0.137)	(0.138)
1–2 km	0.271**	0.268**	0.318***	0.300**	0.271**	0.258**	0.301**	0.262**
	(0.121)	(0.121)	(0.120)	(0.118)	(0.123)	(0.121)	(0.120)	(0.123)
2–3 km	0.164**	0.171**	0.197**	0.181**	0.167**	0.155**	0.175**	0.153*
	(0.078)	(0.078)	(0.079)	(0.076)	(0.079)	(0.077)	(0.078)	(0.079)
3–4 km	0.102	0.107	0.119	0.101	0.104	0.091	0.097	0.090
	(0.083)	(0.086)	(0.086)	(0.085)	(0.085)	(0.085)	(0.084)	(0.085)
4–5 km	0.098	0.102	0.112	0.083	0.099	0.078	0.081	0.076
	(0.074)	(0.074)	(0.075)	(0.071)	(0.076)	(0.072)	(0.074)	(0.074)
R^2	0.084	0.084	0.084	0.084	0.084	0.085	0.085	0.085
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644	136,644	136,644	136,644

Notes: Each column of each panel is one regression with ten one-kilometer ring indicators entered jointly, of which the first five are shown; Panel A has exposure to any protest as the treatment and Panel B exposure to a violent protest. The columns differ only in the earlier-protest controls, the ring indicators for the protests of January 2017 to May 2020. Column (1) is the specification of Tables A5 and A6, with one indicator per ring for an earlier protest. Columns (2) to (8) replace them with the same indicators built separately for the earlier protests coded left-wing (L), right-wing (R) and neither (N) in the Crowd Counting Consortium's coding of each event's political orientation, entered in the combination named at the head of the column. The indicators are matched to the panel's treatment: in Panel B, in every column, they count violent earlier protests only. The comparison of interest is whether each coefficient of columns (2) to (8) lies inside the 95 percent confidence interval of the column (1) coefficient on the same ring. The specification is otherwise that of Figure 2; $N = 136{,}644$ in every column. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A7: Exposure to violent and to nonviolent protests, by distance ring



Notes: The figure plots two ring profiles side by side: circles are the coefficients on ten one-kilometer ring indicators for exposure to a violent protest, squares the coefficients for exposure to a nonviolent protest; shaded bands are 95% confidence intervals. The specification is that of Figure 2, with each profile one regression with the ten ring indicators of one family entered jointly. The nonviolent count in a ring is the number of protests minus the number of violent protests, and the indicator equals one when it is positive. The sample is the same for both profiles, $N = 136{,}644$; 344 clusters. The difference between the two profiles is not statistically established. The violent coefficients for the first five rings are in Table A6.

D.3 Which votes move, and the controls ladder

Tables A8 and A9 show the 0–1 km estimate on five outcomes as each class of control enters, for any protest and for violent protests (Sections 4.2 and 4.3).

Table A8: Exposure to any protest across five outcomes, as the controls enter

	Treatment: any protest within 0–1 km (ten-ring specification)					
	(1)	(2)	(3)	(4)	(5)	(6)
	No fixed effects	Baseline	+ Demographic	+ Distance to city center	+ Previous protest	+ Pre-trend control
Δ Republican vote share (pp)	0.312**	0.442***	0.173***	0.189***	0.193***	0.187***
	(0.157)	(0.090)	(0.064)	(0.063)	(0.060)	(0.062)
Δ Democratic vote share (pp)	-0.159	-0.217**	-0.072	-0.093	-0.129**	-0.125*
	(0.206)	(0.100)	(0.067)	(0.067)	(0.064)	(0.065)
Δ Republican votes (log)	0.014	0.011*	0.005	0.006	0.008*	0.007*
	(0.012)	(0.006)	(0.004)	(0.004)	(0.004)	(0.004)
Δ Democratic votes (log)	-0.024***	-0.021***	-0.008**	-0.008**	-0.007**	-0.007**
	(0.005)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)
Δ Turnout (pp)	-1.320***	-0.776***	-0.524***	-0.503***	-0.397***	-0.395***
	(0.179)	(0.196)	(0.158)	(0.157)	(0.153)	(0.153)
Demographic controls	No	No	Yes	Yes	Yes	Yes
Distance to city center	No	No	No	Yes	Yes	Yes
Previous protest controls	No	No	No	No	Yes	Yes
Pre-trend control	No	No	No	No	No	Yes
Zip code fixed effects	No	Yes	Yes	Yes	Yes	Yes
N (vote / log votes)	136,644	136,644	136,644	136,644	136,644	136,644
N (turnout)	124,971	124,971	124,971	124,971	124,971	124,971
R^2 (Republican vote share)	0.024	0.003	0.084	0.084	0.085	0.097

Notes: Each row is one outcome and each column adds one class of controls to the regression. As in Figure 2, with the treatment in every cell any protest (0/1) in the 0–1 km ring, the innermost of the ten ring indicators, the other nine rings entered jointly. Column (1) has no fixed effects and no controls; column (2) adds zip code fixed effects; column (3) adds the demographic controls; column (4) adds distance to city center; column (5) adds the previous protest controls and is the specification of Table A5; column (6) adds the 2012–2016 change in the same outcome as a control (the pre-trend control). The outcomes are changes between the 2016 and 2020 presidential elections: the Republican and Democratic shares of all votes cast, in percentage points; the natural logarithms of Republican and Democratic votes; and turnout, votes cast per voting-age resident, in percentage points. N = 136,644 block groups; the turnout rows use the block groups with a voting-age population estimate, N = 124,971. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10. Education × race interactions are centered at sample means from the mobilization estimation sample; level coefficients are therefore read at those means. Table A9 is the same ladder for violent protests.

Table A9: Exposure to a violent protest across five outcomes, as the controls enter

	Treatment: any violent protest within 0–1 km (ten-ring specification)					
	(1)	(2)	(3)	(4)	(5)	(6)
	No fixed effects	Baseline	+ Demographic	+ Distance to city center	+ Previous protest	+ Pre-trend control
Δ Republican vote share (pp)	0.156	0.591***	0.337**	0.372***	0.336**	0.339**
	(0.235)	(0.148)	(0.137)	(0.137)	(0.135)	(0.136)
Δ Democratic vote share (pp)	0.104	-0.282*	-0.179	-0.227	-0.198	-0.200
	(0.279)	(0.170)	(0.139)	(0.138)	(0.141)	(0.141)
Δ Republican votes (log)	-0.001	0.015	0.010	0.011	0.010	0.010
	(0.019)	(0.010)	(0.011)	(0.011)	(0.011)	(0.011)
Δ Democratic votes (log)	-0.018*	-0.030***	-0.021***	-0.021***	-0.019***	-0.019***
	(0.009)	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)
Δ Turnout (pp)	-1.542***	-1.394***	-1.112***	-1.072***	-0.834**	-0.836**
	(0.326)	(0.328)	(0.363)	(0.365)	(0.387)	(0.387)
Demographic controls	No	No	Yes	Yes	Yes	Yes
Distance to city center	No	No	No	Yes	Yes	Yes
Previous protest controls	No	No	No	No	Yes	Yes
Pre-trend control	No	No	No	No	No	Yes
Zip code fixed effects	No	Yes	Yes	Yes	Yes	Yes
N (vote / log votes)	136,644	136,644	136,644	136,644	136,644	136,644
N (turnout)	124,971	124,971	124,971	124,971	124,971	124,971
R^2 (Republican vote share)	0.020	0.001	0.083	0.084	0.084	0.096

Notes: The table repeats Table A8 with any violent protest (0/1) in the 0–1 km ring as the treatment, so that column (5) is the specification of Table A6. $N = 136{,}644$ block groups, and 124,971 for turnout. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Education × race interactions are centered at sample means from the mobilization estimation sample; level coefficients are therefore read at those means.

D.4 A second research design

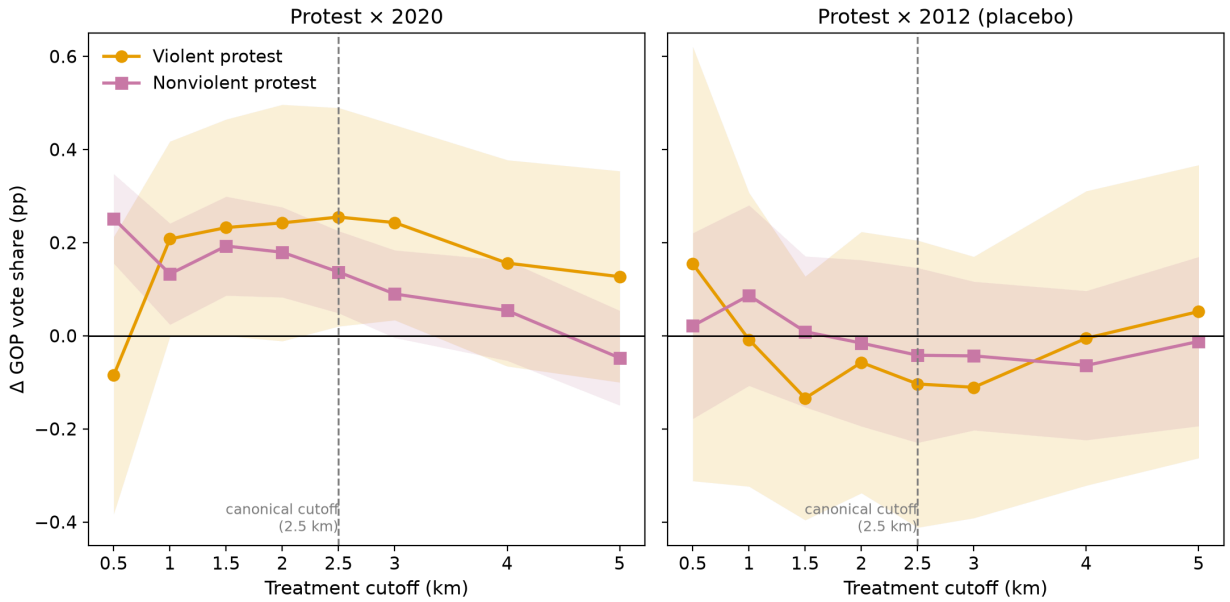
Table A10 reports the event-study coefficients of Figure 4, with two further columns in which the comparison group is restricted, and Figure A8 moves the 2.5 km cutoff (Section 4.4).

Table A10: Spatial difference-in-differences estimates of exposure within 2.5 km, violent and nonviolent

Dependent variable: Republican vote share (pp)	(1) Violent (full sample)	(2) Violent (≤10 km)	(3) Nonviolent (full sample)	(4) Nonviolent (≤10 km)
Treated / control (km)	0–2.5 / all others	0–2.5 / 2.5–10	0–2.5 / all others	0–2.5 / 2.5–10
Protest × 2020	0.255**	0.154*	0.137***	0.097**
	(0.119)	(0.087)	(0.045)	(0.040)
Protest × 2024	0.378***	0.233**	0.128**	0.072
	(0.106)	(0.091)	(0.063)	(0.062)
Protest × 2012 (placebo)	-0.104	-0.066	-0.042	-0.023
	(0.157)	(0.157)	(0.095)	(0.095)
Placebo p (2012)	0.509	0.672	0.661	0.807
Controls × year	Yes	Yes	Yes	Yes
Block group fixed effects	Yes	Yes	Yes	Yes
Zip code × year fixed effects	Yes	Yes	Yes	Yes
Block groups (treated)	10,267	10,267	41,162	41,162
Identifying zip codes	898	898	3373	3367
Observations (block group × year)	542,288	200,452	542,288	417,484

Notes: The table reports the event-study coefficients of Figure 4, which plots columns (1) and (3), and two further columns; the specification is that of Figure 4, the rows Protest × 2020, Protest × 2024 and Protest × 2012 being the exposure indicator interacted with the election-year indicators, relative to 2016. Treated block groups have at least one George Floyd protest of the column's family, violent in columns (1) and (2) and nonviolent in columns (3) and (4), within 2.5 km. In columns (1) and (3) the controls are all other block groups; in columns (2) and (4) the control group is restricted to block groups whose nearest protest of the family lies between 2.5 and 10 km. Block groups (treated) is the number of treated block groups; Identifying zip codes is the number of zip codes that contain both a treated and a control block group, the only ones that contribute to the estimate; Observations (block group × year) is the number of block group by year observations, four per block group in a balanced panel. Standard errors in parentheses, clustered by metropolitan area; 339, 142, 339 and 321 clusters in columns (1) to (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Placebo p (2012) is the p-value of the Protest × 2012 coefficient, the pre-period test. Controls here are standardized within the panel, so these interaction terms are not comparable in magnitude with the cross-sectional tables.

Figure A8: Spatial difference-in-differences estimates across treatment cutoffs, 2020 and the 2012 placebo

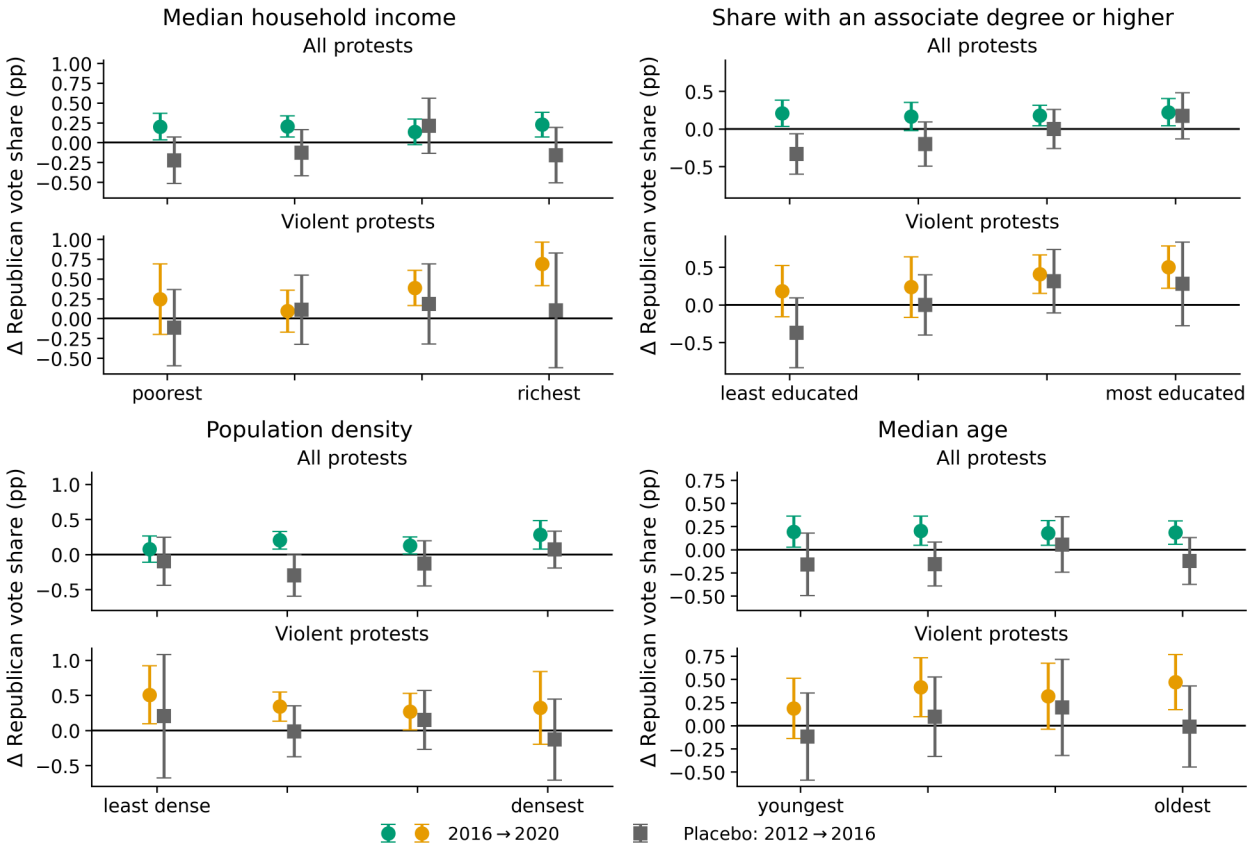


Notes: The figure re-estimates the specification of Figure 4 with the distance that defines exposure moved across eight cutoffs from 0.5 to 5 km; the dashed line marks the 2.5 km cutoff used in that figure. The left panel plots the Protest × 2020 coefficient and the right panel the Protest × 2012 placebo coefficient, at each cutoff, for exposure to a violent protest (circles) and to a nonviolent protest (squares); bands are 95% confidence intervals. At every cutoff the control group is all other block groups; the balanced panel has 542,288 block group by year observations; the number of treated block groups rises with the cutoff; 339 clusters. Treated block groups lie within the stated cutoff of a protest, and the controls are all other block groups. Controls here are standardized within the panel, so these interaction terms are not comparable in magnitude with the cross-sectional tables. Table A10 reports the 2.5 km estimates with the restricted control group.

D.5 The remaining heterogeneity dimensions

Figure A9 repeats Figure 5 for median household income, education, density and median age (Section 4.5).

Figure A9: Exposure by quartile of median household income, education, population density and median age



Notes: The figure repeats Figure 5 for four further characteristics, one per panel: median household income, the share of residents aged 25 and over with an associate degree or higher, population density and median age. In each panel block groups are split into four quartiles of the characteristic measured relative to the other block groups of the same zip code, income and density in logarithms, so that the leftmost group is block groups in the bottom quarter of their zip code and the rightmost those in the top quarter. Within each panel the upper plot is exposure to any protest and the lower plot exposure to a violent protest. Circles are the 0–1 km coefficient for each quartile, from one regression per characteristic and treatment in which the ten ring indicators are interacted with the four quartile indicators, with the specification of Figure 5; squares are the same coefficients for the 2012–2016 change (the placebo); whiskers are 95% confidence intervals. $N = 136{,}644$ in every fit; standard errors are clustered by metropolitan area; 344 clusters. Whether the four quartile coefficients differ is decided by a joint test of the quartile interactions, whose p-values are, for any protest and for violent protests: median household income 0.7648 and 0.0008; education 0.8824 and 0.1462; population density 0.2768 and 0.8187; median age 0.9804 and 0.2439. The joint test of the placebo interactions rejects for education under any protest ($p = 0.04$), so the pre-period change differs across education quartiles under any protest and that panel is read with this in mind.

D.6 Persistence to 2024

Table A11 re-estimates the ring profiles with the 2020–2024 and the cumulative 2016–2024 change as outcomes (Section 4.6).

Table A11: Ring coefficients over four outcome windows, nonviolent and violent exposure, first five rings

	Nonviolent protest				Violent protest			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2016–20	Persistence	2020–24, lagged	Cumulative	2016–20	Persistence	2020–24, lagged	Cumulative
Dep. variable:	$\Delta_{16–20}$	$\Delta_{20–24}$	$\Delta_{20–24}$	$\Delta_{16–24}$	$\Delta_{16–20}$	$\Delta_{20–24}$	$\Delta_{20–24}$	$\Delta_{16–24}$
0–1 km	0.176***	-0.051	-0.010	0.126	0.336**	0.182	0.257*	0.508***
	(0.062)	(0.071)	(0.065)	(0.079)	(0.135)	(0.163)	(0.149)	(0.150)
1–2 km	0.177***	-0.008	0.033	0.169**	0.271**	0.124	0.184	0.385***
	(0.052)	(0.062)	(0.055)	(0.067)	(0.121)	(0.133)	(0.116)	(0.113)
2–3 km	0.059	0.028	0.043	0.091	0.164**	0.154	0.190**	0.310***
	(0.044)	(0.045)	(0.045)	(0.059)	(0.078)	(0.094)	(0.091)	(0.099)
3–4 km	0.033	0.040	0.048	0.077*	0.102	0.052	0.075	0.152
	(0.036)	(0.035)	(0.034)	(0.042)	(0.083)	(0.083)	(0.081)	(0.105)
4–5 km	-0.031	-0.024	-0.031	-0.054	0.098	-0.028	-0.005	0.070
	(0.035)	(0.034)	(0.033)	(0.043)	(0.074)	(0.082)	(0.073)	(0.076)
$\Delta_{16–20}$ (lag)	–	–	-0.229***	–	–	–	-0.230***	–
			(0.046)				(0.046)	
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint F (inner-5) p	0.004	0.588	0.270	0.001	0.222	0.477	0.257	0.006
Observations	136,644	135,572	135,572	135,572	136,644	135,572	135,572	135,572
R^2	0.085	0.082	0.126	0.185	0.084	0.083	0.126	0.186

Notes: The table reports the ring profiles for nonviolent protests, columns (1) to (4), and violent protests, columns (5) to (8), over four outcome windows; the first five of ten rings are shown. 2016–20 is the 2016–2020 change, as in Figure A7 and Table A6; Persistence is the 2020–2024 change; the column labeled 2020–24, lagged is the 2020–2024 change with the block group's own 2016–2020 change added as a control, whose coefficient is the $\Delta_{16–20}$ (lag) row; Cumulative is the 2016–2024 change. Every column is the specification of Figure 2, with the outcome window changed and the ten ring indicators of the family entered jointly. Observations is N from the fit: the 2016–20 columns use the 136,644 block groups of Table A5; the three columns that use the 2024 returns use the 135,572 of them with a 2024 return. Standard errors in parentheses, clustered by metropolitan area; 344 clusters in the 2016–20 columns and 339 in the six columns that use the 2024 returns. *** p $<$ 0.01, ** p $<$ 0.05, * p $<$ 0.10. Joint F (inner-5) p is the p-value of a test that the five ring coefficients shown are all zero.

D.7 Robustness of the ring design

This subsection reports the three exercises of Section 4.7 in the same order.

Random relocation of the venues. In each of 1,000 draws the venues of a metropolitan area are moved to block groups of the same metropolitan area drawn at random, the rings are rebuilt and the specification is re-estimated, which gives the distribution of the 0–1 km coefficient when the venues are as many as observed but placed elsewhere (Table A12). Random placement does not replicate the headline: a 0–1 km coefficient as large as observed arises in 2 of 1,000 draws for any protest and 21 of 1,000 for violent protests, and the observed decay across rings is steeper than the decay random geometry itself produces. Venues placed at random also give a profile that falls with distance, and the observed profile falls faster than that.

Equal-area rings. The one-kilometer rings widen with distance, so an outer ring covers far more area, and more protests, than an inner one; with boundaries at $\sqrt{k}$ kilometers the 0–1 km ring is unchanged by construction and only the outer profile is carved differently (Table A13). The 0–1

km estimates stay positive and significant for violent and for nonviolent exposure, 0.38 and 0.20.

Single states and metropolitan areas. None of the 72 drops, each of the 42 states and each of the 30 largest metropolitan areas, moves the violent 0–1 km estimate outside the full-sample confidence interval (Figure A10).

Table A12: The 0–1 km estimates against random relocation of the venues

	(1) Any protest	(2) Violent protest
0–1 km coefficient against random venue relocation (1,000 draws per null; permutation $p = (r+1)/(B+1)$)		
Observed coefficient	0.1931	0.3363
Null mean, relocation within metro	0.0808	0.1080
Permutation p, 0–1 km ring	0.003	0.022
Permutation p, max over all 10 rings	0.003	0.022
Permutation p, population-weighted placement	0.005	0.032
Decay steepness: drop from 0–1 km to the far field (>5 km), in pp; same p convention		
Observed drop	0.2049	0.3254
Null mean	0.0847	0.0823
Permutation p	0.004	0.018
Companion p: slope over inner rings	0.001	0.028
Companion p: inner-vs-outer mean contrast	0.046	0.017
Share of random placements where the conventional t-test fires at 0–1 km ($\lvert t\rvert \geq 1.96$)		
Relocation within metro	48.1%	13.5%
Population-weighted placement	53.4%	11.7%
Placement on actual venue tracts (exposure preserved)	99.9%	66.7%
Observations (each regression)	136,644	136,644

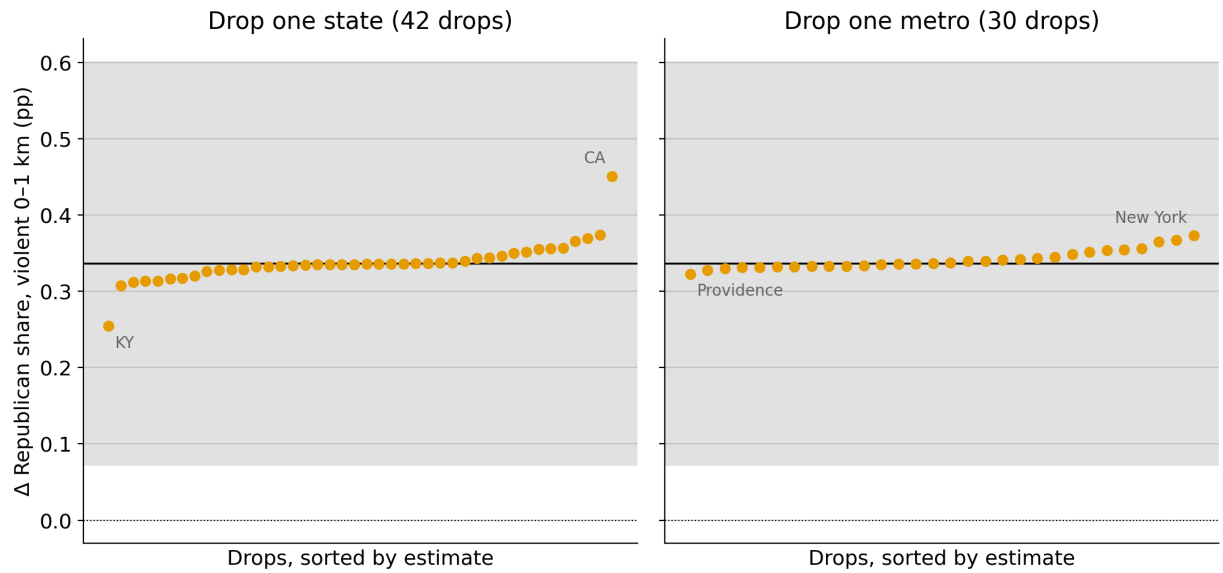
Notes: The table compares the observed 0–1 km coefficients of Tables A5 and A6 with the distribution of the same coefficient when the protest venues are relocated at random, for any protest in column (1) and violent protests in column (2). Each null is built from 1,000 draws: in every draw the venues of a metropolitan area are moved to block groups of the same metropolitan area drawn at random, the ten ring indicators are rebuilt, and the specification of Figure 2 is re-estimated on the same 136,644 block groups. Relocation within metro draws the receiving block groups uniformly; population-weighted placement draws them with probability proportional to population; placement on actual venue tracts restricts the draws to the census tracts that hosted a venue. The permutation p is $(r+1)/(B+1)$, where r is the number of draws whose statistic is at least as large as the observed one and $B = 1{,}000$; the max over all 10 rings row takes the largest coefficient across rings in each draw, so that it accounts for the ten rings tested. The decay steepness panel applies the same construction to the drop from the 0–1 km coefficient to the mean coefficient beyond 5 km; the two companion rows test the slope over the inner rings and the contrast between the inner and outer mean. The third panel is the share of draws on which a conventional t-test on the 0–1 km coefficient is significant at the 5% level. Random placement does not replicate the headline: a 0–1 km coefficient as large as observed arises in 2 of 1,000 draws for any protest and 21 of 1,000 for violent protests. The observed decay across rings is steeper than the decay random geometry itself produces. Permutation p-values are used because the conventional t-test is not a usable yardstick here: under random placement it reports significance on 48.1% of draws for any protest and 13.5% for violent protests, because the null profile is not centered at zero (the null mean at 0–1 km is positive in both columns).

Table A13: The 0–1 km estimates under equal-area ring boundaries

	(1) Violent protest	(2) Nonviolent protest
0–1 km	0.382*** (0.129)	0.198*** (0.063)
Demographic controls	Yes	Yes
Distance to city center	Yes	Yes
Previous protest controls	Yes	Yes
Zip code fixed effects	Yes	Yes
Observations	136,644	136,644

Notes: The table re-estimates the 0–1 km coefficient with the distance rings redrawn so that every ring covers the same area as the innermost one: boundaries at $\sqrt{k}$ km for $k = 1, \ldots, 25$, twenty-five equal-area rings out to 5 km and one band from 5 to 10 km, all entered jointly. Column (1) is exposure to a violent protest and column (2) exposure to a nonviolent protest. The specification and sample are otherwise those of Figure 2; Observations is N from the fit. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** p $<$ 0.01, ** p $<$ 0.05, * p $<$ 0.10. Re-drawing boundaries so every ring covers equal area (boundaries at $\sqrt{k}$ km; the 0–1 km ring unchanged by construction) is a robustness column beside the canonical rings: the profile is robust to dose-comparable ring boundaries. The innermost cell is identical in both carvings, so the difference between the coefficient here and the 0–1 km coefficient of Table A6 measures only the sensitivity of that estimate to how the outer profile is carved; neither carving is a truer estimate.

Figure A10: The violent 0–1 km estimate with one state or metropolitan area dropped at a time



Notes: The figure re-estimates the violent 0–1 km coefficient of Table A6 with one unit of the sample removed at a time. In the left panel each dot is the estimate after dropping one state, for each of the 42 states in the sample; in the right panel each dot is the estimate after dropping one of the 30 largest metropolitan areas. Dots are sorted by size; the horizontal line is the full-sample estimate and the gray band its 95% confidence interval; the two drops that move the estimate furthest in each direction are named; the dotted line is zero. Each dot is the specification of Figure 2, with exposure to a violent protest as the treatment, re-estimated on the remaining block groups; the full sample is N = 136,644 with 344 clusters.

E Protest participation

This appendix gives the participation estimates of Section 5 in full: the coefficients behind the figures of that section, the controls ladder, other outcomes, the remaining heterogeneity dimensions, persistence, the robustness exercises and the single-state exercise.

E.1 The headline estimates and the two splits

Tables A14 and A17 print the coefficients of Figures 6 and 7. Table A15 splits the placebo of column (3) of Table A14 by the political orientation of the earlier protests, and Table A16 adds earlier attendance as a control (Section 5.2). The 2,285 block groups that are the only sample block group of their zip code are absorbed entirely by their fixed effect and contribute nothing to the estimates.

Table A14: Participation and the change in the Democratic vote share, coefficients and placebo tests

	(1) Δ Dem share 2016–2020	(2) Placebo: Δ Dem share 2012–2016	(3) Placebo: pre-protest attendance, same venues
Any participants (extensive margin)	0.0730***	0.0352	0.0234
	(0.0178)	(0.0353)	(0.0187)
Participants, log (intensive margin)	0.0141	0.0400	-0.0147
	(0.0138)	(0.0322)	(0.0171)
Wald p (extensive margin)		0.3258	0.0268
Wald p (intensive margin)		0.4811	0.1940
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2	0.1089	0.0700	0.1094
Observations	135,691	135,691	132,482

Notes: Each column reports an extensive margin, whether the block group sent anyone to a protest, and an intensive margin, how much it sent. The regression is of the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections, in percentage points, on the two terms, with the venue-traffic controls, the demographic controls, the neighborhood mobility controls, indicators for deciles of the block group's panel devices, and zip code fixed effects, so that every comparison is between block groups in the same zip code; the regression is weighted by votes cast in 2020. A resident participates when a device homed in the block group is detected within 200 m of a protest venue during a George Floyd protest; regulars at those venues are counted, and the venue-traffic controls hold fixed their presence at the same venues in the three weeks before the first event. The intensive margin is the log of attendances per panel device, zero where nobody attended. The demographic controls are listed in the note to Figure 2; the mobility controls are four measures of how far residents' devices ranged on an ordinary day before the pandemic. The sample is the block groups of metropolitan areas with a protest venue that have vote returns in 2012, 2016 and 2020 and a device panel; $N = 135{,}691$ in columns (1) and (2), 85,399 of which sent at least one resident, and 132,482 in column (3). Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (2) runs the specification on the 2012–2016 change and column (3) replaces the treatment with attendance at earlier protests at the same venues; the Wald rows test each against the column (1) estimate rather than against zero. Column (3)'s test uses the treatment's centering and the column (1) specification re-estimated on the 132,482 block groups of column (3); under the placebo's own centering its p-value is 0.0472.

Table A15: Earlier protest attendance and the change in the Democratic vote share, by political orientation

	Dep. variable: Δ Dem share 2016–2020 (pp)		
	(1)	(2)	(3)
	Pooled	Left-coded	Right-coded
	Pre-protest attendance, same venues, 2019–May 2020		
Any participants (extensive margin)	0.0234	0.0243	-0.0557*
	(0.0187)	(0.0180)	(0.0306)
Participants, log (intensive margin)	-0.0147	-0.0603***	-0.1164***
	(0.0171)	(0.0206)	(0.0445)
Block groups with any attendance (%)	46.03	33.46	6.74
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2	0.1094	0.1095	0.1095
Observations	132,482	132,482	132,482

Notes: The table repeats the placebo of column (3) of Table A14, split by the political orientation of the earlier protests. In every column the participation treatment is replaced by the residents' pre-protest attendance: their attendance, measured in the same way, at the protests of 2019 and of 1 January to 26 May 2020 held at venues that later hosted a George Floyd protest, entered as an extensive and an intensive margin: at all of those protests in column (1), which is column (3) of Table A14; only at those coded left-wing in column (2); and only at those coded right-wing in column (3), in the Crowd Counting Consortium's coding of each event's political orientation. Block groups with any attendance is the percentage of the sample with at least one such attendance. The specification is otherwise that of Table A14, and the sample is that of its column (3), $N = 132{,}482$. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A16: Participation and the change in the Democratic vote share, with earlier protest attendance as a control

	Dep. variable: Δ Dem share 2016–2020 (pp)				
	(1)	(2)	(3)	(4)	(5)
Any participants (extensive margin)	0.0730***	0.0740***	0.0748***	0.0747***	0.0736***
	(0.0178)	(0.0182)	(0.0182)	(0.0183)	(0.0182)
Participants, log (intensive margin)	0.0141	0.0157	0.0142	0.0163	0.0140
	(0.0138)	(0.0138)	(0.0139)	(0.0139)	(0.0139)
Pre-wave attendance, left-coded	No	Yes	No	Yes	No
Pre-wave attendance, right-coded	No	No	Yes	Yes	No
Pre-wave attendance, any	No	No	No	No	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.1097	0.1096	0.1098	0.1095
Observations	135,691	132,482	132,482	132,482	132,482

Notes: Column (1) is column (1) of Table A14. Columns (2) to (5) add to that specification the residents' attendance at earlier protests (pre-wave attendance in the table), the attendance of the placebo in column (3) of Table A14, entered in the same two parts as the treatment, an extensive and an intensive margin: attendance at the protests coded left-wing in column (2), at those coded right-wing in column (3), at both, as separate terms, in column (4), and at all earlier protests in column (5), in the Crowd Counting Consortium's coding of each event's political orientation. The coefficients on the earlier attendance are not shown. Attendance at the earlier protests coded neither left- nor right-wing is not available on its own: the attendance data give only all earlier protests and the left- and right-coded ones, and because a venue can host protests of both orientations it cannot be recovered by subtraction. Columns (2) to (5) are estimated on the 132,482 block groups whose earlier attendance is observed, the sample of column (3) of Table A14; in the other 3,209 part of that count was withheld before it left the secure server (Table A1). The comparison of interest is whether each coefficient of columns (2) to (5) lies inside the 95 percent confidence interval of the column (1) coefficient. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: Participation split by protest violence and by whether the attender was a regular, coefficients and Wald tests

	By violence		By habituality	
	Violent	Nonviolent	Habitual	Non-habitual
Attendance (extensive margin)	−0.0190 (0.0307)	0.0723*** (0.0178)	−0.0721** (0.0332)	0.0781*** (0.0173)
Attendance (intensive margin)	−0.0903*** (0.0290)	0.0114 (0.0145)	−0.1326*** (0.0274)	0.0250* (0.0141)
Wald p, extensive margin	0.022		<0.0001	
Wald p, intensive margin	0.003		<0.0001	
Observations	135,689		135,691	

Notes: Each pair of columns is one split of the participation treatment, estimated in one regression with all four resulting terms entered jointly: the first pair splits attendance by whether the protest attended turned violent, the second by whether the attender was already a regular at that venue. The first row is each half's extensive margin and the second its intensive margin; each half enters as its own extensive margin and its own intensive margin, so a block group whose residents attended both kinds of venue-day carries both pairs of terms, and the halves are not mutually exclusive categories. The Wald rows test whether the two halves of a split load equally; they are not nested-model tests. The specification is otherwise that of Table A14. $N = 135{,}689$ for the violence split and 135,691 for the habitual split; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment is split and the venue-traffic controls are not.

E.2 The controls ladder

Table A18 shows the estimate as each class of control enters (Section 5.2).

Table A18: Participation and the change in the Democratic vote share, as the controls enter

	(1) Treatment only	(2) + Zip-code fixed effects	(3) + Demographics except race	(4) + Race	(5) + Device-data	(6) + Venue presence
Any participants (extensive margin)	1.1584***	0.1946***	0.1494***	0.1105***	0.0753***	0.0730***
	(0.2112)	(0.0259)	(0.0209)	(0.0183)	(0.0188)	(0.0178)
Participants, log (intensive margin)	-0.2939	0.0176	0.0024	-0.0058	0.0154	0.0141
	(0.2344)	(0.0182)	(0.0152)	(0.0142)	(0.0146)	(0.0138)
Venue-traffic controls	No	No	No	No	No	Yes
Neighborhood mobility controls	No	No	No	No	Yes	Yes
Device-panel size deciles	No	No	No	No	Yes	Yes
Race controls	No	No	No	Yes	Yes	Yes
Demographic controls (except race)	No	No	Yes	Yes	Yes	Yes
Zip code fixed effects	No	Yes	Yes	Yes	Yes	Yes
R^2	0.0163	0.0009	0.0546	0.1072	0.1089	0.1089
Observations	135,691	135,691	135,691	135,691	135,691	135,691

Notes: Each column adds one class of control to the previous one and is otherwise as in Table A14. Column (1) is the treatment alone. Column (2) adds zip code fixed effects. Column (3) adds the demographic controls other than race: median age, median household income, unemployment rate, education, renter share, log population, log density, and distance to city center. Column (4) adds race, which here means the five racial-composition shares of the block group and the four products of education with those shares; the products are part of what race means in this specification and the table carries no separate row or header for them. Column (5) adds the device-panel controls: the mobility controls and the panel-size indicators. Column (6) adds the venue-traffic controls and is the specification of Table A14. The sample is held at $N = 135{,}691$ in every column, so movement across columns is conditioning and never a change of sample; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Education $\times$ race interactions are centered at sample means from the mobilization estimation sample; level coefficients are therefore read at those means. Comparing within zip codes removes most of the raw association, and the extensive margin then falls to about two-fifths of its within-zip-code size as the remaining controls enter.

E.3 Other outcomes

Table A19 estimates the specification on four further outcomes (Section 5.2).

Table A19: Participation across five outcomes: vote shares, turnout and party votes per voting-age person

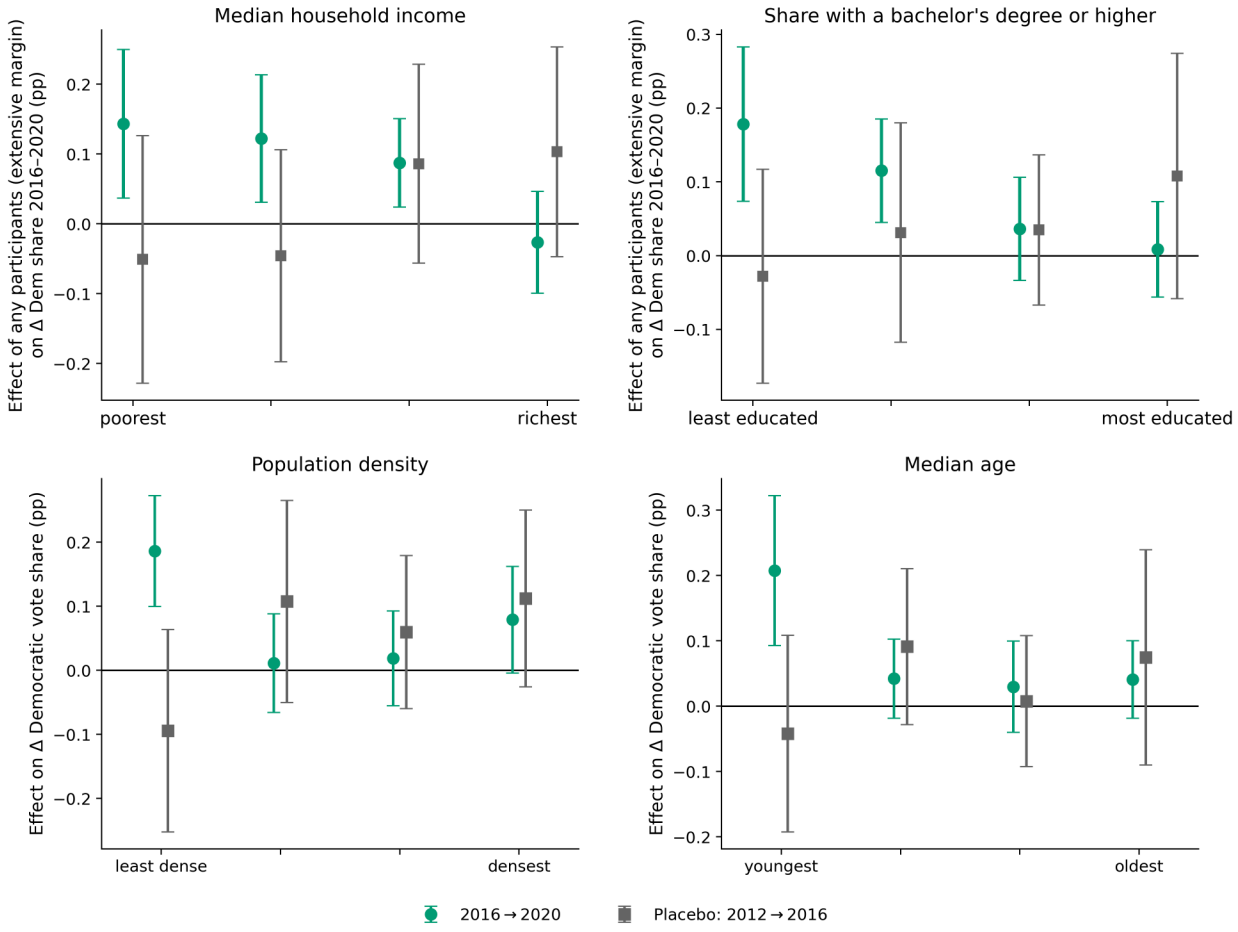
	(1) Δ Democratic share	(2) Δ Republican share	(3) Δ Turnout	(4) Δ Democratic votes per voting-age person	(5) Δ Republican votes per voting-age person
Any participants (extensive margin)	0.0730***	-0.0664***	0.0068	0.0353	-0.0374
	(0.0178)	(0.0172)	(0.1731)	(0.1019)	(0.0808)
Participants, log (intensive margin)	0.0141	-0.0114	-0.0262	-0.0792	0.0456
	(0.0138)	(0.0132)	(0.1423)	(0.0808)	(0.0653)
Placebo p (extensive / intensive margin)	0.32 / 0.21	0.17 / 0.48	0.89 / 0.22	0.57 / 0.87	0.74 / 0.04
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.0896	0.0136	0.0169	0.0118
Observations	135,689	135,689	135,689	135,689	135,689

Notes: Each column is the same specification with a different outcome. Columns (1) and (2) are vote shares: the change between 2016 and 2020 in the Democratic and in the Republican share of all votes cast for president, in percentage points; the denominator is all votes cast, so the two are not exact mirrors and the gap between them is third-party and blank ballots. Columns (3) to (5) are per-capita outcomes: the change in turnout, in Democratic votes and in Republican votes, each per voting-age person, in percentage points of the voting-age population. The placebo row gives, for each column, the p-value of the same specification run on the 2012–2016 change, for the extensive margin and for the intensive margin in that order. As in Table A14, except that the sample is one constant set of block groups across all five columns, $N = 135{,}689$; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** p $<$ 0.01, ** p $<$ 0.05, * p $<$ 0.10. Neither per-capita extensive-margin estimate is distinguishable from zero, and that null is too imprecise to separate a change in the composition of the vote from a change in its volume. The shift in shares is not evidence that particular votes moved from one party to the other: it is equally consistent with individuals switching, with a change in who turned out, and with differential migration.

E.4 The remaining heterogeneity dimensions

Figure A11 repeats Figure 8 for median household income, education, density and median age (Section 5.4), and Table A20 reports the specification with the participation terms left out, the regression behind the race check of that section.

Figure A11: Participation by quartile of median household income, education, population density and median age



Notes: The figure repeats Figure 8 for four further characteristics, one per panel: median household income, the share of residents with a bachelor's degree or higher, population density in residents per square kilometer, and median age. In each panel block groups are sorted into four quartiles of the characteristic, ranked across the whole estimation sample, and the specification of Table A14 is estimated with the extensive margin interacted with the quartile. The education share is the bachelor's-degree share that also enters the specification's education-by-race terms, not the associate-degree share among its demographic controls. Circles are the quartile extensive-margin estimates; squares are the same estimates when the 2012–2016 change is the outcome, the placebo; whiskers are 95 percent confidence intervals; the vertical axis is in percentage points of the Democratic share of all votes cast. $N = 135{,}691$ for income and education and 135,689 for density and age; standard errors are clustered by metropolitan area, 333 clusters. Whether the four quartile estimates differ is decided by a joint test of the quartile interactions: median household income $\chi^2(3) = 7.50$, $p = 0.057$; education $\chi^2(3) = 7.78$, $p = 0.051$; population density $\chi^2(3) = 12.86$, $p = 0.005$; median age $\chi^2(3) = 7.20$, $p = 0.066$.

Table A20: The change in the Democratic vote share on the controls alone

Dep. variable: Δ Democratic vote share 2016–2020 (pp)	(1)
Black, non-Hispanic (share)	-3.5958***
	(0.3403)
Hispanic (share)	-2.8401***
	(0.4036)
Asian, non-Hispanic (share)	-4.8457***
	(0.7023)
White, non-Hispanic (share)	0.0385
	(0.4320)
White alone (share)	0.8633**
	(0.3532)
Median age (years)	-0.0086***
	(0.0023)
Median household income ($10,000)	0.0467***
	(0.0069)
Unemployment rate (share)	-1.3234***
	(0.2016)
Associate degree or higher (share)	1.8742***
	(0.2410)
Renter-occupied (share)	0.1354
	(0.2000)
log population	-0.3930***
	(0.0478)
log(1 + density)	0.1479***
	(0.0286)
Bachelor's degree or higher (%)	0.0105***
	(0.0031)
× Black share (centered)	0.0733***
	(0.0191)
× Hispanic share (centered)	0.1372***
	(0.0165)
× Asian share (centered)	0.0891***
	(0.0277)
× White share (centered)	0.0235
	(0.0143)
Distance to city center (km)	-0.0270***
	(0.0045)
Participation (extensive and intensive margins)	No
Ordinary-presence control (two-part)	Yes
Neighborhood mobility controls	Yes
Device-panel size deciles	Yes
Zip code fixed effects	Yes
Within R^2	0.1088
Observations	135,691

Notes: The table reports the specification of Table A14 with the two participation terms left out, so that it shows how the controls alone relate to the outcome; the demographic controls are listed row by row. The racial shares are shares of the block group's population; the share with a bachelor's degree or higher enters on its own and interacted with the four racial shares, the four products centered at their sample means, so the coefficient on a racial share is its partial slope at the mean level of education. The sample is that of Table A14, $N = 135{,}691$. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

E.5 Persistence to 2024

Table A21 gives the participation estimate over the 2016–2020, 2020–2024 and 2016–2024 windows (Section 5.5).

Table A21: Participation and the change in the Democratic vote share, 2016–2020, 2020–2024 and 2016–2024, coefficients

Δ Dem share	(1) 2016–2020	(2) 2020–2024	(3) 2016–2024 (cumulative)
Extensive margin (any participants)	0.0732***	0.0490***	0.1222***
	(0.0179)	(0.0184)	(0.0275)
Intensive margin (log participants)	0.0139	-0.0070	0.0069
	(0.0139)	(0.0165)	(0.0200)
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2	0.1097	0.0795	0.1947
Observations	134,570	134,570	134,570

Notes: Each column is the same specification with a different outcome window: the 2016–2020 change, the 2020–2024 change, and the cumulative 2016–2024 change. As in Table A14, except that the sample is the block groups that have a vote return in 2024 as well, so $N = 134{,}570$ in all three columns; 328 clusters, five fewer than elsewhere in the arm because five metropolitan areas have no block group with a 2024 return. Standard errors in parentheses, clustered by metropolitan area. *** p < 0.01, ** p < 0.05, * p < 0.10.

E.6 Robustness of the participation measure

This subsection reports four of the five exercises of Section 5.6; the single-state exercise is in Section E.7.

Controls for ordinary presence. Nine measurements of ordinary visiting at the same venues, under conditions in which no protest was happening, can enter in any combination, 512 in all, and each combination was estimated with the 2016–2020 outcome and with the placebo (Table A22). The extensive margin is positive and significant at the 1 percent level in every combination, between 0.06 and 0.10, and the placebo's p-value is below 0.05 in none of them; the intensive margin is positive in every combination and reaches the 5 percent level in about three-quarters of them.

Detection radius. The extensive margin is positive and significant at all four radii of Table A23. It grows with the radius, but a wider radius also admits more ordinary passers-by, so the growth is not read as a larger effect.

Neighboring block groups. Terms for the neighboring block groups enter in steps (Table A24): the share of neighbors that sent anyone and their mean participation, then their ordinary presence at the venues, then their everyday movement, up to a column in which every own control has a neighbor counterpart. The block group's own extensive margin stays positive and significant at the 1 percent level in every column. The neighbor-share term is positive and significant in columns (2) to (4), but its placebo is not clean in columns (2) and (3), and in the symmetric column the term is not distinguishable from zero, so no reading of spillovers rests on that table.

Denominators. The extensive margin is positive and significant at the 1 percent level under all five denominators, between 0.07 and 0.09; the intensive margin is positive under all five and significant under four, and is a null only under the denominator the paper uses, the panel devices homed in the block group (Table A25). The reading that heavier participation adds nothing detectable is therefore specific to that rate and is not a property of the participant count itself. One of the ten placebo p-values in that table, the intensive margin's under land area, is 0.01 and not clean.

Table A22: Participation across the 512 combinations of the venue-traffic controls

	Any participants (extensive margin)	Participants, log (intensive margin)
All 512 subsets of the nine baseline-visit controls (1,024 regressions)		
Coefficient positive	512 of 512	512 of 512
Range across subsets	0.0556 to 0.0975	0.0004 to 0.0738
Significant at 5%	512 of 512	377 of 512
Significant at 1%	512 of 512	293 of 512
Placebo $p < 0.05$ (Δ 2012–2016)	0 of 512	0 of 512
Subsets with a singular fit	0 of 512	0 of 512
Three reference cells inside the census		
No baseline-visit controls	0.0753	0.0154
Pooled baseline-visit control alone	0.0743	0.0179
All nine controls together	0.0790	0.0673
Neighborhood mobility controls	Yes	Yes
Device-panel size deciles	Yes	Yes
Demographic controls	Yes	Yes
Zip code fixed effects	Yes	Yes
Observations (each regression)	132,482	132,482

Notes: The nine venue-traffic controls of the main specification can be included or excluded independently, giving 512 combinations; each was estimated with the 2016–2020 outcome and again with the 2012–2016 placebo outcome, so the table summarizes 1,024 regressions. The upper block reports, across those 512 fits, how many gave a positive coefficient, the range of the coefficients, how many reached significance, how many had a placebo below 0.05, and how many were singular; the left column is the extensive margin and the right the intensive margin. The lower block gives three named cells inside the census: the fit with no venue-traffic control at all, the fit with the single pooled control that the main specification uses, and the fit with all nine entered together. Every fit is otherwise the specification of Table A14. The nine are measurements of ordinary visiting at the same venues under conditions in which no protest was happening, each built with the treatment's own denominator and functional form so that it is the same measurement under a different condition: visits by the venue's own regulars on the day it hosted; visits to the same venue that morning, from 08:00 to 12:00; visits to it overnight, from 02:00 to 06:00; visits to the protest venues on days they hosted nothing, over the protest window; and the same background traffic in five further windows — before the pandemic, during the lockdown, in the weeks just before the protests began, and in the matching protest and pre-protest windows of the previous year. The sample is held fixed at $N = 132{,}482$ in every one of the 1,024 regressions, below the 135,691 of Table A14 because the widest combination needs a window that some block groups lack; 333 clusters. Significance is at the levels marked, on standard errors clustered by metropolitan area.

Table A23: Participation by detection radius, 50 to 400 m

	(1) Extensive margin	(2) Intensive margin	(3) Placebo p (ext.)	(4) Placebo p (int.)
50 m	0.0503***	-0.1585***	0.6832	0.8153
	(0.0188)	(0.0277)		
100 m	0.0624***	-0.0601***	0.3441	0.8396
	(0.0199)	(0.0227)		
200 m	0.0730***	0.0141	0.3206	0.2150
	(0.0178)	(0.0138)		
400 m	0.1044***	0.0226	0.6363	0.1470
	(0.0302)	(0.0144)		
Venue-traffic controls	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	135,691	135,691	135,691	135,691

Notes: Each row re-estimates the specification of Table A14 with the detection radius changed, and reports both treatment terms and both placebo p-values from that row's own fit; 200 m is the radius used everywhere else in the paper and is set in bold. Columns (1) and (2) are the extensive margin and the intensive margin with standard errors in parentheses beneath; columns (3) and (4) are p-values from the same specification run on the 2012–2016 change, one for each term, given as their own columns rather than in parentheses. The sample is $N = 135{,}691$ at every radius; standard errors are clustered by metropolitan area, 333 clusters. *** p $<$ 0.01, ** p $<$ 0.05, * p $<$ 0.10.

Table A24: Participation as terms for neighboring block groups enter

	(1) Own only	(2) + neighbor participation	(3) + neighbor traffic controls	(4) + neighbor mobility controls	(5) full symmetric mirror
Own: any participants (extensive margin)	0.0730***	0.0867***	0.0847***	0.0633***	0.0602***
	(0.0178)	(0.0183)	(0.0182)	(0.0181)	(0.0183)
Own: participants, log (intensive margin)	0.0141	0.0106	0.0112	-0.0012	0.0011
	(0.0138)	(0.0142)	(0.0142)	(0.0138)	(0.0140)
Neighbor share with any participants	—	0.2454***	0.2146***	0.1194**	0.0472
		(0.0565)	(0.0547)	(0.0543)	(0.0571)
placebo p (Δ 2012–16)	—	0.0001	0.0008	0.3156	0.2866
Neighbor mean log participants	—	-0.1388***	-0.1340***	-0.2411***	-0.1808***
		(0.0512)	(0.0515)	(0.0462)	(0.0470)
placebo p	—	0.3061	0.7520	0.2659	0.6412
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.1094	0.1095	0.1273	0.1285
Observations	135,691	135,680	135,680	135,673	135,673

Notes: Each column adds one more thing about the surrounding neighborhood to the specification of Table A14, and reports the block group's own extensive and intensive margins above the neighborhood terms. Column (1) is that specification unchanged. Column (2) adds two neighbor terms: the share of neighboring block groups that sent anyone, and the mean log participation among those that did. Column (3) adds the neighbors' venue traffic, column (4) the neighbors' everyday movement measures, and column (5) enters every own-block-group control in a matching form for the neighbors. Each neighbor term carries its own placebo row, the p-value of the same specification run on the 2012–2016 change, printed as a row and not in parentheses. Standard errors in parentheses, clustered by metropolitan area; the sample falls slightly as neighbor terms enter, from $N = 135{,}691$ in column (1) to 135,680 in columns (2) and (3) and 135,673 in columns (4) and (5), because a block group with no neighbor in the sample has no neighbor term; 333 clusters throughout. *** p $<$ 0.01, ** p $<$ 0.05, * p $<$ 0.10. Each column is a different comparison group, holding progressively more of the neighborhood fixed; the five are estimands, never merged or read as a trend, and column (1) remains the paper's estimate. The symmetric column is preferred on the ground that it treats the two sides of the comparison the same way, not on the ground that its placebo is cleaner.

Table A25: Participation under five denominators of the participation rate

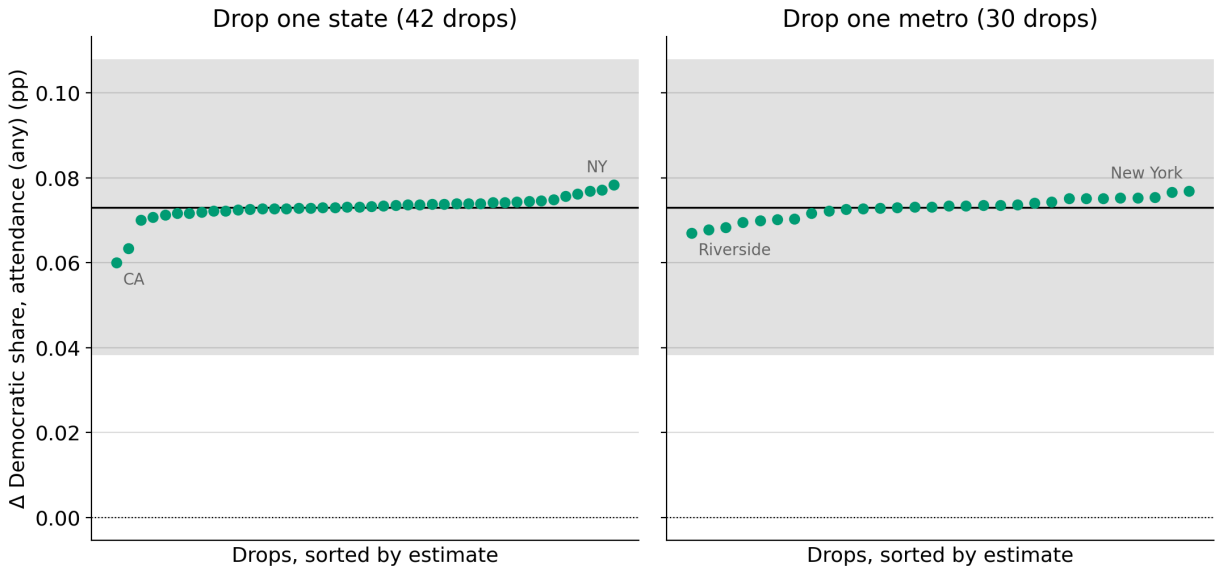
	(1) Extensive margin	(2) Intensive margin	(3) Placebo p (ext./int.)
Panel devices resident in the block group	0.0742*** (0.0178)	0.0151 (0.0138)	0.27 / 0.18
Resident population	0.0865*** (0.0193)	0.0844*** (0.0169)	0.23 / 0.11
Voting-age population	0.0848*** (0.0187)	0.0431*** (0.0134)	0.24 / 0.29
Land area	0.0683*** (0.0177)	0.0264** (0.0115)	0.17 / 0.01
None (raw participant count)	0.0921*** (0.0193)	0.1001*** (0.0176)	0.18 / 0.06
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
Observations	135,691	135,691	135,691

Notes: The intensive margin of participation is a log rate, and a rate needs a denominator. Each row re-estimates the specification of Table A14 with a different denominator for that rate, and reports the extensive margin, the intensive margin, and the two placebo p-values from its own fit. The first row is the denominator used everywhere else in the paper, the number of panel devices resident in the block group; the others divide the same participant count by the resident population, by the voting-age population, by the block group's land area, and by nothing at all, which makes the intensive margin a log count. The extensive margin does not depend on the denominator, since it is whether the block group sent anyone. Coefficients are given with standard errors in parentheses on the same line; the third column gives the p-values of the same specification run on the 2012–2016 change, extensive margin first and intensive margin second, as a column and not in parentheses. The one difference from Table A14 is that the neighborhood mobility controls enter as their two measures, not split by weekday and weekend into the four of that table, which is why the first row differs slightly from its column (1). The sample is $N = 135{,}691$ in every row; 333 clusters. Standard errors clustered by metropolitan area. *** p < 0.01, ** p < 0.05, * p < 0.10.

E.7 Sensitivity to single states and metropolitan areas

Each of the 42 states and each of the 30 largest metropolitan areas is dropped in turn (Figure A12): every one of the 72 estimates lies inside the full-sample confidence interval and none crosses from significant to not significant.

Figure A12: The extensive margin of participation with one state or metropolitan area dropped at a time



Notes: Each panel re-estimates the specification of Table A14 with one geography dropped from the sample, and plots the resulting extensive-margin estimate. The left panel drops one state at a time, forty-two of them; the right drops one metropolitan area at a time, the thirty largest. Markers are sorted by the size of the estimate, so the horizontal axis carries no meaning beyond that ordering; the two extreme drops in each panel are labeled. The solid horizontal line is the full-sample estimate and the shaded band is its 95 percent confidence interval. The full sample is $N = 135{,}691$, with 333 metropolitan-area clusters; a drop removes both the block groups and the clusters of that geography.

F Exposure and participation together

This appendix reports the joint regressions and the aggregate of Section 6 in full.

F.1 The joint regressions in full

Tables A26 and A27 print the joint regressions behind Figure 9, pooled and split by protest violence, every exposure ring with its joint test (Section 6.1).

Table A26: Participation and exposure in one regression, across four outcomes

	(1) Δ Dem (pp)	(2) Δ Rep (pp)	(3) Net margin (Dem − Rep, pp)	(4) Δ Turnout (pp)
Participation (device attendance, two-part)				
Extensive margin	0.075*** (0.018)	-0.072*** (0.017)	0.146*** (0.034)	0.031 (0.173)
Intensive margin (log rate)	0.018 (0.013)	-0.019 (0.013)	0.037 (0.026)	-0.004 (0.145)
Exposure: any protest, 1 km rings				
0–1 km	-0.151** (0.065)	0.209*** (0.061)	-0.360*** (0.124)	-0.467 (0.344)
1–2 km	-0.097* (0.051)	0.161*** (0.051)	-0.258** (0.101)	-0.485* (0.284)
2–3 km	-0.033 (0.042)	0.039 (0.042)	-0.072 (0.083)	-0.226 (0.258)
3–4 km	-0.015 (0.039)	0.020 (0.036)	-0.034 (0.074)	-0.184 (0.209)
4–5 km	0.042 (0.034)	-0.058* (0.033)	0.099 (0.067)	-0.211 (0.217)
5–6 km	0.002 (0.028)	-0.024 (0.026)	0.026 (0.052)	0.034 (0.232)
6–7 km	0.003 (0.038)	-0.002 (0.037)	0.005 (0.073)	-0.161 (0.233)
7–8 km	0.043 (0.029)	-0.047* (0.026)	0.090* (0.054)	-0.120 (0.154)
8–9 km	-0.014 (0.033)	0.017 (0.032)	-0.031 (0.064)	-0.195 (0.164)
9–10 km	0.020 (0.026)	-0.017 (0.026)	0.037 (0.052)	-0.024 (0.181)
Joint test over the ring family				
0–2 km, F	2.90	7.02	4.76	1.95
p	0.0564	0.0010	0.0092	0.1446
0–5 km, F	1.85	4.53	3.09	1.13
p	0.1028	0.0005	0.0096	0.3464
0–10 km, F	1.64	3.37	2.45	0.79
p	0.0950	0.0003	0.0078	0.6367
Pre-period rings	Yes	Yes	Yes	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes
Device-range controls	Yes	Yes	Yes	Yes
Panel-size controls	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	133,406	133,406	133,406	133,404
R^2	0.110	0.091	0.101	0.014

Notes: The table reports the regression that carries both of the paper's treatments at once, so each is estimated holding the other fixed. Each column is one outcome and each row one estimated term. The specification is $\Delta Y_i = \beta_D D_i + \beta_L (L_i - c) D_i + \sum_{r=1}^{10} \beta_r D_{ir} + \sum_{r=1}^{10} \gamma_r P_{ir} + A_i'\theta + X_i'\delta + M_i'\lambda + S_i'\kappa + \alpha_{z(i)} + \varepsilon_i$. The four columns are the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections; the change in the Republican share on the same denominator; the net margin, the first minus the second; and the change in turnout, votes cast per voting-age person. All four are in percentage points, and the three vote-share columns are shares of all votes cast, never of the two-party vote. D_i is the participation extensive margin and equals one when at least one resident of the block group attended a George Floyd protest. $(L_i - c)\, D_i$ is the participation intensive margin, the second participation row: the log of attendances per homed device minus c for block groups that sent anyone, and zero for those that sent nobody, as in Section 5.1, with c the mean of that log among the sending block groups of the joint sample. D_{ir} equals one when at least one George Floyd protest had its venue in ring r, one kilometer wide, and P_{ir} is the same indicator for the protests of January 2017 to May 2020, the pre-period rings. A_i, M_i and S_i are the venue-traffic, device-range and panel-size controls of the rows near the foot, and X_i the demographics row, all as in Table A14; $\alpha_{z(i)}$ are zip code fixed effects, so that every comparison is between block groups in the same zip code. The regression is weighted by votes cast in 2020. The sample is the 133,406 block groups on which both treatments are observed together, 133,404 for turnout. Standard errors are in parentheses beside each coefficient and are clustered by metropolitan area; the fit carries 333 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10. The joint-test block reports, for the ring family at three widths, the F statistic and beneath it its p-value, each in its own labeled row.

Table A27: Participation and exposure in one regression, split by protest violence

	(1) Δ Dem (pp)	(2) Δ Rep (pp)	(3) Net margin (Dem − Rep, pp)	(4) Δ Turnout (pp)
Participation (device attendance, two-part, split by venue violence)				
Violent: Ext. margin	-0.015 (0.032)	0.001 (0.032)	-0.016 (0.063)	0.496 (0.336)
Violent: Int. margin (log rate)	-0.087*** (0.030)	0.072** (0.029)	-0.159*** (0.058)	1.110*** (0.399)
Nonviolent: Ext. margin	0.074*** (0.018)	-0.072*** (0.017)	0.146*** (0.034)	-0.129 (0.178)
Nonviolent: Int. margin (log rate)	0.016 (0.014)	-0.014 (0.014)	0.030 (0.027)	0.026 (0.154)
Exposure: violent protest, 1 km rings (overlapping sets; not a decomposition)				
0–1 km	-0.209 (0.144)	0.289** (0.134)	-0.498* (0.269)	-1.275** (0.609)
1–2 km	-0.165 (0.124)	0.243** (0.118)	-0.409* (0.239)	-0.885** (0.442)
2–3 km	-0.132 (0.093)	0.163** (0.080)	-0.295* (0.168)	0.312 (0.455)
3–4 km	-0.127 (0.088)	0.137 (0.084)	-0.264 (0.169)	0.119 (0.307)
4–5 km	-0.133* (0.072)	0.148** (0.071)	-0.280** (0.140)	-0.333 (0.306)
5–6 km	-0.102* (0.062)	0.101* (0.059)	-0.202* (0.119)	0.390 (0.378)
6–7 km	-0.078 (0.068)	0.088 (0.060)	-0.165 (0.126)	0.339 (0.313)
7–8 km	-0.069 (0.054)	0.037 (0.049)	-0.106 (0.101)	0.116 (0.250)
8–9 km	-0.009 (0.065)	-0.020 (0.063)	0.011 (0.127)	-0.151 (0.301)
9–10 km	-0.013 (0.059)	-0.008 (0.057)	-0.005 (0.115)	0.320 (0.260)
Exposure: nonviolent protest, 1 km rings (overlapping sets; not a decomposition)				
0–1 km	-0.095 (0.066)	0.149** (0.060)	-0.244* (0.125)	-0.304 (0.371)
1–2 km	-0.068 (0.051)	0.125** (0.049)	-0.194** (0.098)	-0.377 (0.294)
2–3 km	0.000 (0.041)	0.009 (0.041)	-0.009 (0.081)	-0.215 (0.288)
3–4 km	0.030 (0.038)	-0.023 (0.035)	0.053 (0.072)	-0.210 (0.211)
4–5 km	0.059* (0.032)	-0.074** (0.030)	0.133** (0.061)	-0.208 (0.209)
5–6 km	0.029 (0.029)	-0.049* (0.028)	0.077 (0.056)	-0.100 (0.235)
6–7 km	0.018 (0.034)	-0.021 (0.032)	0.040 (0.065)	-0.244 (0.258)
7–8 km	0.059** (0.030)	-0.058** (0.026)	0.117** (0.055)	-0.125 (0.163)
8–9 km	-0.014 (0.035)	0.017 (0.033)	-0.031 (0.067)	-0.134 (0.179)
9–10 km	0.024 (0.028)	-0.017 (0.028)	0.040 (0.055)	-0.142 (0.190)
Pre-period rings (split)	Yes	Yes	Yes	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes
Device-range controls	Yes	Yes	Yes	Yes
Panel-size controls	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	133,406	133,406	133,406	133,404
R^2	0.111	0.092	0.102	0.015

Notes: The table repeats the regression of Table A26 with both treatments split by whether the protest was violent. Every other element of that specification is unchanged, on the same joint sample of 133,406 block groups, 133,404 for turnout. Participation enters as four terms rather than two, an extensive margin and an intensive margin for violent protests and an extensive margin and an intensive margin for nonviolent ones, so a block group whose residents attended both kinds carries both; the two intensive margins are centered at the same constant c as in Table A26. Ext. margin = extensive margin; Int. margin = intensive margin. Exposure enters as two families of ten one-kilometer ring indicators, one for violent protests and one for nonviolent ones. The two ring families are overlapping sets and not a decomposition: a block group with a violent and a nonviolent protest in the same ring appears in both families there, so the two coefficients at a ring neither add to the pooled coefficient of Table A26 nor are shares of it. The same holds of the participation split. The earlier-protest ring indicators are split by protest violence as the exposure rings are; the other controls are unsplit, and the venue-traffic controls carry no violence split. Standard errors are in parentheses beside each coefficient and are clustered by metropolitan area; the fit carries 333 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10. On the net-margin column the two participation extensive margins differ between violent and nonviolent protests ($p = 0.046$) and so do the two intensive margins ($p = 0.005$), each a single contrast on 332 degrees of freedom, while the two exposure ring families are not statistically distinguishable, jointly over all ten rings ($F(10, 332) = 1.41$, $p = 0.174$) or at the first ring alone ($p = 0.399$), so the pattern in the exposure rows is a pattern in point estimates.

F.2 The average effects and the persuasion rates

Table A28 gives the two average effects of Section 6.2, beside the effects on a vote share that seven studies of other political shocks report, and Table A29 the persuasion rates that Figure 10 draws for this paper.

Table A28: The paper's two average effects and the vote-share effects reported by seven studies of other political shocks

Paper	Treatment	Dependent variable	Sample / population	Reported effect
Madestam et al. 2013	Rain-free Tax-Day Tea Party rally in county	Republican vote share, 2010	US counties; all voters	1.55 pp toward Republicans in the county vote share (as a share of county population, 1.04 pp); the vote-share result alone has p about 0.09 under the paper's own placebo inference
Adena et al. 2015	1 standard deviation radio subscription rate (panel; cross-sectional instrumental variables)	NSDAP vote share, 1933	German electoral districts	1–2 pp toward the NSDAP (sign reverses with content: pro-government radio 1930 gave −1.8 to −2.9 per standard deviation)
Autor et al. 2020	Interquartile increase in Chinese import penetration	Republican two-party presidential vote share	US counties; all voters	0.91 pp toward Republicans (two-stage least squares t about 1.9, marginally significant — the paper's own framing)
Pons 2018	Precinct randomly assigned for canvassing	Hollande presidential vote share, 2012, first round (intention to treat)	French precincts; field experiment	0.44–0.63 pp toward Hollande (intention to treat)
Spenkuch & Toniatti 2018	1 standard deviation partisan-ad differential (border-county pairs)	Two-party vote-share margin (Democratic − Republican)	US border-county pairs; all voters	0.49–0.67 pp in the margin, toward the candidate with more advertising
Martin & Yurukoglu 2017	Fox News viewing: 2.5 additional minutes per week, induced by cable channel position	2008 McCain two-party vote share by zip code (two-sample instrumental variables)	US zip codes	0.3 pp toward Republicans among induced viewers (the 0.46–6.34 pp figures are national counterfactual removals, not per-exposed-unit effects)
DellaVigna & Kaplan 2007	Town got Fox News on local cable, 1996–2000	Change in two-party Republican share, 1996–2000	9,256 towns, 28 states	0.4–0.7 pp toward Republicans
This paper (exposure)	Block group with a violent protest within 1 km (average effect on the treated; full violent-protest exposure)	Δ Republican vote share, 2016→2020 (pp)	US block groups in metropolitan areas; 333 clusters	**0.623** pp toward Republicans [0.133, 1.112]
This paper (participation)	Block group sending attenders to any protest (average effect on the treated)	Δ Democratic vote share, 2016→2020 (pp)	US block groups in metropolitan areas; 333 clusters	**0.091** pp toward Democrats [0.058, 0.124]

Notes: The two lower rows give this paper's two average effects (Section 6.2): a point estimate and a 95 percent confidence interval, on block groups in metropolitan areas, with standard errors clustered by metropolitan area and the 333 clusters the table prints. They are average effects over treated block groups computed from the violence-split joint regression of Table A27, on its sample of 133,406 block groups, and not the arms' own estimates of Sections 4 and 5: the backlash row is the average of that fit's ten violent-ring coefficients over the block groups with a violent protest within one kilometer, and the participation row the average of its four participation terms over the block groups that sent anyone. The seven upper rows give, for seven published studies of other political shocks, what the treatment is, what the outcome is, which population it was estimated on, and the effect on a vote share the paper reports: the span it reports across its own specifications, or in three cases a single reported estimate, not a confidence interval. The dependent variables of this paper are shares of all votes cast for president, never of the two-party vote; where another row names a two-party share, that is that paper's denominator and not this one's. The qualifications in the last column are the ones the papers themselves state or that the verification of each entry established. The effects in this table are in percentage points of a vote share, each for its own treatment, and are not persuasion rates; Figure 10 compares persuasion rates, and five of the seven studies appear there with the rate they publish.

Table A29: Persuasion rates of exposure and participation

		(1) $y_T - y_C$ (pp)	(2) y_0	(3) $e_T - e_C$	(4) f (%)
Exposure, any protest	Reach assumed	0.193	0.230	1.00	0.25
		(0.060)			(0.08)
	Reach assumed			0.50	0.50
					(0.16)
	Reach assumed			0.25	1.00
					(0.31)
Exposure, violent protest	Reach assumed	0.336	0.183	1.00	0.41
		(0.135)			(0.17)
	Reach assumed			0.50	0.82
					(0.33)
	Reach assumed			0.25	1.65
					(0.66)
Participation	Measured, per voting-age person, coverage-scaled, $c = 0.04$	0.073	0.538	0.0723	2.19
		(0.018)			(0.53)
	Measured, per voting-age person, coverage-scaled, $c = 0.19$			0.0152	10.39
					(2.53)

Notes: The table gives the persuasion rates that Figure 10 draws for this paper (DellaVigna and Gentzkow, 2010): among the people a treatment reached who were not already voting the way it pushes, the share whose vote it changed, $f = (y_T - y_C)/[(e_T - e_C)(1 - y_0)]$, in percent with $y_T - y_C$ in percentage points. Column (1), $y_T - y_C$, is the effect on the vote share: for exposure, the 0–1 km coefficient of Table A5 (any protest) or of Table A6 (violent protest), on the change in the Republican share; for participation, the extensive margin of Table A14, on the change in the Democratic share. Column (2), y_0, is the share already voting that way: the mean 2016 Republican share of the block groups with such a protest within one kilometer, for exposure, and the mean 2016 Democratic share of the block groups that sent anyone, for participation. Column (3), $e_T - e_C$, is the reach, the share of the treated group the treatment reached, the comparison group's being zero. For exposure it is assumed, since who among a block group's residents saw the protest is not observed: one, every resident within one kilometer, which gives the lower bound on the rate; one half; and one quarter. For participation it is measured: attendances per voting-age resident of the sending block groups, divided by c, the share of a protest's reported crowd that the device panel detects, at the median of that share over the 4,211 events with a reported crowd size, $c = 0.04$, and at its 75th percentile, $c = 0.19$. Column (4) is the rate. The effect, y_0 and their standard error are those of a block's first row and enter every row of the block. Standard errors in parentheses: beneath column (1), the coefficient's own, clustered by metropolitan area, on the samples of the three tables named; beneath column (4), that standard error carried through the formula with y_0 and the reach held fixed.

F.3 The aggregate decomposition

Table A31 is the decomposition behind Figure 11; Table A30 draws the exposure rings 2.5 km wide, and Table A32 extends the window to 2024 (Section 6.3).

Table A30: The aggregate decomposition with exposure rings 2.5 km wide

	Dependent variable: Net margin (Dem − Rep)		
	pp	Votes	95% CI (votes)
Total	0.155	165,892	[-32,973, 364,757]
Participation	0.126	134,900	[87,072, 182,728]
violent	0.036	38,190	[14,743, 61,637]
nonviolent	0.091	96,710	[51,539, 141,880]
Exposure	0.029	30,992	[-159,352, 221,337]
0–2.5 km	-0.049	-52,702	[-109,743, 4,339]
2.5–10 km	0.078	83,694	[-81,667, 249,056]
violent	-0.111	-118,228	[-248,063, 11,607]
0–2.5 km	-0.030	-32,272	[-63,070, -1,473]
2.5–10 km	-0.081	-85,956	[-199,953, 28,040]
nonviolent	0.140	149,220	[-6,923, 305,363]
0–2.5 km	-0.019	-20,431	[-64,947, 24,086]
2.5–10 km	0.159	169,651	[23,560, 315,742]
Observations (block groups)		133,406	

Notes: The table repeats the aggregate decomposition of Table A31 with the exposure rings drawn differently. Each row is a contribution to the aggregate net margin, reported in percentage points, in votes, and with a 95 percent confidence interval on the votes. The rows labeled Participation are the participation treatment and the rows labeled Exposure the exposure treatment; the indented rows split each by distance band, by protest violence, and by both. The rows within each treatment are a decomposition of that treatment and add to it; the two treatments add to the total. Here exposure enters as four rings two and a half kilometers wide and the near band is cut at two and a half kilometers; in Table A31 it enters as ten rings one kilometer wide and the near band is cut at three. Everything else is the same. Every interval is built as in Table A31, and each is a 95 percent interval for its own row, with no adjustment for the number of rows. The total has an interval that covers zero and is therefore not distinguishable from zero. Table A31 is the decomposition of record.

Table A31: The aggregate decomposition of the net margin, 2016–2020

	Dependent variable: Net margin (Dem − Rep), 2016–2020		
	pp	Votes	95% CI (votes)
Total	0.098	104,428	[-101,587, 310,444]
Participation	0.129	137,320	[89,379, 185,260]
violent	0.036	38,156	[14,581, 61,731]
nonviolent	0.093	99,164	[53,757, 144,571]
Exposure	-0.031	-32,891	[-225,455, 159,672]
0–3 km	-0.081	-86,868	[-164,841, -8,896]
3–10 km	0.051	53,977	[-104,766, 212,719]
violent	-0.100	-106,825	[-217,157, 3,507]
0–3 km	-0.033	-35,204	[-70,456, 48]
3–10 km	-0.067	-71,622	[-166,370, 23,126]
nonviolent	0.069	73,934	[-95,863, 243,731]
0–3 km	-0.048	-51,664	[-121,905, 18,576]
3–10 km	0.118	125,598	[-21,793, 272,990]
Observations (block groups)		133,406	

Notes: The table is the full decomposition of the aggregate net effect that Figure 11 plots, with every row the figure omits. Each row is a contribution to the aggregate, reported in percentage points of the net margin, in votes, and with a 95 percent confidence interval on the votes. The net margin is the change in the Democratic share of all votes cast for president minus the change in the Republican share, between the 2016 and 2020 elections; a positive number is a shift toward the Democrats. The aggregate is built by applying the estimated coefficients of the joint regressions, with the controls and zip code fixed effects of Figure 9, to the treated block groups of the estimation sample and summing the implied vote shift, so it is a statement about the electorate in the sample and not about the national electorate; the percentage-point column is the vote total divided by the votes cast in that electorate and is a mean, not a sum. The rows labeled Participation are the participation treatment and the rows labeled Exposure the exposure treatment, at least one protest with its venue within the stated distance band of the block group. Exposure enters as ten rings one kilometer wide, and the rows report them grouped into a near band, zero to three kilometers, and an outer band, three to ten. The indented rows under each treatment split it by protest violence, and under the violence split by band as well; the rows within a group add to the group and the two treatments add to the total, within the rounding of the printed integers, which differ by one vote in three places. Every interval is built the same way. The quantity is a weighted combination of the fit's own coefficients; its standard error is computed from the fit's own cluster-robust covariance matrix, clustered by metropolitan area, over the whole block of treatment terms, so that the covariances between those terms enter and not only their variances; and the interval is the estimate plus and minus a t critical value on the fit's own clusters less one, 332 degrees of freedom on 333 metropolitan-area clusters. Each is a 95 percent interval for its own row, with no adjustment for the number of rows, of which there are thirteen. The total has an interval that crosses zero and is therefore not distinguishable from zero; a positive point total with an interval that crosses zero is not evidence that the movement gained votes on net. The difference between the violent and the nonviolent distance profiles is not statistically established on this fit's own test over all ten rings ($F(10, 332) = 1.41$, $p = 0.174$), so the pattern in which the violent family is negative in both bands is a description of point estimates and not a demonstrated difference.

Table A32: The aggregate decomposition of the net margin, 2016–2024

	Dependent variable: Net margin (Dem − Rep), 2016–2024		
	pp	Votes	95% CI (votes)
Total	0.191	202,807	[-87,640, 493,254]
Participation	0.177	188,008	[107,920, 268,096]
violent	0.035	37,554	[5,660, 69,447]
nonviolent	0.142	150,454	[75,348, 225,561]
Exposure	0.014	14,799	[-253,149, 282,746]
0–3 km	-0.103	-108,955	[-203,166, -14,744]
3–10 km	0.117	123,754	[-84,040, 331,548]
violent	-0.141	-149,813	[-277,916, -21,710]
0–3 km	-0.066	-69,747	[-106,668, -32,825]
3–10 km	-0.076	-80,066	[-192,809, 32,677]
nonviolent	0.155	164,612	[-86,685, 415,908]
0–3 km	-0.037	-39,208	[-124,561, 46,144]
3–10 km	0.192	203,820	[12,416, 395,224]

Notes: The table repeats the decomposition of Table A31 over a longer window, from the 2016 election to the 2024 election rather than to the 2020 election. Each row is a contribution to the aggregate net margin over that window, in percentage points, in votes, and with a 95 percent confidence interval on the votes. The weights are votes cast in 2020 in both windows. The outcome here is constructed as the sum of the two windows it spans, the change from 2016 to 2020 and the change from 2020 to 2024, added inside the estimator, so one regression produces these intervals and no covariance between the two windows is added by hand. This table needs a 2024 return as well, so it is fitted on $N = 132{,}336$ block groups on 328 metropolitan-area clusters, and carries its own critical value at 327 degrees of freedom; every interval is built as in Table A31, on this fit's own coefficients and covariance matrix. Each is a 95 percent interval for its own row, with no adjustment for the number of rows. The difference between the violent and the nonviolent distance profiles is not statistically established on this fit's own test over all ten rings either ($F(10, 327) = 1.43$, $p = 0.165$, on this fit's own degrees of freedom), so that pattern remains a pattern in point estimates.

F.4 The swing-state exercise

Table A33 applies the national coefficients to the six states where the 2020 election was closest (Section 6.3).

Table A33: The aggregate net shift in the swing states against their 2020 margins

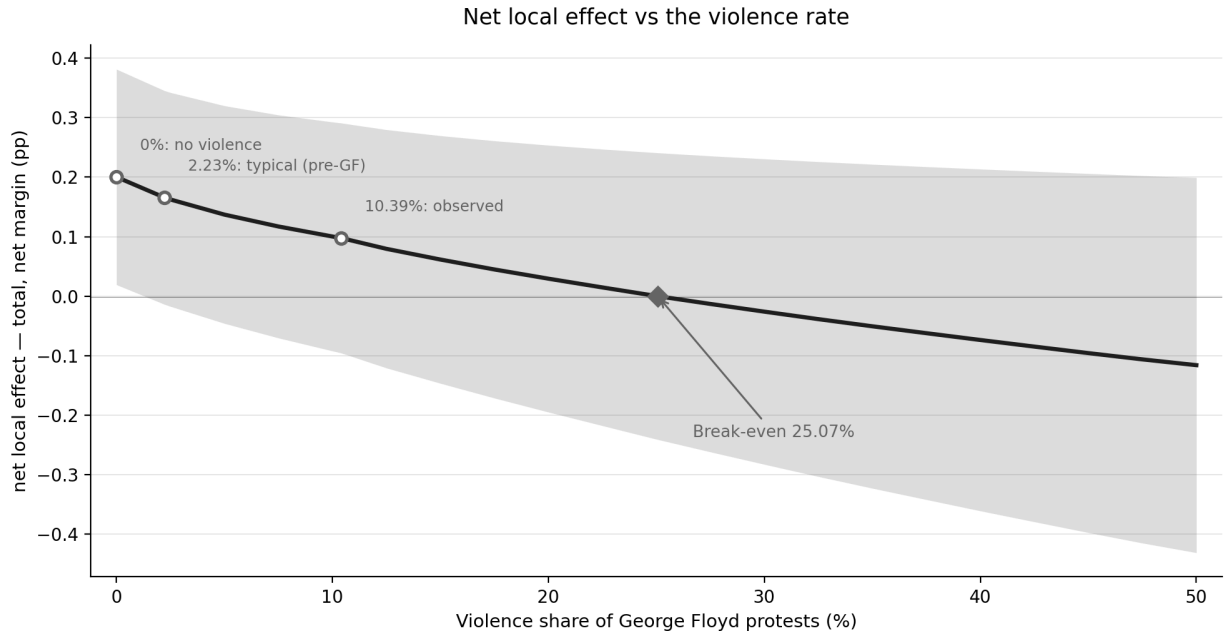
	2020 margin (D − R)	Estimated shift (votes)	Participation (votes)	Exposure (votes)
AZ	10,515	1,718 (1,092)	1,911*** (355)	-194 (970)
GA	12,670	5,753*** (1,962)	5,411*** (1,039)	342 (1,459)
WI	20,699	210 (2,889)	2,908*** (516)	-2,697 (2,784)
PA	81,723	6,216 (5,517)	6,801*** (1,244)	-586 (5,160)
NC	-74,481	4,535* (2,390)	5,630*** (990)	-1,095 (2,030)
MI	154,212	4,773 (3,339)	4,117*** (772)	656 (3,093)
AZ+GA+WI	43,884	7,681 (5,704)	10,231*** (1,853)	-2,549 (5,097)
Observations (block groups, national fit)		133,406	133,406	133,406

Notes: The table sets the estimated net shift in the swing states beside their 2020 margins. Each row is one state, and the last row is the three closest states taken together. The columns are the state's final 2020 margin in votes, the estimated net shift in votes, and the same shift split into the part coming from participation and the part coming from exposure. A positive shift is toward the Democrats. The shift is computed by applying the estimated coefficients to each treated block group in the state and summing, exactly as the national aggregate of Figure 11 is built, with the specification and controls listed in the note to Figure 9. The national coefficients of record are applied to each state's treated block groups, so the per-state split shows where the treated block groups are and how violent that state's protests were, not a state-specific response. The two contributions sum to the net shift by construction. The printed integers are rounded independently of one another, and in four of the seven rows the two printed contributions differ from the printed net shift by one vote as a result. Standard errors in parentheses, in votes, from the delta method on the fit's cluster-robust covariance; $N = 133{,}406$ block groups in the national fit. Significance stars are from the aggregator's own t statistic on 332 degrees of freedom: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Eight states and the District of Columbia are absent from the estimation sample altogether because they have no 2012 returns: Alaska, Arkansas, the District of Columbia, Hawaii, Louisiana, Nebraska, Nevada, South Dakota and Washington, so Nevada, one of the closest states in 2020, cannot be reported here. The vote columns add across states, because the block group sets are disjoint; the standard errors behind the stars are correlated across states and do not. The national aggregate this applies is Figure 11; the full decomposition behind it is Table A31.

F.5 The alternative-violence-rate exercise and the venue counterfactual

Figure A13 and Table A34 vary the share of violent protests, and Table A35 moves every protest to a venue that had hosted protests before the wave (Section 6.4).

Figure A13: The aggregate net effect as the share of violent protests varies



Notes: The figure traces the aggregate net effect as the share of protests that are violent is varied, holding everything else at what was observed. The horizontal axis is that share; the vertical axis is the aggregate net margin in percentage points, on the same construction and the same 133,406 block groups as Figure 11. The line is the point path. Landmarks are marked along the axis: the rate observed in the George Floyd wave, the rate typical of American protests in the three years before it, and the share at which the point path crosses zero. This is a model-based reallocation and not a causal claim about where violence occurred or why. It reallocates the exposure channel only: the distance rings are recomputed under counterfactual violence labels. The mobilization channel is held at its observed level at every point on the path: the attendance response to a protest's character is not observed and is not modeled. Every share on the axis above the observed rate is outside the range the data contain, so the right-hand part of the path is an extrapolation. The crossing is at a violent share of 25.1 percent, with a 95 percent confidence interval running from −20.0 to 70.2 percent, wider than the whole range of shares the figure draws, so the crossing locates where the central estimate changes sign and establishes nothing about where the sign actually changes; it is a landmark on the point path and not an estimated threshold. The scenarios behind the path are in Table A34.

Table A34: The aggregate net effect under three violence scenarios

	Dependent variable: Net margin (Dem − Rep), pp			
	Participation	Exposure	Total	Δ vs baseline
Baseline (observed violence)	0.129	−0.031	0.098	—
Violence at the typical rate (2.23%)	0.129	0.037	0.166	0.068 [−0.001, 0.137]
No violence	0.129	0.072	0.200	0.102 [−0.002, 0.207]
Observations (block groups)	133,406			

Notes: The table reports the aggregate net effect under three assumptions about how violent the protest wave was, and is the exercise Figure A13 traces continuously. Each row is one scenario and each column a contribution to the aggregate net margin in percentage points: the mobilization contribution, the exposure contribution, their total, and the difference between that total and the observed baseline. The specification is that of Table A31. The first row is what was observed. The second sets the violent share of protests to 2.23 percent, which is the share of protest events coded violent in the three years before the wave, 826 of 36,974 events. The third sets it to zero. The counterfactual is executed by relabeling violent protest days as nonviolent at random until the target share is reached, in 500 seeded draws, and recomputing the distance-ring exposures under the new labels. It is a model-based reallocation and not a causal claim about where violence occurred or why. The mobilization contribution is held at its observed level in every row: only exposure is reallocated, and the attendance response to a protest's character is not observed and is not modeled. The difference against the baseline is its own weighted combination of the fit's coefficients, so its standard error is that combination's own and not a difference of two separately computed errors; otherwise every interval is built as in Table A31. One source of variation sits outside the interval: which events are relabeled is drawn 500 times, and the spread of the exposure contribution across those draws, 0.002 percentage points, is a separate quantity and is not folded in. The columns are rounded independently of one another, so a total can sit one unit in the last place away from the sum of the two contributions printed beside it. Both counterfactual rows set the violent share below what was observed, so both sit inside the range of the data; a share above the observed rate would not, and Figure A13 marks where that begins.

Table A35: The aggregate net effect with every protest moved to a pre-existing venue

	Dependent variable: Net margin (Dem − Rep), pp				
	Participation	Exposure	Total	Δ	95% CI
Baseline (all venues)	0.129	−0.031	0.098	—	—
Pre-existing venues, attendance survives	0.129	−0.022	0.107	0.009	[−0.086, 0.104]
Pre-existing venues, attendance scales	0.098	−0.022	0.077	−0.021	[−0.119, 0.077]
Observations (block groups)	133,406				

Notes: The table reports the aggregate with every protest moved to a venue that already existed as a protest site before the wave, rather than to the new sites the wave created. Each row is one scenario and the columns are the mobilization contribution, the exposure contribution, their total, the difference from the observed baseline, and a 95 percent confidence interval on that difference, all in percentage points of the aggregate net margin. The specification is that of Table A31. The first row is what was observed. The exposure side of both counterfactual rows is computed by recomputing the distance-ring exposures on the pre-existing venue set alone. The participation side cannot be computed the same way, because whether a given block group's residents would have traveled to a different venue is not observed, so it is reported under two bounds rather than one: in the second row attendance survives the move unchanged, and in the third it scales with the venues that remain. This is a model-based reallocation of where protests were held and not a causal claim about why venues spread as they did. Every interval is built as in Table A31, and each change against the baseline is its own weighted combination, so its standard error is that combination's own and not a difference of two separately computed errors; each is a 95 percent interval for its own row. Neither bound is the estimate; the two are the range within which the participation side must lie under the exercise's own assumptions.

G North Carolina

This appendix reports the North Carolina estimates of Section 7 in full. The voter files and the measures built from them are described in Section 2.4 and Appendix A.4.

G.1 The paper's specifications in North Carolina

Section 7 reports the headline reproduction; this subsection carries the detail. The North Carolina sample is the state's block groups of the national samples, 3,655 in the exposure specification and 3,649 in the participation and joint specifications (Section 2.4); it is neither the national sample nor a random sample of the state, and block groups outside it are unobserved rather than untreated. The shift toward the Republicans near a violent protest within one kilometer, 1.11 percentage points in North Carolina against 0.34 nationally, is a point estimate about three times the national one on a much wider interval; that the North Carolina estimate is the larger is a comparison of two point estimates, and Section 4.7 found the dispersion of the state estimates consistent with a common effect. The channels are the three measures of Section 2.4. Turnout is the share of the people on the roll at both the 2016 and the 2020 election who cast a ballot, so its change is a change in the behavior of a fixed set of people and not in who is on the roll. The composition of the roll is measured by the change in the Democratic share of registered voters, which is not a vote share: it counts the party people registered with, whether or not they voted; and the five movements that change it, new registrations, removals from the roll, moves in from elsewhere in the state, moves out, and party changes, are counted person by person and sum to that change exactly.

Table A36: Exposure, participation and the joint specification in North Carolina and nationally

	Urban NC	National
Exposure specification: Δ Republican share (pp); national column is the exposure fit, N 136,644		
Violent 0–1 km	1.11** (0.42)	0.34** (0.14)
Violent 1–2 km	0.21 (0.32)	0.27** (0.12)
Violent 2–3 km	0.59* (0.29)	0.16** (0.08)
Violent 3–4 km	0.64*** (0.22)	0.10 (0.08)
Participation specification: Δ Democratic share (pp); national column is the participation fit, N 135,691		
Attendance (extensive margin)	−0.048 (0.086)	0.073*** (0.018)
Attendance (intensive margin)	0.068 (0.051)	0.014 (0.014)
Joint specification: Δ net margin (pp); national column is the pooled joint fit, N 133,406		
Violent 0–1 km	−2.436** (0.853)	–
Attendance (extensive margin)	−0.043 (0.160)	0.146*** (0.034)
Attendance (intensive margin)	0.176 (0.099)	0.037 (0.026)
Block groups: exposure	3,655	136,644
Block groups: participation	3,649	135,691
Block groups: joint	3,649	133,406

Notes: The table runs the paper's three vote specifications on urban North Carolina alone, beside their national estimates. The three blocks are the three specifications. The exposure specification's outcome is the change in the Republican share of all votes cast for president between 2016 and 2020, in percentage points. Its treatment differs between the two columns: in the North Carolina column it is twenty one-kilometer ring indicators, ten for a violent and ten for a nonviolent George Floyd protest, entered jointly, of which the first four violent rings are printed; in the national column it is the ten violent-protest ring indicators of Figure 3, of which the same four are printed. The participation specification's outcome is the change in the Democratic share of all votes cast, and its treatment the participation extensive and intensive margins of Table A14. The joint specification carries both treatments at once, with the change in the net margin as its outcome, the Democratic share change minus the Republican share change, so a negative number is a shift toward the Republicans. All three carry the same controls in both columns: the venue-traffic, mobility, panel-size and demographic controls of Table A14, the earlier-protest ring indicators of Figure 2, and zip code fixed effects, so that every comparison is between block groups in the same zip code. Weights are votes cast in 2020. The lower rows give each specification's number of block groups. Each block's second column carries that block's own national estimate, from three national fits and not one. The North Carolina sample is the North Carolina block groups of the national samples. Standard errors in parentheses are cluster-robust throughout, clustered by North Carolina county in the first column, 44 clusters, and by metropolitan area in the second, 344 clusters for the exposure specification and 333 for the participation and joint specifications. Stars come from each column's own inference of record: a wild cluster bootstrap on the county clusters, Rademacher weights and 9,999 draws, for North Carolina, because 44 clusters are too few for the asymptotic standard error to be trusted, and the asymptotic test for the national column, so a coefficient divided by the standard error beside it does not always reproduce the North Carolina stars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A37: North Carolina channel estimates in the joint and the single-treatment specifications

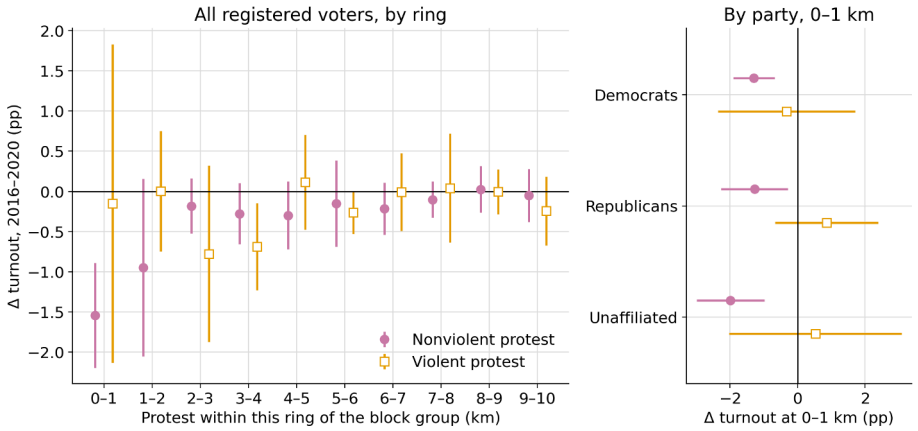
	Joint	Exposure arm	Participation arm
Turnout, nonviolent 0–1 km	−1.54*** (0.33)	−1.59*** (0.37)	–
Turnout, violent 0–1 km	−0.15 (1.01)	−0.26 (0.98)	–
Turnout, attendance (extensive margin)	−0.04 (0.11)	–	−0.08 (0.11)
Dem. registration share, nonviolent 0–1 km	−0.03 (0.40)	0.08 (0.41)	–
Dem. registration share, violent 0–1 km	−1.14* (0.51)	−1.04* (0.51)	–
Dem. registration share, attendance (extensive margin)	−0.09 (0.10)	–	−0.11 (0.10)
Net Dem→Rep switching, nonviolent 0–1 km	−0.891 (0.561)	−1.022* (0.565)	–
Net Dem→Rep switching, violent 0–1 km	−0.365 (1.039)	−0.564 (1.032)	–
Net Dem→Rep switching, attendance (extensive margin)	0.350 (0.213)	–	0.369* (0.206)
Dem. new registration, nonviolent 0–1 km	−0.002 (0.004)	0.001 (0.005)	–
Dem. new registration, violent 0–1 km	0.040*** (0.013)	0.041*** (0.015)	–
Dem. new registration, attendance (extensive margin)	0.002* (0.001)	–	0.002 (0.001)
Block groups	3,649	3,655	3,649

Notes: This table estimates each channel three times: once in the joint specification, once in the exposure specification alone on that arm's own sample and control set, and once in the participation specification alone on its own. A dash marks a cell the specification does not contain, because the exposure arm has no attendance term and the participation arm has no ring geometry. The four channels are the rows, in groups. Turnout is the change between the 2016 and 2020 general elections in the share of a block group's registered voters who cast a ballot, in percentage points, as defined in Section 2.4. The Democratic registration share is the change in the Democratic share of its registered voters, in percentage points. Net Democratic-to-Republican switching counts people already registered at both ends who changed the party they are registered with, as the number who moved from Democratic to Republican minus the number who moved the other way, per thousand of the block group's baseline cohort. Democratic new registration counts people who appear on the roll in 2020 and not in 2016 at that block group and are registered as Democrats, per unit of the 2016 registered base. The three treatments named in the rows are an indicator for at least one nonviolent George Floyd protest within one kilometer of the block group, the same for a violent protest, and the participation extensive margin of Table A14. The participation intensive margin is carried in every one of these regressions and is not reported. All three specifications carry zip code fixed effects, so that every comparison is between block groups in the same zip code, and each carries its own control battery, as in Table A36: the joint and exposure specifications the venue-traffic controls, the four ranging measures, the device-decile indicators, the demographic controls and the earlier-protest ring indicators; the participation specification its own baseline-visit battery on the same demographic block. Each regression is weighted by its outcome's own baseline denominator. The joint and participation columns are 3,649 block groups and the exposure column 3,655, because each arm's sample and its own listwise deletion drop different rows. Standard errors in parentheses are cluster-robust, clustered by North Carolina county, 44 clusters in every column; stars come from a wild cluster bootstrap on those clusters with Rademacher weights and 9,999 draws, which is the inference of record, so the stars will not always match the ratio of a coefficient to the standard error printed next to it. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Decimal places follow each channel's own units, two for the two share outcomes and three for switching and new registration. The three columns are not estimated on the same block groups, so they are compared on sign and rough magnitude, and a cell is never subtracted from the cell beside it. Figure A16 decomposes the registration share of the second group of rows.

G.2 The turnout channel

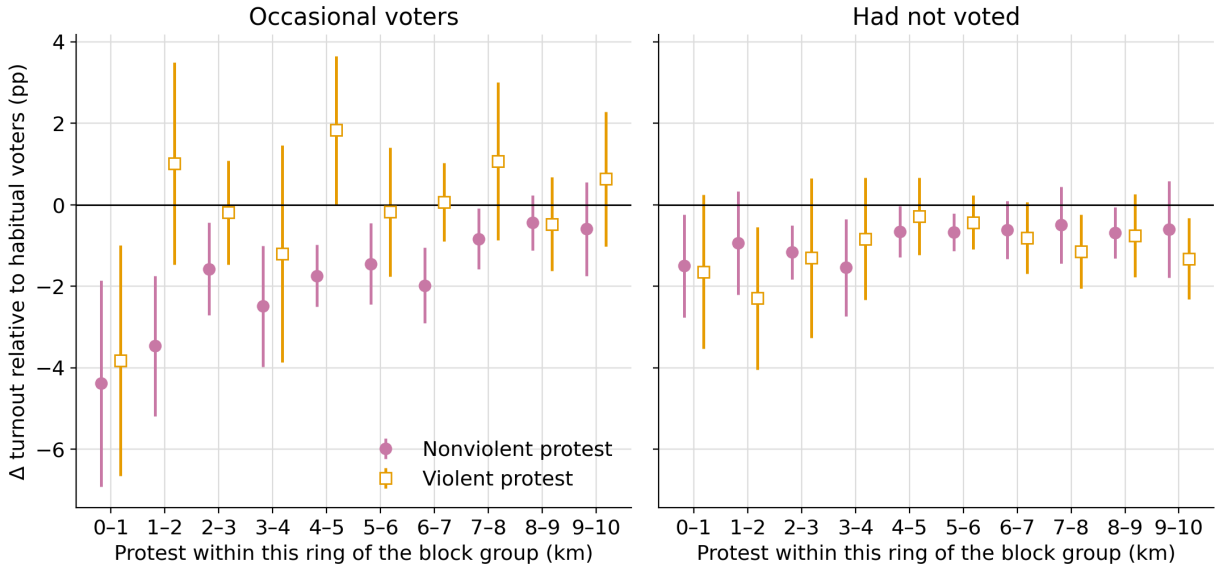
Figure A14 shows a decline in turnout near nonviolent protests and no detectable change near violent ones, and the joint test that the two families of rings are equal, $F(10, 43) = 1.81$, $p = 0.09$, does not establish the difference at the 5 percent level. Its right panel repeats the estimate within one kilometer for registered Democrats, Republicans and unaffiliated voters separately: near nonviolent protests turnout fell for each of the three, each with an interval below zero, and near violent protests none of the three intervals excludes zero; the three are separate regressions, so the larger fall among the unaffiliated is a difference of point estimates.

Figure A14: Turnout near violent and nonviolent protests in North Carolina, by distance and by party



Notes: The outcome in both panels is the change between the 2016 and 2020 general elections in the share of the block group's registered voters who cast a ballot, in percentage points, as defined in Section 2.4. Each marker is an estimated coefficient and each whisker a 95 percent confidence interval. The left panel gives the effect for all registered voters together, by distance. Its treatment is two sets of ten indicators, one set for violent and one for nonviolent protests, in one-kilometer rings. The right panel repeats the within-one-kilometer estimate on four separate regressions, one for each party a voter can be registered with at the start of the period, and shows three of them: registered Democrats, registered Republicans and registered unaffiliated voters. The specification is otherwise the one used throughout the section, with the controls and zip code fixed effects of Table A36, together with the two-part participation treatment. Each regression is weighted by the number of people in its own baseline group. The sample in the left panel is 3,649 block groups of urban North Carolina in 325 zip-code cells; in the right panel each party's regression runs on the block groups with registrants of that party, 3,649 for Democrats and 3,648 each for Republicans and the unaffiliated. Standard errors are clustered by North Carolina county, 44 clusters, with a wild cluster bootstrap on those clusters, Rademacher weights and 9,999 draws, as the inference of record. The violent estimate within one kilometer is −0.15 percentage points with an interval that covers zero and is wide enough to contain the nonviolent estimate as well, so the figure shows a decline near nonviolent protests and no detectable change near violent ones rather than a demonstrated difference between the two, and the joint test of the two families gives $F(10, 43) = 1.81$, $p = 0.089$.

Figure A15: Turnout of occasional voters and past non-voters against habitual voters, by distance

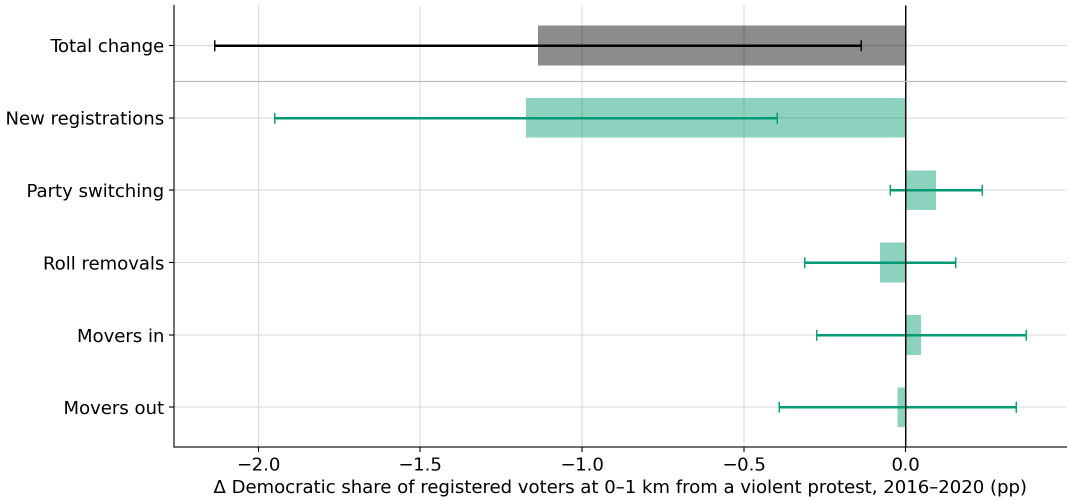


Notes: The unit here is a person and not a block group: the estimation uses the 3,606,754 people registered at both the 2016 and the 2020 snapshots and placed in one of the 3,649 block groups in 2016. The outcome is whether that person voted in the 2020 general election minus whether they voted in 2016. Each person is sorted, on their record before the protests, into one of three groups by the two pre-2020 elections the state's voter-history file still covers: habitual voters, who voted in both the 2016 general election and the 2018 midterm; occasional voters, who were registered by the 2018 midterm and voted in exactly one of the two; and registrants who had not voted, who were registered by the 2018 midterm and voted in neither (Appendix A.4). Each panel plots one group against the habitual voters, by distance: the left panel occasional voters, the right panel registrants who had not voted. The markers are estimated coefficients on the interaction of the group indicator with each one-kilometer ring indicator, filled circles for nonviolent and open squares for violent protests, and the whiskers are 95 percent confidence intervals. The regression carries block group fixed effects, so that every comparison is between people in the same block group, together with the group indicators themselves and the two-part participation treatment interacted with group; the exposure and participation controls of the block-group specifications enter through the block group fixed effect. Standard errors are clustered by North Carolina county, 44 clusters, with a wild cluster bootstrap on those clusters, Rademacher weights and 9,999 draws, as the inference of record. Whether a protest was nearby is a property of the block group, not of the person, so its own effect is absorbed by the block group fixed effect: every coefficient plotted is how much further one group's turnout moved than the habitual voters' in the same block group. No individual record left the secure environment in which the voter file is held; the estimation was run there and only coefficients, standard errors and counts were returned.

G.3 The registration channel

In the five-movement split of Figure A16, party switching runs the other way from new registration and adds 0.09 percentage points, and removals and moves in and out are each not distinguishable from zero.

Figure A16: The change in the Democratic registration share near violent protests, by component



Notes: The outcome is the change between the 2016 and 2020 general elections in the Democratic share of the block group's registered voters, in percentage points, and the figure reports it for block groups with a violent George Floyd protest within one kilometer. The top bar is that total change. The five bars below it are the five movements of Appendix A.4 each written as its own contribution to the change in the Democratic share over a common denominator, the block group's 2016 registered base, so that the five sum to the total exactly. Each bar is a separate regression of that component on the specification of the section, with the controls and zip code fixed effects of Table A36, each weighted by the block group's 2016 registered base. The sample is 3,649 block groups of urban North Carolina in 325 zip-code cells. Whiskers are 95 percent confidence intervals; standard errors are clustered by North Carolina county, 44 clusters, and the inference of record is a wild cluster bootstrap on those clusters with Rademacher weights and 9,999 draws. In the 50 block groups with a violent protest within one kilometer, the 2016 roll was 51.2 percent Democratic and the new registrants of 2016 to 2020 are 41.8 percent Democratic, against 31.5 percent for new registrants in the rest of the urban North Carolina sample, outside this ring. Every one of the 50 block groups with a violent protest within one kilometer is in the upper half of the sample's distribution of residential turnover measured over 2012 to 2016, and none is in the lower half, while of the 151 block groups with a nonviolent protest within one kilometer, 136 are in the upper half and 15 in the lower.

G.4 The composition placebo

In the composition placebo of Table A38, at two to four kilometers the check is not clean: moves out, and at one band each new registrations and moves in, load on the 2020 treatment, and the total row with them, so whatever the 2016-to-2020 estimates pick up at those distances was already under way in the earlier window, and they are not read as effects of the 2020 protests.

Table A38: Components of the North Carolina registration-share change over 2012–2016, on the 2020 violent-protest rings

Component (2012–2016 change)	Violent 0–1 km	Violent 2–3 km	Violent 3–4 km
Total Dem. share change	−0.64 (0.61)	−0.76* (0.24)	−0.85* (0.35)
New registrations	−0.67 (0.36)	−0.60** (0.16)	−0.38 (0.26)
Party switching	0.11 (0.08)	−0.04 (0.05)	−0.04 (0.06)
Roll removals	0.05 (0.14)	−0.10 (0.09)	−0.01 (0.09)
Movers in	−0.05 (0.29)	−0.21 (0.18)	−0.57** (0.11)
Movers out	−0.08 (0.15)	0.19* (0.09)	0.16* (0.06)

Notes: This table takes the five components of the registration-share change and computes them over the earlier window from the 2012 to the 2016 general election, then regresses each of them on the 2020 treatment. The rows are the total change in the Democratic share of a block group's registered voters over 2012 to 2016, in percentage points, and the five movements of Appendix A.4 that sum to it exactly: new registrations, party switching among people on the roll at both ends, removals from the roll, people moving in and people moving out. The columns are three of the violent-protest ring indicators of 2020, at zero to one, two to three and three to four kilometers. The specification is the one used throughout the section, with the controls and zip code fixed effects of Table A36, weighted by the 2012 registered base. The sample is the one used throughout the section, 3,649 block groups of urban North Carolina; each component is one regression, the three printed rings are three terms of that regression, and the total row is its own regression on the same basis as the components and not their sum. Standard errors in parentheses are cluster-robust, clustered by North Carolina county, 44 clusters; stars come from a wild cluster bootstrap on those clusters with Rademacher weights and 9,999 draws, the inference of record. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.